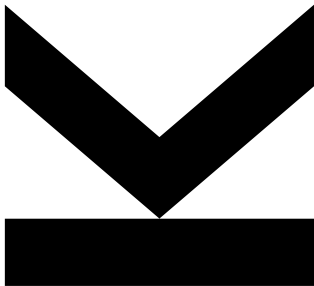


AN ENHANCED ALGORITHM FOR SIMULATING NON- UNITARY DYNAMICS ON QUANTUM COMPUTERS



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STATUTORY DECLARATION

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Abstract

This thesis introduces an algorithm for efficiently simulating the non-unitary dynamics of open quantum systems on quantum computers. Open quantum systems interact with their surrounding environment, resulting in non-unitary dynamics. This creates a fundamental challenge for quantum simulation, since quantum hardware is inherently restricted to unitary operations. Building on the Kraus operator sum representation, this work explores dilation techniques that embed non-unitary maps into larger unitary operations, suitable for implementation on quantum processors. In addition to reproducing recent results from the literature, an enhanced algorithm that reuses the quantum state from the previous time step is presented. This reuse strategy increases efficiency, permits repeated application of the same quantum circuit, and shifts the dominant computational workload toward the quantum processor. The method is implemented and tested on models of the Fenna–Matthews–Olson light-harvesting complex and compares the obtained results against existing approaches. Importantly, the algorithm is broadly applicable to a wide class of open quantum systems. The proposed scheme provides a practical pathway for studying transport, dynamics, and spectral properties of interacting quantum systems on a quantum simulator, while keeping the demands on classical resources modest. The thesis concludes with a discussion of implementation details, numerical results, and prospects for scaling the method to larger systems.

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1 Introduction

Quantum computing represents an emerging field of research with the potential to tackle problems beyond the reach of today’s most powerful supercomputers [1][2]. Over the past years, substantial advances in building and operating quantum processors have been made, turning the subject from a mainly theoretical idea into a practical technology [3]. Unlike classical computing, quantum computing processes information in a fundamentally different way, which makes it particularly effective for certain problem classes that are hard to simulate on conventional hardware [4].

Many systems of central scientific interest are themselves quantum. Their state spaces grow exponentially with system size, making an exact brute-force simulation on classical supercomputers infeasible [5]. Richard Feynman first articulated this observation in 1982 and argued that a controllable quantum device could simulate quantum dynamics more naturally than classical computers [6].

Classical computers store and process information using bits, which can only be in one of two states, $|0\rangle$ or $|1\rangle$. A register with n bits holds exactly one n -bit string at a time, and classical algorithms manipulate these strings step-by-step using logic gates. In contrast, quantum computers use qubits. A single qubit can exist in a superposition of the classical states $|0\rangle$ and $|1\rangle$, and a register of n qubits spans a 2^n -dimensional Hilbert space. This implies that a quantum state is specified by 2^n complex amplitudes. The exponential growth of the state space is a key reason why quantum computing is so promising: quantum devices can naturally encode and manipulate information that would require exponentially large memory to store on a classical machine [7].

However, exploiting this advantage is not without its challenges. Quantum states are fragile, and operations on current hardware are noisy. Furthermore, information cannot be read out in the same way as in a classical register. Instead, measurements are required that yield probabilistic outcomes according to Born’s rule [8]. While this allows us to estimate expectation values or probabilities with great accuracy through repeated executions of the same circuit, it is not sufficient to reconstruct the full quantum state, as its complex amplitudes cannot be directly captured. In practice, reconstructing the full quantum state is a very challenging endeavor that requires advanced methods such as quantum tomography [9][10].

In practice, quantum computers do not replace classical computers entirely but work in close synergy [11]. As schematically illustrated in Figure 1.1, a classical processor is essential at every stage of the workflow. It translates algorithms into sequences of quantum gates, controls their timing and execution on the quantum hardware, and collects the measurement data. Following the execution of the quantum algorithm, the classical processor interprets the results, often applying statistical analysis or optimization routines [12]. This division of labor allows the quantum processor to explore the exponentially large Hilbert space, while the classical processor guides the computation and adjusts parameters.

An intuitive way to understand the advantage of quantum computers is through the many-particle problem as discussed by Fauseweh [13]. Consider a system of many interacting quantum particles described by a microscopic Hamiltonian. Classically, the exact state of the whole system is encoded by a table whose size grows exponentially with the number of particles. Simulating its time evolution by direct numerical methods quickly becomes impossible as the system’s size increases. A quantum computer bypasses this rapid scaling: the same Hamiltonian that defines an exponentially large Hilbert space for a classical description is

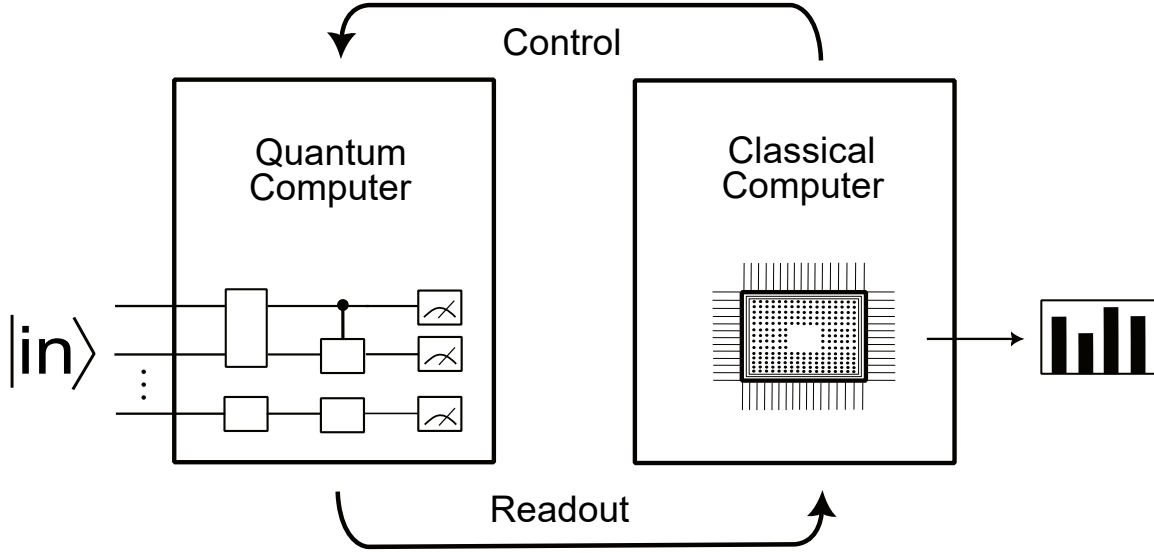


Figure 1.1: Synergy between classical and quantum computers. A quantum computer executes a programmable sequence of elementary quantum operations on an initial state, producing a final state that can then be measured. While the quantum computer performs the core calculations, a classical computer is required to decompose algorithms into sequences of quantum gates and collect the measurement data.

simply the generator of the unitary evolution on the qubits, which the quantum device can directly follow. Intuitively, the quantum processor itself is a quantum system and therefore can avoid reconstructing the full classical description of the evolution [6]. These capabilities make quantum devices particularly attractive for studying transport, dynamics, and spectral properties of large interacting systems, where classical simulations become infeasible [4][14].

Building on the outlined fundamental advantages, this work addresses a critical challenge in the field of quantum simulations: the efficient and accurate simulation of open quantum systems on quantum computers. To implement and test the developed methods, simulations were carried out using the quantum computer simulator provided by the Python library Qiskit.

Open quantum systems interact with and are influenced by their external environment. They appear across a broad spectrum of systems, from biological processes to applications in quantum technologies. Since they are not isolated, open quantum systems cannot be described by a unitary evolution like closed systems. Instead, their dynamics is governed by a non-unitary evolution. However, a quantum computer itself ideally represents a closed system and thus only supports unitary operations that preserve the norm of the qubits [15]. Hence, implementing non-unitary operations on a quantum computer proves challenging. To address this limitation, this work discusses methods proposed by Seneviratne et al. [16], Hu et al. [17], and Dan et al. [18], that “dilate” a non-unitary matrix into a unitary one, which can be implemented on a quantum computer. The term “dilate” refers to the embedding of a smaller non-unitary matrix into a larger unitary one that acts on a higher-dimensional Hilbert space. In addition, this work introduces an enhanced version of the method that reuses the quantum state from the previous time step, providing a saving in computational cost and ensuring that the vast majority of the workload is shifted to the quantum computer. To this end, the thesis focuses on the implementation of the method to simulate the dynamics of

open quantum systems, specifically demonstrating its application to the Fenna-Matthews-Olson complex. The core of the method is an algorithm based on the Kraus operator sum representation, which is designed to be generally applicable to a wide class of open quantum systems. It offers a promising path toward understanding and controlling their behavior.

The structure of this thesis is organized as follows. Chapter 2 provides a detailed introduction to open quantum systems and light-harvesting complexes, with a specific focus on the FMO complex and its governing Frenkel Exciton Hamiltonian. Chapter 3 introduces the necessary mathematical concepts for describing and analyzing open quantum systems, including the density operator, composite systems and the partial trace. Chapter 4 examines the dynamics of open quantum systems, discussing fundamental theoretical frameworks such as the Lindblad master equation, the Kraus operator sum representation and the hierarchical equations of motion. Chapter 5 offers a brief introduction to quantum computation and quantum circuits, which are essential for understanding the practical implementation. Finally, Chapter 6 presents the results for the simulations of the FMO complex, including reproduced results from Seneviratne et al. [16], Hu et al., [17] and Dan et al. [18], as well as novel results generated using an enhanced algorithm developed in this thesis.

2 Open Quantum Systems and Light-Harvesting Complexes

Light-harvesting complexes (LHCs) are extended pigment–protein assemblies, consisting of light-absorbing molecules embedded in a protein structure that holds them in place [19]. Although these systems are complex and large, we are often only interested in a certain small part of the entire LHC. This subsystem, which is connected to its environment, is called an open quantum system. In fact, any quantum system in contact with another quantum system can be considered as an open quantum system [20]. Motivated by the interest in only a small part of the entire system, we introduce the system-bath picture. Here, the degrees of freedom of the subsystem of interest are referred to as the “system”, and the remaining degrees of freedom as the “bath” or “environment” [21].

The interaction between these two components usually significantly influences the system, whereas the bath is typically viewed as a reservoir. This leads to processes such as decoherence and energy dissipation [22]. Decoherence refers to the gradual decay of quantum superpositions intrinsic to the system into classical statistical mixtures, which destroys the phase relation between the components of the quantum state and thereby eliminates quantum interference effects [23][24]. Dissipation is the process through which a system loses energy to its environment and relaxes towards the thermal equilibrium. It is typically associated with changes in the populations of the energy levels, i.e., with the diagonal terms of the system density matrix, which is introduced in Section 3.1. In contrast, decoherence mainly affects the off-diagonal elements.

LHCs consist of molecules composed of numerous nuclei surrounded by electrons. The electronic degrees of freedom of the electrons surrounding the nuclei form the system, while the bath is modeled as an infinite set of harmonic oscillators, representing the vibrational modes of the nuclei [25][26]. Thus, the Hamiltonian of an open quantum system generally consists of three terms [27]:

$$H = H_S + H_B + H_{SB}. \quad (2.1)$$

Here, H_S is the electronic Hamiltonian of the system, H_B the vibrational or phononic Hamiltonian of the bath, and H_{SB} is the interaction Hamiltonian between the electronic system and the phononic bath. A detailed explanation of the Hamiltonian follows in Section 2.2.

2.1 The Fenna-Matthews-Olson Complex

The Fenna-Matthews-Olson (FMO) complex plays a vital role in photosynthesis, since it converts sunlight into chemical energy. The photons of the sunlight are absorbed by an antenna, creating excitations in the system which are referred to as excitons. The FMO complex then transfers these excitons to the reaction center with an efficiency of nearly 100%. The transfer occurs typically on the order of picoseconds [28].

The FMO complex is the first LHC whose molecular structure was resolved. As determined by Fenna in 1975 [29], it is a trimer consisting of three identical monomers with a nearest neighbor distance of about 24 Å. The monomers have a global C3 symmetry, which implies that they are arranged around a three-fold symmetry axis. For a long time, it was believed that each monomer contains seven bacteriachlorophyll molecules (BChls) with a nearest neighbor distance of 11.3 to 14.4 Å [28]. However, in 2011, an eight BChl molecule was discovered [30], which enhances the interaction with the antenna [31]. Since the simulations in this work treat the FMO complex as isolated from both the antenna and the reaction center, only seven

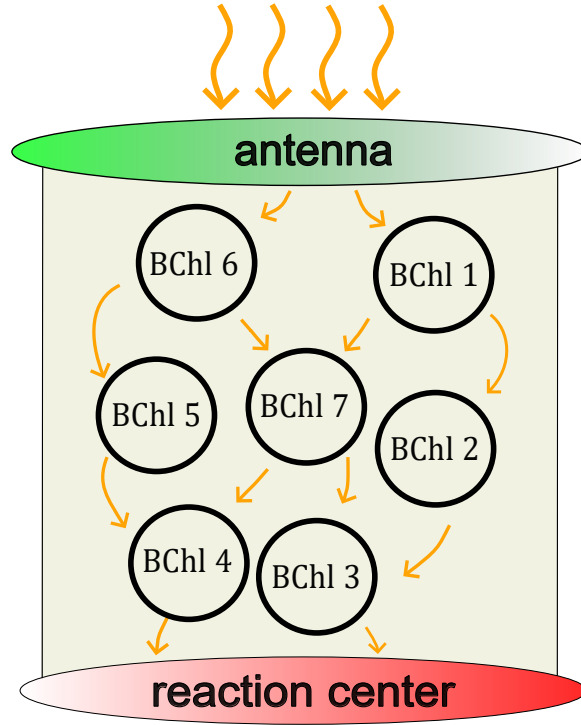


Figure 2.1: Schematic drawing of a single monomer of the FMO complex, consisting of seven BChl molecules. The excitation is absorbed at the antenna and transported to the reaction center via the BChl molecules, where it is stored as chemical energy. BChl 1 and 6 are strongly coupled to the antenna, while BChl 3 and 4 connect to the reaction center. The arrows indicate the excitation transfer through the complex, but do not represent all possible transitions.

BChl molecules need to be considered. The monomers are embedded in a polypeptide protein environment, and since individual monomers interact only weakly, the analysis can be restricted to a single monomer. The basic structure of the FMO complex is shown in Figure 2.1.

The wavelength to excite the electronic configuration of the BChl molecules is around 800 nm and thus in the infrared range. If an excitation enters a monomer of the FMO complex, it delocalizes over the seven BChl molecules as it is transported towards the reaction center. The delocalization is mediated by a dipole-dipole interaction between the electronic configurations of the individual BChls, which is introduced in Section 2.2.1. As shown in Figure 2.1, BChl 1 and 6 are located near the antenna [32][33], and BChl 3 and 4 near the reaction center [34][35]. It is therefore believed that the corresponding BChls interact strongly with the antenna and the reaction center, respectively. Consequently, the excitation is assumed to be initially localized primarily at BChl 1 and 6 and shifts towards BChl 3 and 4 as the excitation is transported towards the reaction center, where it is trapped and stored as chemical energy [36].

2.2 The Frenkel Exciton Hamiltonian

The Frenkel exciton model describes the excitation dynamics in light-harvesting complexes such as the FMO complex. In this system, the electronic configurations localized on different BChls can interact via the Coulomb interaction, enabling energy transfer between the individual BChls. However, no electrons can be exchanged since the BChls are spatially separated [36]. In the following, we derive the Frenkel exciton Hamiltonian for the FMO complex, following Kreisbeck [36] and Chapters 2 and 8 of May and Kühn [37].

2.2.1 The System Hamiltonian

The system Hamiltonian describes the degrees of freedom of the electronic configuration of the BChl molecules and is given as:

$$H_S = \sum_i H_i^{el} + \sum_{i < j} V_{ij}. \quad (2.2)$$

Here, H_i^{el} corresponds to the electronic Hamiltonian at the i -th BChl molecule and V_{ij} describes the coupling between the electronic configurations at different BChl molecules via Coulomb interaction. Thus, it is given as:

$$V_{ij} = \frac{1}{4\pi\epsilon} \int d^3r \int d^3r' \frac{\varrho_i(\vec{r}) \varrho_j(\vec{r}')}{|\vec{r} - \vec{r}'|}, \quad (2.3)$$

where $\varrho_i(\vec{r})$ and $\varrho_j(\vec{r}')$ correspond to the charge density on BChl i and j . The distance between the BChls is sufficiently large to consider the charge distributions as centered around the individual BChls, which makes a dipole approximation viable. Thus, we can approximate the electronic coupling V_{ij} as:

$$V_{ij} \approx \frac{1}{4\pi\epsilon} \left[\frac{\vec{\mu}_i \cdot \vec{\mu}_j}{r_{ij}^3} - \frac{3(\vec{\mu}_i \cdot \vec{r}_{ij})(\vec{\mu}_j \cdot \vec{r}_{ij})}{r_{ij}^5} \right]. \quad (2.4)$$

The distance between BChl i and j is denoted as $r_{ij} = |\vec{r}_j - \vec{r}_i|$ and the dipole moments of the electronic configuration on site i can be written as $\vec{\mu}_i = \int d^3r \varrho_i(\vec{r}) (\vec{r} - \vec{r}_i)$.

In the following, we introduce a basis to describe the effect of an excitation on the electronic structure. We first note that the BChls are spatially separated, which leads to a stronger interaction of the electrons at a certain BChl molecule with the protein environment than with neighboring BChl molecules. Thus, the coupling V_{ij} can be treated as a perturbation, and the electronic states of the BChl molecules can be used to introduce a basis for the electronic configuration of the coupled system. Therefore, we start by:

$$H_i^{el} |\phi_i^e\rangle = \epsilon_i^e |\phi_i^e\rangle, \quad (2.5)$$

$$H_i^{el} |\phi_i^g\rangle = \epsilon_i^g |\phi_i^g\rangle, \quad (2.6)$$

where H_i^{el} is the electronic Hamiltonian of the BChl molecule i , and ϕ_i^g and ϕ_i^e denote the ground and excited state of the electronic configuration of the BChl molecule. This indicates that each BChl molecule can be modeled as a two-level system with the ground state corresponding to the highest occupied molecular orbital and the excited state to the lowest unoccupied molecular orbital.

Since the coupling between the BChl molecules is weak, we can assume that the electronic wave function of the coupled BChls factorizes. In addition, we assume that at any time, there is at most one excitation in the LHC. With these simplifications, we can introduce the so-called site basis as:

$$|i\rangle = |\phi_1^g\rangle \dots |\phi_i^e\rangle \dots |\phi_N^g\rangle. \quad (2.7)$$

A site basis state $|i\rangle$ implies that the electronic configuration at the BChl molecule i is excited, whereas

the electronic configuration at the remaining BChl molecules is in the ground state, which is given as $|0\rangle = \prod_i |\phi_i^g\rangle$. By using the site basis, we exploit the weak coupling between the BChls, allowing us to construct the basis of the coupled system from the basis functions of the individual BChl molecules. Expanding the electronic Hamiltonian from Equation 2.2 in the site basis yields:

$$H_S = \sum_{mij} \langle i | H_m^{el} | j \rangle |i\rangle \langle j| + \sum_{m < n, i, j} \langle i | V_{mn} | j \rangle |i\rangle \langle j|. \quad (2.8)$$

Using the definition of the dipole matrix elements:

$$\langle i | \frac{1}{4\pi\epsilon} \left[\frac{\vec{\mu}_m \cdot \vec{\mu}_n}{r_{mn}^3} - \frac{3(\vec{\mu}_m \cdot \vec{r}_{mn})(\vec{\mu}_n \cdot \vec{r}_{mn})}{r_{mn}^5} \right] | j \rangle = J_{mn} \delta_{im} \delta_{jn}. \quad (2.9)$$

and the eigenvalue Equations 2.5 and 2.6 for the site basis states, we obtain:

$$H_S = \sum_m \epsilon_m^e |m\rangle \langle m| + \sum_{m \neq n} J_{mn} |m\rangle \langle n|, \quad (2.10)$$

where the dipole-dipole coupling coefficient J_{mn} is assumed to be real, which means that $J_{mn} = J_{nm}$. It describes the transition of an excitation from site m to site n and thus allows for the delocalization of the excitation over the coupled system. The electrons, however, are not exchanged and remain at the corresponding site.

2.2.2 The Bath Hamiltonian

In the last section, we discussed the degrees of freedom of the electronic system of the BChl molecules. Next, we consider the degrees of freedom of the bath, in particular the vibrations of the nuclei. We start by distinguishing between two types of vibrations: the intra-BChl and inter-BChl vibrations. The inter-BChl vibrations describe collective vibrations between all atoms of all BChl molecules. However, since the BChl molecules are spatially separated and interact only weakly, these vibrations can be neglected. The intra-BChl vibrations refer to the motion of the nuclei within a certain BChl molecule and are the dominant part. The vibrations of the nuclei also affect the electronic charge distribution, which leads to fluctuations in the dipole-dipole coupling J_{mn} . However, this effect is negligible.

The dynamics of the nuclei are determined by potential energy surfaces (PES). The PES of the electronic ground state, as well as the excited state, can be approximated with a harmonic potential around the equilibrium position of the nuclei. Thus, the vibrations of the nuclei are described by normal modes created by the PES. This is illustrated in Figure 2.2, where we assume that the PES in the ground and excited states are nearly identical, which is a good approximation for LHCs. Nevertheless, in the excited state, the equilibrium position of the PES is shifted, and the frequencies of the vibrations may also be different.

In Figure 2.2, an excitation process is schematically depicted. Since the electronic dynamics are much faster than the nuclear dynamics, the nuclei can be treated as static during the excitation and thus perceive the change in the electronic state as instantaneous. This is called a vertical Frank-Condon transition, which is indicated by the red arrow. The nuclei gain not only the energy difference between the ground and

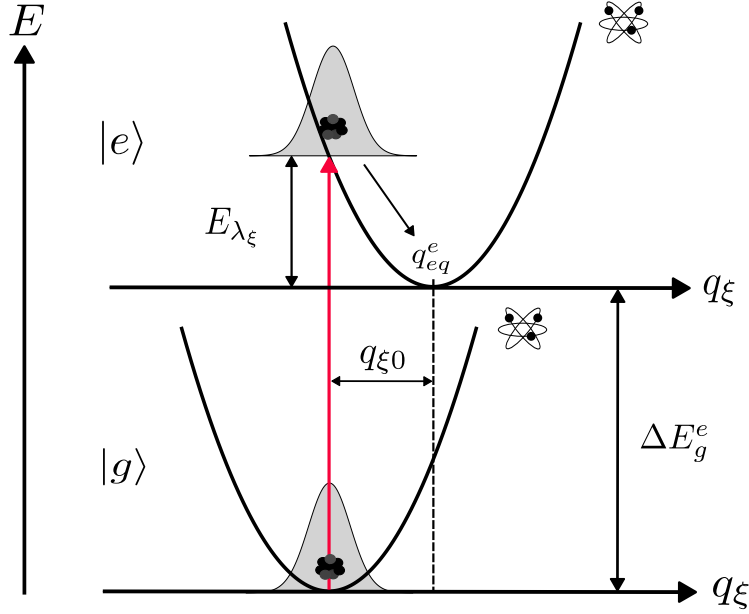


Figure 2.2: The PES of the electronic ground and excited state determines the vibrational modes of the nuclei. The excited state is indicated by $|e\rangle$ and its equilibrium position is shifted relative to the ground state $|g\rangle$. The red arrow indicates an excitation of the electronic system around the nuclei, which tunes the nuclei out of equilibrium. Due to interactions with the environment, the nuclei relax to their new equilibrium position. The figure is adapted from reference [36].

excited state ΔE_g^e , but also the so-called reorganization energy E_{λ_ξ} , since the equilibrium position of the ground and excited state is shifted by $q_{\xi 0}$. Due to interactions with the environment, the nuclei dissipate the reorganization energy and vibrate into the new equilibrium position q_{eq}^e .

Since the nuclei experience a harmonic oscillator potential, we expect the same potential in the Hamiltonian describing the nuclear dynamics. Indeed, the phononic Hamiltonian of the bath is given as:

$$H_B = \sum_{\xi} \left(\frac{p_{\xi}^2}{2m_{\xi}} + \frac{1}{2} m_{\xi} \omega_{\xi}^2 q_{\xi}^2 \right) = \sum_{\xi} \hbar \omega_{\xi} \left(b_{\xi}^{\dagger} b_{\xi} + \frac{1}{2} \right),$$

and describes the vibrations of the nuclei in the electronic ground state, where $b_{\xi}^{\dagger}$ and b_{ξ} are bosonic creation and annihilation operators. Thus, $b_{\xi}^{\dagger}$ creates a phonon with energy $\hbar \omega_{\xi}$ for the normal mode q_{ξ} , whereas b_{ξ} annihilates one.

2.2.3 The System-Bath Interaction Hamiltonian

In this section, we show that the reorganization process described in Figure 2.2 leads to a coupling between the electronic system and the vibrations of the nuclei, which gives rise to the system-bath interaction Hamiltonian. This is motivated by the insight that the nuclei, which form the bath, dissipate energy in response to the electronic excitation. We start by considering the shift of the equilibrium position by $q_{\xi 0}$ in the excited bath Hamiltonian:

$$\begin{aligned}
H_B^{\text{excited}} &= \sum_{\xi} \left(\frac{p_{\xi}^2}{2m_{\xi}} + \frac{1}{2} m_{\xi} \omega_{\xi}^2 (q_{\xi} - q_{\xi 0})^2 \right) = \sum_{\xi} \left(\frac{p_{\xi}^2}{2m_{\xi}} + \frac{1}{2} m_{\xi} \omega_{\xi}^2 q_{\xi}^2 + \frac{1}{2} m_{\xi} \omega_{\xi}^2 q_{\xi 0}^2 - m_{\xi} \omega_{\xi}^2 q_{\xi 0} q_{\xi} \right) \\
&= H_B + \sum_{\xi} \left(\frac{1}{2} m_{\xi} \omega_{\xi}^2 q_{\xi 0}^2 - m_{\xi} \omega_{\xi}^2 q_{\xi 0} q_{\xi} \right). \tag{2.11}
\end{aligned}$$

The coordinate q_{ξ} can be expressed with the bosonic creation and annihilation operators, which yields $q_{\xi} = \sqrt{\frac{\hbar}{2m_{\xi}\omega_{\xi}}} (b_{\xi}^{\dagger} + b_{\xi})$. Furthermore, we introduce the displacements $d_{\xi} = -\sqrt{\frac{m_{\xi}\omega_{\xi}}{2\hbar}} q_{\xi 0}$, and rewrite Equation 2.11 as:

$$H_B^{\text{excited}} = H_B + \sum_{\xi} \hbar \omega_{\xi} d_{\xi}^2 + \sum_{\xi} \hbar \omega_{\xi} d_{\xi} (b_{\xi}^{\dagger} + b_{\xi}). \tag{2.12}$$

The first term, the bath Hamiltonian H_B , corresponds to the vibrations of the nuclei in the electronic ground state and does not induce a coupling. However, the second and the third term depend on the displacement d_{ξ} and are thus only non-zero if the electronic configuration is excited. Therefore, the excited electronic state leads to a coupling, expressed by the last two terms. We can thus write:

$$H_{SB} = \sum_{\xi} \hbar \omega_{\xi} d_{\xi}^2 |\phi^e\rangle \langle \phi^e| + \sum_{\xi} \hbar \omega_{\xi} d_{\xi} (b_{\xi}^{\dagger} + b_{\xi}) \otimes |\phi^e\rangle \langle \phi^e| \tag{2.13}$$

for the coupling Hamiltonian between the electronic system of a BChl molecule and the vibrations of its nuclei. The first expression in H_{SB} is called the reorganization energy, which is described in Figure 2.2. It is the energy dissipated into the protein environment as the nuclei relax to their new equilibrium position. If we generalize to a system of N coupled BChl molecules, we use the site basis states $|m\rangle$ and obtain:

$$H_{SB} = \sum_{m=1}^N \sum_{\xi} \hbar \omega_{\xi m} d_{\xi m} (b_{\xi m}^{\dagger} + b_{\xi m}) \otimes |m\rangle \langle m|. \tag{2.14}$$

Here, the sum runs over all sites N and all vibrational modes ξ , to account for the coupling of each mode of the environment to each site of the system. Notice that we split off the reorganization Hamiltonian, which can also be written in the site basis:

$$H_{\text{reorg}} = \sum_{m=1}^N \sum_{\xi} \hbar \omega_{\xi m} d_{\xi m}^2 |m\rangle \langle m| = \sum_{m=1}^N \lambda_m |m\rangle \langle m|,$$

where $\lambda_m = \sum_{\xi} \hbar \omega_{\xi m} d_{\xi m}^2$ is the reorganization energy associated with site m . The reorganization process is local on site m and the reorganization Hamiltonian contains no off-diagonal terms, which would indicate a coupling between the sites. Hence, H_{reorg} only produces a shift in the energy levels, which does not affect the site coupling.

3 Mathematical Concepts for Open Quantum Systems

In quantum mechanics, the density matrix is essential for describing systems that are not in a well-defined pure state. Consider a composite system consisting of a system of interest, which is entangled with its bath. Then, the state of the composite system cannot be represented by a pure state, and a density matrix must be employed. Typically, the object of interest is the density matrix of the system, which is obtained by tracing out the environmental degrees of freedom by applying a so-called partial trace. Hence, the density operator, composite systems, and the partial trace are central concepts for the study of open quantum systems. Thus, we follow reference [38] to further investigate them in this section.

3.1 The Density Operator

To deepen our understanding of the density operator, let us consider a pure state ensemble $\{q_i, |\psi_i\rangle\}_{i=1}^N$, where the state $|\psi_i\rangle$ occurs with probability q_i . Suppose we measure the state $|\psi_i\rangle$ with a set of measurement operators M_k . The measurement transforms $|\psi_i\rangle$ as:

$$|\psi_i\rangle \rightarrow \frac{M_k |\psi_i\rangle}{\sqrt{p_{k|i}}} = |\psi_i^k\rangle, \quad (3.1)$$

where $p_{k|i} = \langle \psi_i | M_k^\dagger M_k | \psi_i \rangle$ is the probability of outcome k , given a state $|\psi_i\rangle$. The division by $\sqrt{p_{k|i}}$ ensures that $|\psi_i^k\rangle$ has a norm of 1.

Now suppose that we forget the input state and only know that it is part of the pure state ensemble $\{q_i, |\psi_i\rangle\}_{i=1}^N$. Then the probability of obtaining outcome k from a measurement is given by the sum of the probabilities for outcome k for each individual state $|\psi_i\rangle$ in the ensemble. We thus have:

$$p_k = \sum_i p_{k|i} q_i = \sum_i q_i \langle \psi_i | M_k^\dagger M_k | \psi_i \rangle = \text{Tr} \left[M_k^\dagger M_k \left(\sum_i q_i |\psi_i\rangle \langle \psi_i| \right) \right]. \quad (3.2)$$

We define the expression in parentheses of equation 3.2 as the density operator or density matrix:

$$\rho = \sum_i q_i |\psi_i\rangle \langle \psi_i|. \quad (3.3)$$

The density operator is equivalent to the pure state ensemble $\{q_i, |\psi_i\rangle\}_{i=1}^N$ and, in contrast to a state vector, can also describe mixed states. Hence, it is more frequently used to determine, for example, the expectation value of an observable $\langle A \rangle = \text{Tr}[\rho A]$ or to compute the partial trace over a specific subsystem.

3.1.1 Properties of the Density Operator

The density operator is of great importance for the description of quantum systems. Thus, we will investigate a few of its central properties.

The diagonal elements of the density operator are given by the probabilities of obtaining the corresponding basis state. Therefore, to ensure that the probabilities sum to 1, it must have a unit trace:

$$\text{Tr}[\rho] = \sum_i q_i \text{Tr}[|\psi_i\rangle \langle \psi_i|] = \sum_i q_i = 1. \quad (3.4)$$

Here, we used that $\text{Tr}[|\psi_i\rangle\langle\psi_i|] = \sum_n \langle n|\psi_i\rangle\langle\psi_i|n\rangle = 1$, where $|n\rangle$ is an arbitrary orthonormal basis and the probabilities q_i sum to 1.

The density operator is Hermitian since:

$$\rho^\dagger = \sum_i q_i^* (|\psi_i\rangle\langle\psi_i|)^\dagger = \sum_i q_i |\psi_i\rangle\langle\psi_i| = \rho, \quad (3.5)$$

where we used that the q_i are real since they are probabilities, which implies that $q_i = q_i^*$ and that $|\psi\rangle^\dagger = \langle\psi|$. The density operator is positive since:

$$\langle\nu|\rho|\nu\rangle = \sum_i q_i |\langle\psi_i|\nu\rangle|^2 \geq 0, \quad (3.6)$$

for all vectors $|\nu\rangle$ of the Hilbert space on which ρ is defined. Since the q_i are probabilities, they are non-negative and sum to 1. Therefore, the eigenvalues of ρ are non-negative and at least one is non-zero, making ρ a positive operator.

3.1.2 Unitary Dynamics and the Density Operator

Unitary dynamics govern the time evolution of closed quantum systems, i.e, of quantum systems that are not in contact with an environment. To describe the non-unitary dynamics of open quantum systems, additional terms are required, as explained in Section 4.

The unitary dynamics of pure quantum states can be expressed with the unitary time evolution operator $U(t) = e^{-\frac{i}{\hbar}Ht}$, which propagates from time 0 to time t :

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad (3.7)$$

where H is the Hamiltonian of the system of interest. Equivalently, the dynamics can be described with the Schrödinger equation as:

$$\frac{d}{dt} |\psi\rangle = -\frac{i}{\hbar} H |\psi\rangle. \quad (3.8)$$

Using the definition of the density operator from Equation 3.3, we can calculate its time evolution as:

$$\rho(t) = \sum_i q_i |\psi_i(t)\rangle \langle\psi_i(t)| = \sum_i q_i U(t) |\psi_i(0)\rangle \langle\psi_i(0)| U^\dagger(t) = U(t) \rho(0) U^\dagger(t). \quad (3.9)$$

Similarly, we can derive the Schrödinger equation for the density operator using Equation 3.8:

$$\begin{aligned} \frac{d}{dt} \rho(t) &= \frac{d}{dt} \sum_i q_i |\psi_i(t)\rangle \langle\psi_i(t)| = \sum_i q_i \left[\left(\frac{d}{dt} |\psi_i(t)\rangle \right) \langle\psi_i(t)| + |\psi_i(t)\rangle \left(\frac{d}{dt} \langle\psi_i(t)| \right) \right] \\ &= \sum_i q_i \left(-\frac{i}{\hbar} H |\psi_i(t)\rangle \langle\psi_i(t)| + \frac{i}{\hbar} |\psi_i(t)\rangle \langle\psi_i(t)| H \right) \\ &= -\frac{i}{\hbar} \left(H \sum_i q_i |\psi_i(t)\rangle \langle\psi_i(t)| - \sum_i q_i |\psi_i(t)\rangle \langle\psi_i(t)| H \right) \\ &= -\frac{i}{\hbar} (H\rho - \rho H) = -\frac{i}{\hbar} [H, \rho] \equiv \mathcal{L}\rho, \end{aligned} \quad (3.10)$$

where $\mathcal{L}$ is the Liouvillian of the system. Here, we used that the Hermitian conjugate of the Schrödinger equation in Equation 3.8 is given as $\frac{d}{dt} \langle \psi | = \frac{i}{\hbar} \langle \psi | H$. Equation 3.10 is also known as the Von-Neumann equation.

3.2 Composite Systems

Let us consider a composite system consisting of a subsystem of interest, referred to as the system, and its surrounding environment, or bath. Together, the system and the bath form a closed total system. We introduce the Hilbert space of the system and the bath as:

$$\mathcal{H}_S = \text{span}\{|i\rangle_S\}, \quad \mathcal{H}_B = \text{span}\{|\mu\rangle_B\}. \quad (3.11)$$

Here, the indices i and μ range from 0 to $d_S - 1$ and $d_B - 1$, where d_S and d_B are the dimension of the Hilbert space of the system and bath, respectively. Typically, the dimension of the bath is considered to be infinite, whereas the dimension of the system stays finite. The composite Hilbert space of the two systems can then be obtained by:

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B = \text{span}\{|i\rangle_S \otimes |\mu\rangle_B\}. \quad (3.12)$$

This indicates that any state $|\Psi_\alpha\rangle \in \mathcal{H}$ of the pure state ensemble $\{|\Psi_\alpha\rangle, q_\alpha\}_{\alpha=1}^N$ can be written as:

$$|\Psi_\alpha\rangle = \sum_{i\mu} c_{\alpha i\mu} |i\rangle_S \otimes |\mu\rangle_B. \quad (3.13)$$

Thus, for the density matrix of the composite system, we obtain:

$$R = \sum_{\alpha} q_{\alpha} |\Psi_{\alpha}\rangle \langle \Psi_{\alpha}| = \sum_{\alpha} q_{\alpha} \left(\sum_{i\mu} c_{\alpha i\mu} |i\rangle_S \otimes |\mu\rangle_B \right) \left(\sum_{j\nu} c_{\alpha j\nu}^* \langle j|_S \otimes \langle \nu|_B \right). \quad (3.14)$$

Defining $\lambda_{ij\mu\nu} = \sum_{\alpha} q_{\alpha} c_{\alpha i\mu} c_{\alpha j\nu}^*$, we can write any density matrix in the composite Hilbert space $\mathcal{H}$ as:

$$R = \sum_{ij\mu\nu} \lambda_{ij\mu\nu} |i\rangle_S \langle j| \otimes |\mu\rangle_B \langle \nu|. \quad (3.15)$$

Note that the simple case $\lambda_{ij\mu\nu} = \lambda_{ij}^S \lambda_{\mu\nu}^B$ allows the composite density matrix to be written as $R = \rho_S \otimes \rho_B$. In this case, the density matrix of the system and bath factorize, and R is consequently called a factorizable state, implying that the system and the bath are not correlated. Conversely, if R does not factorize, correlations between the system and bath are present.

3.3 The Partial Trace

Usually, one is solely interested in the degrees of freedom of the system. Thus, it is desirable to extract the density matrix of the system from the density matrix of the composite system. This can be achieved with the so-called partial trace, which effectively averages out the components of the bath, leaving behind the density matrix of the system. Thus, the partial trace is defined as a linear operator that maps from the composite Hilbert space $\mathcal{H}$ to the Hilbert space $\mathcal{H}_S$ of the system. To further investigate the partial

trace, we introduce an operator $O = \sum_{ij} c_{ij} M_S^i \otimes N_B^j$, which implies that O acts on $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. Then, defining basis vectors $|b\rangle$ of the bath Hilbert space, the partial trace acts as follows:

$$\begin{aligned} \text{Tr}_B[O] &= \sum_{ij} c_{ij} \text{Tr}_B(M_S^i \otimes N_B^j) = \sum_{ij} c_{ij} M_S^i \otimes \sum_b \langle b | N_B^j | b \rangle \\ &= \sum_b \sum_{ij} c_{ij} \langle b | (M_S^i \otimes N_B^j) | b \rangle = \sum_b \langle b | O | b \rangle, \end{aligned} \quad (3.16)$$

where we used the linearity of the partial trace. The basis states $|b\rangle$ act only on the Hilbert space of the bath and thus the matrix element is taken over the operators N_B^j , leaving behind the operators M_S^i that act solely on the system. Applying the partial trace to R in Equation 3.15, by linearity, yields:

$$\begin{aligned} \text{Tr}_B[R] &= \text{Tr}_B \left[\sum_{ij\mu\nu} \lambda_{ij\mu\nu} |i\rangle_S \langle j| \otimes |\mu\rangle_B \langle \nu| \right] = \sum_{ij\mu\nu b} \lambda_{ij\mu\nu} |i\rangle_S \langle j| \langle b | \mu \rangle_B \langle \nu | b \rangle \\ &= \sum_{ijb} \lambda_{ijbb} |i\rangle_S \langle j| = \sum_{ij} \lambda_{ij} |i\rangle_S \langle j|. \end{aligned} \quad (3.17)$$

With the procedure for computing the partial trace in hand, we proceed to justify the expression $\rho_S = \text{Tr}_B[R]$, indicating that the density matrix of the system is obtained via a partial trace over the bath. This relation can be motivated by the following reasoning. Assume that we have a composite system with $\tilde{M} = M_S \otimes I_B \in \mathcal{H}_S \otimes \mathcal{H}_B$. Then, the expectation value of the observable $\tilde{M}$, can be calculate as:

$$\langle \tilde{M} \rangle = \text{Tr}[\tilde{M}R]. \quad (3.18)$$

Since the bath part of $\tilde{M}$ is just the identity, the bath remains unaffected, and we expect that the expectation value of M_S stays unchanged. Thus, we demand that:

$$\text{Tr}[M_S \rho_S] = \text{Tr}[\tilde{M}R]. \quad (3.19)$$

It can be shown, that Equation 3.19 only holds if we set $\rho_S = \text{Tr}_B[R]$. We will outline the proof in one direction by showing that $\langle M_S \rangle = \langle \tilde{M} \rangle$ if $\rho_S = \text{Tr}_B[R]$. To this end, let $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B = \text{span}\{|i\rangle_S \otimes |\mu\rangle_B\}$. Then, applying the partial trace, we obtain:

$$\langle M_S \rangle = \sum_i \langle i | \rho_S M_S | i \rangle_S = \sum_i \langle i | \text{Tr}_B[R] M_S | i \rangle_S = \sum_i \langle i | \sum_{\mu} \langle \mu | R | \mu \rangle_B M_S | i \rangle_S. \quad (3.20)$$

Here, it is important to note that the states $|i\rangle$ act on $\mathcal{H}_S$ and $\mathcal{H}_B$ since they originate from the trace over the entire system. In contrast, the states $|\mu\rangle$ act only on the bath since they come from the partial trace over the latter. If we now extend M_S to the entire system, to show that under the condition $\rho_S = \text{Tr}_B[R]$ the expectation value of M_S in the composite system is still $\langle M_S \rangle$, we need to replace M_S by $M_S \otimes I_B$:

$$\langle M_S \rangle = \sum_{i\mu} \langle i | \otimes \langle \mu | R (M_S \otimes I_B) | i \rangle \otimes |\mu \rangle = \text{Tr}[R(M_S \otimes I_B)] = \langle \tilde{M} \rangle, \quad (3.21)$$

which yields the desired equality $\langle M_S \rangle = \langle \tilde{M} \rangle$ and hence completes the proof.

4 Dynamics of Open Quantum Systems

Closed quantum systems are isolated from their environment and thus evolve unitarily. As discussed in Section 3.1.2, the time dependence of the system density matrix is governed by the Von Neumann equation. Open quantum systems, in contrast, are coupled to their environment, which results in an exchange of information and energy. Hence, their dynamics is governed by a non-unitary evolution where the unitary part is complemented with an additional dissipative term describing the interaction between the system and the bath.

In open quantum systems, we distinguish between Markovian and non-Markovian dynamics. The Markovian dynamics arises when the influence of bath correlations on the system is neglected. This approximation is justified when the dynamics of the system occurs on much larger time scales than the dynamics of the bath. In contrast, non-Markovian dynamics do not rely on a Markovian approximation and also consider the effect of bath correlations on the system, yielding a more accurate description.

In this section, we will derive equations that describe the Markovian and non-Markovian dynamics of open quantum systems. To this end, first, the Markovian Lindblad equation is derived. This is followed by the introduction of the operator sum representation, which is a general formalism that describes both Markovian and non-Markovian dynamics. Lastly, we derive the hierarchical equations of motion technique that allows for an exact description of the time evolution and thus relies on non-Markovian dynamics.

4.1 Lindblad Master Equation

The Lindblad equation represents the most general form of a Markovian master equation. It describes the time evolution of quantum states as a result of interactions with the environment. Hence, it is crucial for describing non-unitary processes, like dissipation and decoherence [39].

The dissipative part of the Lindblad equation involves jump operators, also called Lindblad operators, that induce transitions, which can often be associated with decay rates [40]. Each jump operator corresponds to a specific physical process that the system undergoes due to its interaction with the environment [41].

Starting with the derivation of the Lindblad equation, we follow Kreisbeck [36] and the book of Breuer and Petruccione [27]. The time evolution of the full density matrix $\tilde{R}(t)$ in the interaction picture is given by the Von Neumann equation:

$$\frac{d}{dt}\tilde{R}(t) = -\frac{i}{\hbar} \left[\tilde{H}_{\text{SB}}(t), \tilde{R}(t) \right], \quad (4.1)$$

which was derived in Section 3.1.2. Since the full density matrix $\tilde{R}(t)$ describes the bath and the system as a total closed system, its dynamics is still unitary. Integrating Equation 4.1, we obtain:

$$\tilde{R}(t) = \tilde{R}(0) - \frac{i}{\hbar} \int_0^t ds \left[\tilde{H}_{\text{SB}}(s), \tilde{R}(s) \right]. \quad (4.2)$$

In theory, Equation 4.2 can be used iteratively, i.e, inserted into itself repeatedly. To arrive at Equation 4.3, we perform this procedure only once and obtain a second-order approximation. Since we are usually only interested in the system's degrees of freedom, we take a partial trace over the bath and set $\tilde{R}(0) \equiv \tilde{R}_0$ to obtain an equation for the system density matrix:

$$\frac{d}{dt}\tilde{\rho}_S(t) = -\frac{1}{\hbar^2} \int_0^t ds \operatorname{Tr}_B \left\{ [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(s), \tilde{R}(s)]] \right\}. \quad (4.3)$$

Here, we used $\operatorname{Tr}_B \left[\frac{d}{dt} \tilde{R}(t) \right] = \frac{d}{dt} \tilde{\rho}_S(t)$ and that $\tilde{H}_{SB}$ can always be written such that $\operatorname{Tr}_B \left\{ [\tilde{H}_{SB}(t), \tilde{R}_0] \right\} = 0$ [39]. We continue with a so-called Born approximation, which applies for a weak coupling between the system and the bath, when the influence of the system on the bath is small. Therefore, the interaction barely affects the density matrix of the bath, which means that the full system density matrix factorizes for all times. This yields $\tilde{R}(t) = \tilde{\rho}_S(t) \otimes \rho_B$ where $\tilde{\rho}_S(t)$ denotes the density matrix of the system and ρ_B the density matrix of the bath. Inserting into Equation 4.3 leads to:

$$\frac{d}{dt}\tilde{\rho}_S(t) = -\frac{1}{\hbar^2} \int_0^t ds \operatorname{Tr}_B \left\{ [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(s), \tilde{\rho}_S(s) \otimes \rho_B]] \right\}. \quad (4.4)$$

4.1.1 The Markovian Approximation

To arrive at a Markovian master equation such as the Lindblad equation, we employ the Markov approximation where we first replace $\tilde{\rho}_S(s)$ by $\tilde{\rho}_S(t)$ and obtain the so-called Redfield equation:

$$\frac{d}{dt}\tilde{\rho}_S(t) = -\frac{1}{\hbar^2} \int_0^t ds \operatorname{Tr}_B \left\{ [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(s), \tilde{\rho}_S(t) \otimes \rho_B]] \right\}. \quad (4.5)$$

In the next step, we change variables from s to $t - s$ so that the integral depends on the time difference between interactions. We then extend the upper integration limit to infinity. This is valid when the characteristic timescale, over which the correlations in the environment persist, is much shorter than the system's evolution timescale. In this regime, correlations vanish for sufficiently large time differences, making contributions from large s negligible. The resulting equation is time-local, yielding a Markovian master equation that depends only on the system's current state [42]:

$$\frac{d}{dt}\tilde{\rho}_S(t) = -\frac{1}{\hbar^2} \int_0^\infty ds \operatorname{Tr}_B \left\{ [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(t-s), \tilde{\rho}_S(t) \otimes \rho_B]] \right\}. \quad (4.6)$$

To proceed, we will further investigate the system-bath interaction Hamiltonian $\tilde{H}_{SB}(t)$. Without loss of generality, $\tilde{H}_{SB}$ can be written as a sum of tensor products of a system operator $\tilde{S}_\alpha(t)$ and a bath operator $\tilde{B}_\alpha(t)$. In case of the FMO complex, the subscript α enumerates the different BChl molecules or sites. We thus write:

$$\tilde{H}_{SB}(t) = \sum_\alpha \tilde{S}_\alpha(t) \otimes \tilde{B}_\alpha(t). \quad (4.7)$$

Next, we introduce the eigenbasis of the system Hamiltonian H_S as $H_S|n\rangle = E_n|n\rangle$ and expand the system operator $S_\alpha = a_\alpha^\dagger a_\alpha$ in the Schrödinger picture. The operators $a_\alpha^\dagger$ and a_α are the creation and annihilation operators for an electronic excitation on site α . Hence, the site basis state $|\alpha\rangle = a_\alpha^\dagger |0\rangle$ corresponds to an excitation on the BChl molecule α where the other BChl molecules remain in the ground state. The operator S_α is a diagonal matrix with a 1 in the α^{th} diagonal element and zeros elsewhere. The expansion in the eigenbasis $|n\rangle$ yields:

$$S_\alpha = \sum_{mn} |m\rangle \langle m| S_\alpha |n\rangle \langle n|, \quad (4.8)$$

where S_α is in the Schrödinger picture. A transformation into the interaction picture can be performed by using the eigenvalue equation of H_S :

$$\tilde{S}_\alpha(t) = e^{\frac{i}{\hbar} H_S t} S_\alpha e^{-\frac{i}{\hbar} H_S t} = \sum_{mn} e^{-\frac{i}{\hbar} (E_n - E_m) t} |m\rangle \langle m| S_\alpha |n\rangle \langle n|, \quad (4.9)$$

which yields $\tilde{S}_\alpha$ as a function of time. To obtain $\tilde{S}_\alpha$ as a function of the frequency, we perform a Fourier transformation and use the relations $\int_{-\infty}^{\infty} dt e^{-i(\frac{E_n - E_m}{\hbar} - \omega)t} = 2\pi \delta(\frac{E_n - E_m}{\hbar} - \omega)$ as well as $\delta(\frac{E_n - E_m}{\hbar} - \omega) = \hbar \delta(E_n - E_m - \hbar\omega)$ to arrive at:

$$\begin{aligned} \tilde{S}_\alpha(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} dt e^{i\omega t} \tilde{S}_\alpha(t) = \frac{1}{2\pi} \sum_{mn} \langle m| S_\alpha |n\rangle |m\rangle \langle n| \int_{-\infty}^{\infty} dt e^{i\omega t} e^{-\frac{i}{\hbar} (E_n - E_m) t} \\ &= \hbar \sum_{mn} \langle m| S_\alpha |n\rangle \delta(E_n - E_m - \hbar\omega) |m\rangle \langle n|. \end{aligned} \quad (4.10)$$

Inserting into Equation 4.7, the interaction Hamiltonian in the interaction picture becomes:

$$\begin{aligned} \tilde{H}_{SB}(t) &= \sum_{\alpha} \tilde{S}_\alpha(t) \otimes \tilde{B}_\alpha(t) = \sum_{\alpha} \left(\int_{-\infty}^{\infty} e^{-i\omega t} \tilde{S}_\alpha(\omega) d\omega \right) \otimes \tilde{B}_\alpha(t) \\ &= \sum_{\alpha} \sum_{\omega_{nm}} e^{-i\omega_{nm} t} \tilde{S}_\alpha(\omega_{nm}) \otimes \tilde{B}_\alpha(t), \end{aligned} \quad (4.11)$$

where in the last step we redefined $\tilde{S}_\alpha(w)$ as $\tilde{S}_\alpha(w) = \hbar \sum_{mn} \langle m| S_\alpha |n\rangle |m\rangle \langle n|$ with a sum over all frequencies $\omega_{nm} = \frac{E_n - E_m}{\hbar}$ that correspond to the energy difference $E_n - E_m$. Similarly, for $\tilde{H}_{SB}(t - s)$ we obtain:

$$\tilde{H}_{SB}(t - s) = \sum_{\beta} \sum_{\omega'_{nm}} e^{-i\omega'_{nm}(t-s)} \tilde{S}_\beta(\omega'_{nm}) \otimes \tilde{B}_\beta(t - s). \quad (4.12)$$

In the following, we will drop the subscript nm and just write ω instead of ω_{nm} for brevity.

Revisiting Equation 4.6, we notice the double-commutator structure inside the partial trace. To simplify the expression, we first expand $[\tilde{H}_{SB}(t), [\tilde{H}_{SB}(t'), \tilde{R}(t)]]$, where $t' = t - s$. Omitting the time argument in $\tilde{R}(t) = \tilde{\rho}_S(t) \otimes \rho_B$ we obtain:

$$\begin{aligned} [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(t'), \tilde{R}]] &= \tilde{H}_{SB}(t) \tilde{H}_{SB}(t') \tilde{R} - \tilde{H}_{SB}(t) \tilde{R} \tilde{H}_{SB}(t') \\ &\quad - \tilde{H}_{SB}(t') \tilde{R} \tilde{H}_{SB}(t) + \tilde{R} \tilde{H}_{SB}(t') \tilde{H}_{SB}(t). \end{aligned} \quad (4.13)$$

Substituting Equation 4.11 and 4.12 into each of the four terms of Equation 4.13 yields:

$$\begin{aligned} \tilde{H}_{SB}(t) \tilde{H}_{SB}(t') \tilde{R} &= \left[\sum_{\alpha\omega} e^{-i\omega t} \tilde{S}_\alpha(\omega) \otimes \tilde{B}_\alpha(t) \right] \left[\sum_{\beta\omega'} e^{-i\omega' t'} \tilde{S}_\beta(\omega') \otimes \tilde{B}_\beta(t') \right] \tilde{R} \\ &= \sum_{\alpha\beta} \sum_{\omega\omega'} e^{-i\omega t} e^{-i\omega' t'} \left[\tilde{S}_\alpha(\omega) \tilde{S}_\beta(\omega') \tilde{\rho}_S(t) \right] \otimes \left[\tilde{B}_\alpha(t) \tilde{B}_\beta(t') \rho_B \right] \end{aligned} \quad (4.14)$$

and

$$\begin{aligned}\tilde{H}_{SB}(t)\tilde{R}\tilde{H}_{SB}(t') &= \left[\sum_{\alpha\omega} e^{-i\omega t} \tilde{S}_\alpha(\omega) \otimes \tilde{B}_\alpha(t) \right] \tilde{R} \left[\sum_{\beta\omega'} e^{-i\omega' t'} \tilde{S}_\beta(\omega') \otimes \tilde{B}_\beta(t') \right] \\ &= \sum_{\alpha\beta} \sum_{\omega\omega'} e^{-i\omega t} e^{-i\omega' t'} \left[\tilde{S}_\alpha(\omega) \tilde{\rho}_S(t) \tilde{S}_\beta(\omega') \right] \otimes \left[\tilde{B}_\alpha(t) \rho_B \tilde{B}_\beta(t') \right]\end{aligned}\quad (4.15)$$

and

$$\begin{aligned}\tilde{H}_{SB}(t')\tilde{R}\tilde{H}_{SB}(t) &= \left[\sum_{\beta\omega'} e^{-i\omega' t'} \tilde{S}_\beta(\omega') \otimes \tilde{B}_\beta(t') \right] \tilde{R} \left[\sum_{\alpha\omega} e^{-i\omega t} \tilde{S}_\alpha(\omega) \otimes \tilde{B}_\alpha(t) \right] \\ &= \sum_{\alpha\beta} \sum_{\omega\omega'} e^{-i\omega t} e^{-i\omega' t'} \left[\tilde{S}_\beta(\omega') \tilde{\rho}_S(t) \tilde{S}_\alpha(\omega) \right] \otimes \left[\tilde{B}_\beta(t') \rho_B \tilde{B}_\alpha(t) \right]\end{aligned}\quad (4.16)$$

and

$$\begin{aligned}\tilde{R}\tilde{H}_{SB}(t')\tilde{H}_{SB}(t) &= \tilde{R} \left[\sum_{\beta\omega'} e^{-i\omega' t'} \tilde{S}_\beta(\omega') \otimes \tilde{B}_\beta(t') \right] \left[\sum_{\alpha\omega} e^{-i\omega t} \tilde{S}_\alpha(\omega) \otimes \tilde{B}_\alpha(t) \right] \\ &= \sum_{\alpha\beta} \sum_{\omega\omega'} e^{-i\omega t} e^{-i\omega' t'} \left[\tilde{\rho}_S(t) \tilde{S}_\beta(\omega') \tilde{S}_\alpha(\omega) \right] \otimes \left[\rho_B \tilde{B}_\beta(t') \tilde{B}_\alpha(t) \right].\end{aligned}\quad (4.17)$$

To further simplify the expressions in equations 4.14 to 4.17, we introduce the so-called bath correlation function. It is essentially the autocorrelation function of the bath operator $\tilde{B}_\alpha$ at times t and t' , and hence a measure of the timescale over which fluctuations of the bath remain correlated:

$$C_\alpha(t - t') \equiv \langle \tilde{B}_\alpha(t) \tilde{B}_\alpha(t') \rangle = \text{Tr}_B \left[\tilde{B}_\alpha(t) \tilde{B}_\alpha(t') \rho_B \right]. \quad (4.18)$$

Notice that instead of the subscripts α and β , only one subscript α is used, since we approximate the baths coupled to different sites as independent. Thus, the correlation function of two bath operators stemming from different sites is zero, and we set $\alpha = \beta$. For small time differences t and t' , the correlation function $C_\alpha(t - t')$ generally attains large values, reflecting strong correlations in the environment over short timescales. As the time difference increases, the correlations decay, and $C_\alpha(t - t')$ eventually averages to 0.

Using the cyclicity of the trace we can apply Equation 4.18 to equations 4.14 to 4.17 where in Equation 4.14 and 4.16 we obtain $C_\alpha(t - t') = C_\alpha(s)$ and in Equation 4.15 and 4.17 we obtain $C_\alpha(t' - t) = C_\alpha(-s)$. With Equation 4.18, the trace over the double commutator can be written as:

$$\begin{aligned}\text{Tr}_B \left\{ [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(t'), \tilde{R}]] \right\} &= \sum_{\alpha} \sum_{\omega\omega'} e^{-i\omega t} e^{-i\omega' t'} \left\{ \tilde{S}_\alpha(\omega) \tilde{S}_\alpha(\omega') \tilde{\rho}_S(t) C_\alpha(s) \right. \\ &\quad \left. - \tilde{S}_\alpha(\omega) \tilde{\rho}_S(t) \tilde{S}_\alpha(\omega') C_\alpha(-s) - \tilde{S}_\alpha(\omega') \tilde{\rho}_S(t) \tilde{S}_\alpha(\omega) C_\alpha(s) + \tilde{\rho}_S(t) \tilde{S}_\alpha(\omega') \tilde{S}_\alpha(\omega) C_\alpha(-s) \right\}.\end{aligned}\quad (4.19)$$

The bath operators are encoded in the bath correlation function, and only the system operators $\tilde{S}_\alpha(\omega)$ remain. The $\tilde{S}_\alpha(\omega)$ will later be interpreted as the Lindblad operators, and the bath correlation function contributes to the rates that describe the strength of the dissipative process. Inserting into Equation 4.6 and using $e^{-i\omega t} e^{-i\omega' t'} = e^{i\omega' s} e^{-i(\omega + \omega')t}$ yields:

$$\begin{aligned} \frac{d}{dt} \tilde{\rho}_S(t) = & -\frac{1}{\hbar^2} \sum_{\alpha} \sum_{\omega \omega'} e^{-i(\omega+\omega')t} \int_0^{\infty} ds e^{i\omega' s} \left\{ \tilde{S}_{\alpha}(\omega) \tilde{S}_{\alpha}(\omega') \tilde{\rho}_S(t) C_{\alpha}(s) \right. \\ & \left. - \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}(\omega') C_{\alpha}(-s) - \tilde{S}_{\alpha}(\omega') \tilde{\rho}_S(t) \tilde{S}_{\alpha}(\omega) C_{\alpha}(s) + \tilde{\rho}_S(t) \tilde{S}_{\alpha}(\omega') \tilde{S}_{\alpha}(\omega) C_{\alpha}(-s) \right\}. \end{aligned} \quad (4.20)$$

4.1.2 The Secular Approximation

The last approximation, before we arrive at the Lindblad equation, is the secular approximation. It averages over rapidly oscillating terms and separates the dynamics of the diagonal and off-diagonal elements of the density matrix, so the populations and coherences [43]. The approximation is justified when the dynamics of the system happen at much larger time scales than the inverse of the oscillations [44] and hence, is also known as a rotating wave approximation. Thus, we neglect the exponential in Equation 4.20 and set $\omega' = -\omega$:

$$\begin{aligned} \frac{d}{dt} \tilde{\rho}_S(t) = & -\frac{1}{\hbar^2} \sum_{\alpha \omega} \int_0^{\infty} ds e^{-i\omega s} \left\{ \tilde{S}_{\alpha}(\omega) \tilde{S}_{\alpha}(-\omega) \tilde{\rho}_S(t) C_{\alpha}(s) - \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}(-\omega) C_{\alpha}(-s) \right. \\ & \left. - \tilde{S}_{\alpha}(-\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}(\omega) C_{\alpha}(s) + \tilde{\rho}_S(t) \tilde{S}_{\alpha}(-\omega) \tilde{S}_{\alpha}(\omega) C_{\alpha}(-s) \right\}. \end{aligned} \quad (4.21)$$

Replacing w by $-w$ in the first and third summand and using $C_{\alpha}(s) = C_{\alpha}(-s)^*$ as well as $\tilde{S}_{\alpha}(-\omega) = \tilde{S}_{\alpha}^{\dagger}(\omega)$ as derived in appendix A.1 and A.2, we find:

$$\begin{aligned} \frac{d}{dt} \tilde{\rho}_S(t) = & -\frac{1}{\hbar^2} \sum_{\alpha \omega} \int_0^{\infty} ds e^{-i\omega s} \left\{ \tilde{S}_{\alpha}^{\dagger}(\omega) \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) C_{\alpha}(s) - \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) C_{\alpha}^*(s) \right. \\ & \left. - \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) C_{\alpha}(s) + \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) \tilde{S}_{\alpha}(\omega) C_{\alpha}^*(s) \right\}. \end{aligned} \quad (4.22)$$

The bath correlation function $C_{\alpha}(s)$ can be expanded in its real and imaginary parts, as $C_{\alpha}(s) = \text{Re}[C_{\alpha}(s)] + i \text{Im}[C_{\alpha}(s)]$. This yields:

$$\begin{aligned} \frac{d}{dt} \tilde{\rho}_S(t) = & -\frac{1}{\hbar^2} \sum_{\alpha \omega} \int_0^{\infty} ds e^{-i\omega s} \left\{ \tilde{S}_{\alpha}^{\dagger}(\omega) \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) [\text{Re}(C_{\alpha}(s)) + i \text{Im}(C_{\alpha}(s))] \right. \\ & - \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) [\text{Re}(C_{\alpha}(s)) - i \text{Im}(C_{\alpha}(s))] \\ & - \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) [\text{Re}(C_{\alpha}(s)) + i \text{Im}(C_{\alpha}(s))] \\ & \left. + \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) \tilde{S}_{\alpha}(\omega) [\text{Re}(C_{\alpha}(s)) - i \text{Im}(C_{\alpha}(s))] \right\} \end{aligned} \quad (4.23)$$

Defining the Lamb shift $L_{\alpha}^{\text{Lamb}}(\omega) = \frac{1}{\hbar^2} \text{Im} \left[\int_0^{\infty} ds C_{\alpha}(s) e^{-i\omega s} \right]$ and the rates $\gamma_{\alpha}(\omega) = \frac{2}{\hbar^2} \text{Re} \left[\int_0^{\infty} ds C_{\alpha}(s) e^{-i\omega s} \right]$ we obtain:

$$\begin{aligned} \frac{d}{dt} \tilde{\rho}_S(t) = & \sum_{\alpha \omega} i L_{\alpha}^{\text{Lamb}}(\omega) [\tilde{S}_{\alpha}^{\dagger}(\omega) \tilde{S}_{\alpha}(\omega), \tilde{\rho}_S(t)] \\ & + \sum_{\alpha \omega} \gamma_{\alpha}(\omega) \left(\frac{1}{2} \tilde{S}_{\alpha}^{\dagger}(\omega) \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) - \tilde{S}_{\alpha}(\omega) \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) + \frac{1}{2} \tilde{\rho}_S(t) \tilde{S}_{\alpha}^{\dagger}(\omega) \tilde{S}_{\alpha}(\omega) \right). \end{aligned} \quad (4.24)$$

The Lamb shift corresponds to the shift in the energy levels of an atom and is caused by the coupling of the electrons to an electromagnetic field [45][46], which leads to the absorption and emission of virtual

photons [47]. For a detailed discussion of the Lamb shift and the conditions under which it can be neglected, I refer to Reference [48]. In the following, the Lamb shift $L_\alpha^{\text{Lamb}}(\omega)$ will be neglected, and the full Lindblad master equation in the Schrödinger picture is given by:

$$\frac{d}{dt}\rho_S(t) = -\frac{i}{\hbar}[H_S, \rho_S(t)] + \sum_{\alpha\omega} \gamma_\alpha(\omega) \left[S_\alpha(\omega) \rho_S(t) S_\alpha^\dagger(\omega) - \frac{1}{2} \left\{ S_\alpha^\dagger(\omega) S_\alpha(\omega), \rho_S(t) \right\} \right] \equiv \mathfrak{L}\rho_S(t). \quad (4.25)$$

Here, $\mathfrak{L}$ is the so-called Lindbladian, which yields the formal solution $\rho_S(t) = e^{\mathfrak{L}t}\rho(0)$ and the operators $S_\alpha(\omega)$ are referred to as Lindblad operators. If they are non-Hermitian, the equation can be used to treat dissipation, decoherence, or decays. The $\gamma_\alpha(\omega)$ are the rates associated with the Lindblad operators and describe the strength of the dissipative process [39]. The sum over ω implies that one Lindblad operator with a certain rate exists for each transition frequency ω and hence the Lindblad operators are also called jump operators [40]. Combining the index α for the sites and the index ω for the rates into a single index i , one Lindblad operator for each transition between the sites is obtained. Renaming the $S_\alpha(\omega)$ to L_i we obtain:

$$\frac{d}{dt}\rho_S(t) = -\frac{i}{\hbar}[H, \rho_S(t)] + \sum_i \gamma_i (L_i \rho_S(t) L_i^\dagger - \frac{1}{2} \{L_i^\dagger L_i, \rho_S(t)\}) \equiv \mathfrak{L}\rho_S(t). \quad (4.26)$$

Here, the rates $\gamma_\alpha(\omega)$ are given as [36][49]:

$$\gamma_\alpha(\omega) = \frac{2}{\hbar^2} J(\omega) (n(\omega) + 1), \quad (4.27)$$

where $J(\omega) = 2\nu\lambda\frac{\omega}{\nu^2+\omega^2}$ is the Drude-Lorentz spectral density and $n(\omega) = \frac{1}{e^{\beta\hbar\omega}-1}$ is the Bose-Einstein distribution. For details on the spectral density, I refer to appendix B.1.

4.2 Kraus Operator Sum Representation

The Kraus operator sum representation is a general formalism that describes the time evolution of quantum systems [50]. It applies to both open and closed quantum systems and can thus describe unitary and non-unitary dynamics [51]. The representation is not unique, and hence, different sets of Kraus operators can describe the same dynamics [38].

The Kraus operator sum representation can also be written as a quantum map or quantum channel that propagates the density matrix of a system in time. In fact, Kraus' theorem states that any completely positive, trace preserving map can be expressed as an operator sum representation [52]. In what follows, we present the operator sum representation and its properties, as discussed in Reference [38].

4.2.1 Derivation of the Kraus Operator Sum Representation

To derive the Kraus operator sum representation, we start by considering a system and a bath, described with an initial joint density matrix $R(0)$. The time evolution operator of the joint system is given as $U(t) = e^{-\frac{i}{\hbar}Ht}$, which can be used to calculate the unitary evolution of the joint system with the Schrödinger equation:

$$R(t) = U(t)R(0)U^\dagger(t). \quad (4.28)$$

In general, the joint density matrix $R(0)$ consists of the density matrix ρ_S of the system and the density matrix ρ_B of the bath. Since ρ_B must be physically valid, it is positive and normalized and thus has a spectral decomposition:

$$\rho_B(0) = \sum_{\nu} \lambda_{\nu} |\nu\rangle\langle\nu|. \quad (4.29)$$

The $|\nu\rangle$ are orthogonal eigenvectors and the eigenvalues λ_{ν} must be positive since ρ_B is positive. Usually, one is solely interested in the degrees of freedom of the system, which yields the motivation to take a partial trace of the joint density matrix $R(t)$ over the degrees of freedom of the bath to obtain the system density matrix:

$$\rho_S(t) = \text{Tr}_B[R(t)] = \text{Tr}_B[U(t)R(0)U^\dagger(t)] = \sum_{\mu} \langle\mu| U(t) R(0) U^\dagger(t) |\mu\rangle. \quad (4.30)$$

Here, the $|\mu\rangle$ correspond to orthonormal basis states of the bath. It is often assumed that the full system density matrix initially factorizes [53]. This is justified by the Frank-Condon principle, which was introduced in Section 2.2.2 and states that the nuclear motion is slow compared to the electronic reorganization [54][36]. As a consequence, the bath remains in the equilibrium state when the electronic excitation occurs at time $t = 0$ and we can write $R(0) = \rho_S(0) \otimes \rho_B$. Inserting this in Equation 4.30 yields:

$$\begin{aligned} \rho_S(t) &= \sum_{\mu} \langle\mu| U(t) (\rho_S(0) \otimes \sum_{\nu} \lambda_{\nu} |\nu\rangle\langle\nu|) U^\dagger(t) |\mu\rangle \\ &= \sum_{\mu\nu} \lambda_{\nu} \langle\mu| U(t) |\nu\rangle \rho_S(0) \langle\nu| U^\dagger(t) |\mu\rangle \\ &= \sum_{\mu\nu} M_{\mu\nu}(t) \rho_S(0) M_{\mu\nu}^\dagger(t). \end{aligned} \quad (4.31)$$

Here, we defined the operators $M_{\mu\nu}(t)$ as:

$$M_{\mu\nu}(t) = \sqrt{\lambda_{\nu}} \langle\mu| U(t) |\nu\rangle. \quad (4.32)$$

These operators are called Kraus operators. It is important to note that they act only on the system, as the bath was traced out. Restating Equation 4.31, we obtain the Kraus operator sum representation:

$$\rho_S(t) = \sum_{\mu\nu} M_{\mu\nu}(t) \rho_S(0) M_{\mu\nu}^\dagger(t). \quad (4.33)$$

In the derivation, we only assumed that the full system density matrix initially factorizes, but no Born, Markov, or secular approximation, as for the Lindblad equation, was used. Thus, the operator sum representation is not limited to Markovian dynamics and can also describe non-Markovian dynamics.

Following this derivation, we investigate the properties of the operator sum representation and the Kraus operators over the course of the next sections.

4.2.2 Conservation of Probabilities

We demand that the total probability must be conserved, which leads to the following condition for the Kraus operators:

$$\begin{aligned}\text{Tr}[\rho_S(t)] &= \text{Tr}\left[\sum_{\mu\nu} M_{\mu\nu}(t)\rho_S(0)M_{\mu\nu}^\dagger(t)\right] = \sum_{\mu\nu} \text{Tr}[M_{\mu\nu}(t)\rho_S(0)M_{\mu\nu}^\dagger(t)] \\ &= \sum_{\mu\nu} \text{Tr}[M_{\mu\nu}^\dagger(t)M_{\mu\nu}(t)\rho_S(0)] = \text{Tr}\left[\sum_{\mu\nu} M_{\mu\nu}^\dagger(t)M_{\mu\nu}(t)\rho_S(0)\right] = 1,\end{aligned}\tag{4.34}$$

where we used the cyclicity of the trace. Hence, we see that the probability conservation implies:

$$\sum_{\mu\nu} M_{\mu\nu}^\dagger(t) M_{\mu\nu}(t) = I.\tag{4.35}$$

We can verify this equation by using the definition of the Kraus operators and obtain:

$$\begin{aligned}\sum_{\mu\nu} M_{\mu\nu}^\dagger(t)M_{\mu\nu}(t) &= \sum_{\mu\nu} \lambda_\nu \langle \nu|U^\dagger(t)|\mu\rangle \langle \mu|U(t)|\nu\rangle = \sum_{\nu} \lambda_\nu \langle \nu|U^\dagger(t)\left(\sum_{\mu} |\mu\rangle\langle\mu|\right)U(t)|\nu\rangle \\ &= \sum_{\nu} \lambda_\nu \langle \nu|U^\dagger(t)U(t)|\nu\rangle = \sum_{\nu} \lambda_\nu \langle \nu|\nu\rangle = \sum_{\nu} \lambda_\nu = 1,\end{aligned}\tag{4.36}$$

where we used the unitarity of $U = e^{-\frac{i}{\hbar}Ht}$, the orthonormality of the $|\nu\rangle$ and that the λ_ν sum to 1, since $\text{Tr}_B[\rho_B] = 1$. Lastly, it is interesting to note that a single Kraus operator is forced to be unitary, as in this case $M^\dagger(t)M(t) = 1$.

4.2.3 Relation to the Schrödinger Equation

In the following, we prove that the Kraus operator sum representation contains the unitary evolution of the system in the form of the Schrödinger equation as a special case. Assuming we can write $U = U_S \otimes U_B$, we obtain:

$$M_{\mu\nu} = \sqrt{\lambda_\nu} \langle \mu|U_S \otimes U_B|\nu\rangle = c_{\mu\nu}U_S,\tag{4.37}$$

where $c_{\mu\nu} = \sqrt{\lambda_\nu} \langle \mu|U_S|\nu\rangle$. Using the sum normalization rule from Equation 4.36 it is easy to see that $\sum_{\mu\nu} c_{\mu\nu}^* c_{\mu\nu} = 1$ applies since:

$$\sum_{\mu\nu} M_{\mu\nu}^\dagger M_{\mu\nu} = \sum_{\mu\nu} c_{\mu\nu}^* c_{\mu\nu} U_S^\dagger U_S = \sum_{\mu\nu} c_{\mu\nu}^* c_{\mu\nu} = I,\tag{4.38}$$

where the unitarity of U yields $U_S^\dagger U_S = I$. We can use these insights to write for the time evolution of the system:

$$\rho_S(t) = \sum_{\mu\nu} M_{\mu\nu} \rho_S(0) M_{\mu\nu}^\dagger = \sum_{\mu\nu} c_{\mu\nu} U_S \rho_S(0) c_{\mu\nu}^* U_S^\dagger = U_S \rho_S(0) U_S^\dagger,\tag{4.39}$$

which is the Schrödinger equation for the time evolution of the system density matrix as derived in Equation 3.9. Thus, the Kraus operator sum representation is more general than the Schrödinger equation, since it contains the latter as a special case. It can be seen as a generalization of the Schrödinger equation to open quantum systems.

4.2.4 Kraus Operators and Measurements

In this section, we demonstrate that the Kraus operator sum representation is also suitable for the description of measurements. To this end, we introduce a set of measurement operators F_k which fulfill $\sum_k F_k^\dagger F_k = I$. If a quantum state ρ is subject to a measurement with the operator F_k , the state collapses from ρ to ρ_k :

$$\rho_k = \frac{F_k \rho F_k^\dagger}{p_k}. \quad (4.40)$$

Here, the probability to obtain the state ρ_k is given as $p_k = \text{Tr} [F_k \rho F_k^\dagger]$. Next, we consider a non-selective measurement, which means that we do not learn the outcome of the measurement. Then we obtain:

$$\rho \rightarrow \langle \rho \rangle = \sum_k p_k \rho_k = \sum_k F_k \rho F_k^\dagger, \quad (4.41)$$

where we average over the post-measurement states ρ_k . Equation 4.41 is essentially a Kraus operator sum representation, where the Kraus operators are the measurement operators F_k . This means that both measurements and dynamics can be written in the form of an operator sum representation. Since measurements and dynamics are the only quantum processes, we see that the operator sum representation is truly fundamental.

4.2.5 The Operator Sum Representation as a Map

To better investigate the properties of the operator sum representation, we show that it can be understood as a map. To this end, we define a quantum map or quantum channel Φ from the initial state to the final state:

$$\Phi : \rho_S(0) \mapsto \rho_S(t), \quad (4.42)$$

which can also be written as $\rho_S(t) = \Phi[\rho_S(0)]$. Here, $\Phi[X] = \sum_\alpha M_\alpha X M_\alpha^\dagger$, where the indices μ and ν are merged to one index α .

Moving on to the properties of the map Φ , we first notice that Φ is trace preserving since:

$$\begin{aligned} \text{Tr}[\Phi(\rho_S)] &= \text{Tr} \left[\sum_\alpha M_\alpha \rho_S M_\alpha^\dagger \right] = \sum_\alpha \text{Tr} [M_\alpha^\dagger M_\alpha \rho_S] \\ &= \text{Tr} \left[\left(\sum_\alpha M_\alpha^\dagger M_\alpha \right) \rho_S \right] = \text{Tr} [\rho_S], \end{aligned} \quad (4.43)$$

where we used the probability conservation $\sum_\alpha M_\alpha^\dagger M_\alpha = I$ and the cyclicity of the trace. Next we show that Φ is linear for any scalars a and b :

$$\begin{aligned}
\Phi(a\rho_{S_1} + b\rho_{S_2}) &= \sum_{\alpha} M_{\alpha}(a\rho_{S_1} + b\rho_{S_2})M_{\alpha}^{\dagger} \\
&= a \sum_{\alpha} M_{\alpha}\rho_{S_1}M_{\alpha}^{\dagger} + b \sum_{\alpha} M_{\alpha}\rho_{S_2}M_{\alpha}^{\dagger} = a\Phi(\rho_{S_1}) + b\Phi(\rho_{S_2}).
\end{aligned} \tag{4.44}$$

The third central property of Φ is positivity. Positivity means that Φ maps any positive operator to another positive operator. An operator is positive if it has only non-negative eigenvalues, where at least one eigenvalue is non-zero. Suppose we have a positive operator A with a spectral decomposition of $A = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$. To show positivity it suffices to show that $\langle\nu|\Phi(A)|\nu\rangle \geq 0$ for all $|\nu\rangle \in \mathcal{H}_S$, where $\mathcal{H}_S$ is the Hilbert space of the system. This ensures that the eigenvalues of A are positive, since the scalar products are non-negative. Defining $|w_{\alpha}\rangle = M_{\alpha}^{\dagger}|\nu\rangle$ we thus have:

$$\begin{aligned}
\langle\nu|\Phi(A)|\nu\rangle &= \langle\nu|\sum_{\alpha} M_{\alpha}AM_{\alpha}^{\dagger}|\nu\rangle = \sum_{\alpha} \langle\nu|M_{\alpha}AM_{\alpha}^{\dagger}|\nu\rangle \\
&= \sum_{\alpha} \langle w_{\alpha}|A|w_{\alpha}\rangle = \sum_{\alpha} \lambda_i |\langle w_{\alpha}|i\rangle|^2.
\end{aligned} \tag{4.45}$$

Since the λ_i are positive by definition, Equation 4.45 is positive, and hence Φ is a positive map.

Now the question arises whether every map that is trace preserving, linear, and positive has an operator sum representation. The answer to this question is negative. For a map to have an operator sum representation, positivity is not enough. It must be completely positive while simultaneously being trace preserving and linear. For Φ to be completely positive, it must hold that if $A \geq 0$, then $(\Phi \otimes \mathcal{I}_R^{(k)})(A) \geq 0$ for all $k \in \mathbb{N}$, where k is the dimension of the Hilbert space $\mathcal{H}_R$ and $\mathcal{I}_R$ denotes the identity operation on $\mathcal{H}_R$. In addition, if $\text{Tr}[\Phi(X)] = \text{Tr}[X]$ for all X of the Hilbert space of linear operators acting on $\mathcal{H}_S$, then Φ is said to be a completely positive trace preserving map.

Next, we prove that if Φ has an operator sum representation, it is completely positive. To this end, we show that $\Phi \otimes \mathcal{I}_R^{(k)}$ is positive for all k . Assuming that Φ has an operator sum representation, we obtain:

$$(\Phi \otimes \mathcal{I}_R)(A) = \sum_{\alpha} (M_{\alpha} \otimes \mathcal{I}_R)A(M_{\alpha}^{\dagger} \otimes \mathcal{I}_R) \tag{4.46}$$

and thus:

$$\langle\nu|(\Phi \otimes \mathcal{I}_R)(A)|\nu\rangle = \sum_{\alpha} \langle\nu|(M_{\alpha} \otimes \mathcal{I}_R)A(M_{\alpha}^{\dagger} \otimes \mathcal{I}_R)|\nu\rangle = \sum_{\alpha} \langle w_{\alpha}|A|w_{\alpha}\rangle > 0, \tag{4.47}$$

where we defined $|w_{\alpha}\rangle = (M_{\alpha}^{\dagger} \otimes \mathcal{I}_R)|\nu\rangle$ and $|\nu\rangle \in \mathcal{H}_S \otimes \mathcal{H}_R$. Since A is positive per construction, we conclude that Φ is completely positive. Note that this proof mainly works because we have the same Kraus operators to the left and right of A . If Φ was $\Phi[X] = \sum_{\alpha\beta} M_{\alpha}XM_{\beta}^{\dagger}$, this proof would not have been possible. The proof of the reverse direction is more challenging and thus omitted.

To summarize, we note that the map Φ has an operator sum representation, i.e $\Phi[X] = \sum_{\alpha} M_{\alpha}XM_{\alpha}^{\dagger}$ with $\sum_{\alpha} M_{\alpha}^{\dagger}M_{\alpha} = 1$, if Φ is trace preserving, linear and completely positive.

4.2.6 Relation between the Lindblad Equation and the Operator Sum Representation

In this section, we aim to derive the Lindblad equation from the operator sum representation. To this end, we show that the solution of the differential equation $\dot{\rho}_S(t) = \mathfrak{L}\rho_S(t)$ is a quantum map, where the Lindbladian $\mathfrak{L}$ will play the role of a generator for the dynamics of open quantum systems. To ensure that the evolution generated by $\mathfrak{L}$ is a valid quantum map, we will prove in Section 4.2.7 that it is completely positive.

We start by performing a Taylor expansion of $\rho_S(dt)$ around $t = 0$, where dt is an infinitesimal small timestep:

$$\rho_S(dt) = \rho_S(0) + \dot{\rho}_S|_0 dt + \mathcal{O}(dt^2). \quad (4.48)$$

The error on the order $\mathcal{O}(dt^2)$ enforces the time step to be kept small. On the other hand, the time evolution generated by the operator sum representation yields:

$$\rho_S(dt) = \sum_{\alpha} M_{\alpha}(dt) \rho_S(0) M_{\alpha}^{\dagger}(dt). \quad (4.49)$$

Comparing Equation 4.48 and 4.49, we aim to find the Kraus operators such that the two equations agree up to the order $\mathcal{O}(dt)$. From the comparison, we see that one of the Kraus operators must contain the identity operator, and that a small time step dt is necessary. Thus, we make the simple Ansatz:

$$M_0 = I + L_0 dt, \quad (4.50)$$

which yields:

$$\begin{aligned} M_0 \rho_S(0) M_0^{\dagger} &= (I + L_0 dt) \rho_S(0) (I + L_0^{\dagger} dt) \\ &= \rho_S(0) + [L_0 \rho_S(0) + \rho_S(0) L_0^{\dagger}] dt + \mathcal{O}(dt^2). \end{aligned} \quad (4.51)$$

Clearly, Equation 4.51 already shows a contribution on the order $\mathcal{O}(dt)$. However, more than one Kraus operator is required to reproduce the Lindblad equation, since a single Kraus operator leads only to a unitary evolution, as shown in Equation 4.34. We thus make the simplest possible reasonable Ansatz for the remaining Kraus operators:

$$M_{\alpha} = \sqrt{dt} L_{\alpha}, \quad (4.52)$$

where $\alpha \geq 1$. Inserting Equation 4.52 into the operator sum representation, we end up with:

$$M_{\alpha} \rho_S(0) M_{\alpha}^{\dagger} = dt L_{\alpha} \rho_S(0) L_{\alpha}^{\dagger}. \quad (4.53)$$

Using the condition for the conservation of probability from Equation 4.34 up to order $\mathcal{O}(dt)$, we obtain the following condition:

$$I = M_0^{\dagger} M_0 + \sum_{\alpha \geq 1} M_{\alpha}^{\dagger} M_{\alpha} = I + dt (L_0 + L_0^{\dagger} + \sum_{\alpha \geq 1} L_{\alpha}^{\dagger} L_{\alpha}) + \mathcal{O}(dt^2). \quad (4.54)$$

The operator L_0 can be decomposed into its Hermitian and anti-Hermitian part H and A , which yields

$L_0 = A - \frac{i}{\hbar}H$. Using $A = A^\dagger$ and $H = H^\dagger$ we find $L_0^\dagger = A + \frac{i}{\hbar}H$. Inserting into Equation 4.54 yields:

$$A = -\frac{1}{2} \sum_{\alpha \geq 1} L_\alpha^\dagger L_\alpha. \quad (4.55)$$

Using the result for A and the decomposition of L_0 in Equation 4.49 as well as Equation 4.51, we arrive at:

$$\begin{aligned} \rho_S(dt) &= M_0 \rho_S(0) M_0^\dagger + \sum_{\alpha \geq 1} M_\alpha \rho_S(0) M_\alpha^\dagger \\ &= \rho_S(0) + (A - \frac{i}{\hbar}H)dt \rho_S(0) + \rho_S(0)(A + \frac{i}{\hbar}H)dt + \sum_{\alpha \geq 1} L_\alpha \rho_S(0) L_\alpha^\dagger dt + \mathcal{O}(dt^2) \\ &= \rho_S(0) - \frac{i}{\hbar}[H, \rho_S(0)]dt + \{A, \rho_S(0)\}dt + \sum_{\alpha \geq 1} L_\alpha \rho_S(0) L_\alpha^\dagger dt + \mathcal{O}(dt^2) \\ &= \rho_S(0) - \frac{i}{\hbar}[H, \rho_S(0)]dt + \sum_{\alpha \geq 1} \left(L_\alpha \rho_S(0) L_\alpha^\dagger - \frac{1}{2} \{L_\alpha^\dagger L_\alpha, \rho_S(0)\} \right) dt + \mathcal{O}(dt^2). \end{aligned} \quad (4.56)$$

Taking the limit for small time steps yields:

$$\begin{aligned} \dot{\rho}_S(t)|_{t=0} &= \lim_{dt \rightarrow 0} \frac{\rho_S(dt) - \rho_S(0)}{dt} \\ &= -\frac{i}{\hbar}[H, \rho_S(0)] + \sum_{\alpha \geq 1} \left(L_\alpha \rho_S(0) L_\alpha^\dagger - \frac{1}{2} \{L_\alpha^\dagger L_\alpha, \rho_S(0)\} \right). \end{aligned} \quad (4.57)$$

Equation 4.57 already looks very similar to the Lindblad equation. However, the operators L_α are not dimensionless since they have units of $\frac{1}{\sqrt{\text{time}}}$, as is evident from the Ansatz in Equation 4.52. To resolve this problem, we introduce coefficients γ_α , which have units of $\frac{1}{\text{time}}$ and thus correspond to rates. Replacing L_α with $\sqrt{\gamma_\alpha} L_\alpha$ ensures that the operators L_α are dimensionless. We then have:

$$\dot{\rho}_S(t)|_{t=0} = -\frac{i}{\hbar}[H, \rho_S(0)] + \sum_{\alpha} \gamma_\alpha \left(L_\alpha \rho_S(0) L_\alpha^\dagger - \frac{1}{2} \{L_\alpha^\dagger L_\alpha, \rho_S(0)\} \right), \quad (4.58)$$

which corresponds to a short time expansion of $\dot{\rho}_S(t)$ around $t = 0$. To finally arrive at the Lindblad equation, we must assume that Equation 4.58 is valid for all times $t > 0$. This assumption essentially corresponds to the Markovian limit, in which the system has no memory and the evolution effectively resets after each time step dt . Within this approximation, one arrives at the Lindblad equation:

$$\frac{d}{dt} \rho_S(t) = -\frac{i}{\hbar}[H, \rho_S(t)] + \sum_{\alpha} \gamma_\alpha \left(L_\alpha \rho_S(t) L_\alpha^\dagger - \frac{1}{2} \{L_\alpha^\dagger L_\alpha, \rho_S(t)\} \right) \equiv \mathfrak{L} \rho_S(t). \quad (4.59)$$

Here, the L_α are called Lindblad operators and $\mathfrak{L}$ is the Lindbladian. The operator H was defined in the decomposition of L_0 as an arbitrary Hermitian operator. However, it turns out that this operator H can indeed be interpreted as the Hamiltonian of the system. Analyzing the Lindblad equation further, we see that the first term corresponds to the unitary evolution of the system Hamiltonian, and the second part is the so-called dissipator, which accounts for the influence of the bath on the system and thus gives rise to a non-unitary evolution. The dissipator is then given as:

$$\mathfrak{L}_D[\rho_S(t)] = \sum_{\alpha} \gamma_\alpha \left(L_\alpha \rho_S(t) L_\alpha^\dagger - \frac{1}{2} \{L_\alpha^\dagger L_\alpha, \rho_S(t)\} \right), \quad \gamma_\alpha \geq 0. \quad (4.60)$$

To summarize, we have derived the Lindblad equation from the operator sum representation using a small time expansion around $t = 0$. In the following section, we show that it can be interpreted as a quantum map with Kraus operators given by $M_0 = I + (-iH + A)dt$ and $M_\alpha = \sqrt{\gamma_\alpha dt} L_\alpha$, where $\alpha \geq 1$.

4.2.7 The Lindbladian as a Completely Positive Map

To prove that the solution of the Lindblad equation is a completely positive map, we essentially reverse engineer the proof from the last section, where we outlined that the Lindblad equation can be derived from the Kraus operator sum representation. Showing that a concatenation of completely positive maps is still a completely positive map will complete the proof.

We start from the Lindblad equation and use :

$$A = -\frac{1}{2} \sum_{\alpha \geq 1} L_\alpha^\dagger L_\alpha \quad (4.61)$$

to obtain:

$$\begin{aligned} \dot{\rho}_S(t) &= \mathfrak{L}\rho_S(t) = -\frac{i}{\hbar}[H, \rho_S(t)] + \sum_{\alpha \geq 1} (L_\alpha \rho_S(t) L_\alpha^\dagger - \frac{1}{2}\{L_\alpha^\dagger L_\alpha, \rho_S(t)\}) \\ &= -\frac{i}{\hbar}[H, \rho_S(t)] + \{A, \rho_S(t)\} + \sum_{\alpha \geq 1} L_\alpha \rho_S(t) L_\alpha^\dagger \\ &= (A - \frac{i}{\hbar}H)\rho_S(t) + \rho_S(t)(A + \frac{i}{\hbar}H) + \sum_{\alpha \geq 1} L_\alpha \rho_S(t) L_\alpha^\dagger. \end{aligned} \quad (4.62)$$

Defining $M_0 \equiv I + (A - \frac{i}{\hbar}H)dt + \mathcal{O}(dt^2)$, we find:

$$\begin{aligned} M_0 \rho_S(t) M_0^\dagger &= [I + (A - \frac{i}{\hbar}H)dt] \rho_S(t) [I + (A + \frac{i}{\hbar}H)dt] + \mathcal{O}(dt^2) \\ &= \rho_S(t) + (A - \frac{i}{\hbar}H)dt \rho_S(t) + \rho_S(t)(A + \frac{i}{\hbar}H)dt + \mathcal{O}(dt^2). \end{aligned} \quad (4.63)$$

Finally, using equations 4.62 and 4.63 as well as $M_\alpha \equiv L_\alpha \sqrt{dt}$, we arrive at:

$$\begin{aligned} \rho_S(t + dt) &= \rho_S(t) + \dot{\rho}_S(t)dt + \mathcal{O}(dt^2) \\ &= \rho_S(t) + (A - \frac{i}{\hbar}H)dt \rho_S(t) + \rho_S(t)(A + \frac{i}{\hbar}H)dt + \mathcal{O}(dt^2) + \sum_{\alpha \geq 1} (L_\alpha \sqrt{dt}) \rho_S(t) (L_\alpha^\dagger \sqrt{dt}) \\ &= \sum_{\alpha \geq 0} M_\alpha \rho_S(t) M_\alpha^\dagger + \mathcal{O}(dt^2) \equiv \Phi[\rho_S(t)], \end{aligned} \quad (4.64)$$

which is the Kraus operator sum representation and thus a valid quantum map if the set of $\{M_\alpha\}_{\alpha \geq 0}$ fulfills the probability conservation. Therefore, we calculate:

$$\sum_{\alpha \geq 0} M_\alpha^\dagger M_\alpha = I + dt(2A + \sum_{\alpha \geq 1} L_\alpha^\dagger L_\alpha) + \mathcal{O}(dt^2) = I + \mathcal{O}(dt^2), \quad (4.65)$$

which shows that this condition is fulfilled, and therefore, in the limit $dt \rightarrow 0$, Φ defines a valid quantum

map from $\rho_S(t)$ to $\rho_S(t + dt)$. To prove that Φ is a valid quantum map for all times t , we must show that the concatenation of quantum maps is also a quantum map. This is necessary because, by taking $dt = \lim_{n \rightarrow \infty} t/n$ and applying Φ iteratively on $\rho_S(0)$, we obtain:

$$\lim_{n \rightarrow \infty} \Phi^{on}[\rho_S(0)] = \Phi[\Phi[\dots \Phi[\rho_S(0)]]] = \rho_S(t), \quad (4.66)$$

which is equivalent to the solution of the Lindblad equation, since it maps $\rho_S(0) \rightarrow \rho_S(dt) \rightarrow \rho_S(2dt) \rightarrow \dots \rightarrow \rho_S(t)$. To this end, let Φ_1 and Φ_2 be valid quantum maps. Then:

$$\Phi_2 \circ \Phi_1[\rho_S] = \Phi_2[\Phi_1[\rho_S]] = \sum_{\beta} M'_{\beta} \Phi_1[\rho_S] M_{\beta}^{\dagger} = \sum_{\alpha\beta} M'_{\beta} M_{\alpha} \rho_S M_{\alpha}^{\dagger} M_{\beta}^{\dagger} = \sum_{\gamma} M''_{\gamma} \rho_S M_{\gamma}^{\dagger}, \quad (4.67)$$

where $M''_{\gamma} = M'_{\beta} M_{\alpha}$, and

$$\sum_{\gamma} M''_{\gamma} M_{\gamma}^{\dagger} = \sum_{\alpha} M_{\alpha}^{\dagger} \left(\sum_{\beta} M'_{\beta} M_{\beta}^{\dagger} \right) M_{\alpha} = \sum_{\alpha} M_{\alpha}^{\dagger} M_{\alpha} = I, \quad (4.68)$$

as required, showing that the Lindblad equation gives rise to a completely positive map Φ .

4.3 Hierarchical Equations of Motion

So far, Markovian dynamics have been extensively discussed. The Markovian Lindblad equation is derived under the Born-Markov approximation and the secular approximation. Thus, it is incapable of describing bath relaxations that exist on the timescale of the system's dynamics. Hence, correlations in the bath, that can affect the dynamics of the system, are not accounted for [54]. The operator sum representation can, in general, describe both Markovian and non-Markovian dynamics, depending on the nature of the Kraus operators, as shown in Section 4.2.1.

In this section, the hierarchical equations of motion (HEOM) approach will be introduced. In contrast to the Markovian Lindblad equation, it is a non-Markovian approach and, therefore, neither the Markovian nor the secular approximation is needed for its derivation. In HEOM, the degrees of freedom of the environment are encoded in so-called auxiliary density operators. These operators, together with the system density matrix, form a hierarchy of coupled differential equations. In theory, the hierarchy extends indefinitely and must thus be truncated at a certain hierarchical level. To derive the HEOM formalism, we follow Reference [55] and [54].

As a starting point for the derivation of HEOM, we integrate the Von Neumann equation $\frac{d}{dt} R(t) = \mathcal{L} R(t)$, derived in Section 3.1.2. Here, $R(t)$ is the joint density matrix of the system and the bath. The Liouvillian of the total system is given as $\mathcal{L} = \mathcal{L}_S + \mathcal{L}_B + \mathcal{L}_{SB}$ with $\mathcal{L}_S$, $\mathcal{L}_B$, and $\mathcal{L}_{SB}$ being the Liouvillian of the system, the bath, and the interaction between system and bath. Hence, in the interaction picture, we obtain:

$$\tilde{R}(t) = \tilde{R}(0) - \frac{i}{\hbar} \int_0^t dt_1 \tilde{\mathcal{L}}_{SB}(t_1) \tilde{R}(t_1). \quad (4.69)$$

Equation 4.69 can be used recursively, where, in contrast to the derivation of the Lindblad equation, we do not stop at the second order. This implies that all the density matrices of the past are considered to evaluate the density matrix at present:

$$\tilde{R}(t) = \tilde{R}(0) + \sum_{k=1}^{\infty} \left[\left(-\frac{i}{\hbar} \right)^k \int_0^t dt_1 \cdots \int_0^{t_{k-1}} dt_k \tilde{\mathcal{L}}_{SB}(t_1) \cdots \tilde{\mathcal{L}}_{SB}(t_k) \right] \tilde{R}(0). \quad (4.70)$$

With $\tilde{\mathcal{L}}_{SB}$, the summands describe the action of $\tilde{H}_{SB}$ on the system at different times. In general, the Liouvillians at different times do not commute, which is taken into account by the time ordering operator $\mathcal{T}$. When applied to a product of two operators, it yields:

$$\mathcal{T}O_1(t_1)O_2(t_2) = O_1(t_1)O_2(t_2)\Theta(t_1 - t_2) + O_2(t_2)O_1(t_1)\Theta(t_2 - t_1), \quad (4.71)$$

where $\Theta(t)$ denotes the Heaviside step function. Generalizing Equation 4.71 to a product of n operators gives a total of $n!$ terms. As a result, we must divide by a factorial when applying the time-ordering operator:

$$\begin{aligned} \tilde{R}(t) &= \tilde{R}(0) + \mathcal{T} \sum_{k=1}^{\infty} \left[\frac{1}{k!} \left(-\frac{i}{\hbar} \right)^k \int_0^t dt_1 \cdots \int_0^t dt_k \tilde{\mathcal{L}}_{SB}(t_1) \cdots \tilde{\mathcal{L}}_{SB}(t_k) \right] \tilde{R}(0) \\ &= \mathcal{T} \sum_{k=0}^{\infty} \frac{1}{k!} \left(-\frac{i}{\hbar} \int_0^t dt_1 \tilde{\mathcal{L}}_{SB}(t_1) \right)^k \tilde{R}(0) = \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_0^t dt_1 \tilde{\mathcal{L}}_{SB}(t_1) \right) \tilde{R}(0), \end{aligned} \quad (4.72)$$

where we used the sum representation of the exponential function in the last equality. It is worth emphasizing that up to now, the expression is fully general and no approximations have been employed. However, Equation 4.72 still contains an infinite number of degrees of freedom of the bath, which are numerically impossible to handle. To solve this difficulty, we apply a partial trace over the bath degrees of freedom, and assume that the bath and system density matrix initially factorize. This is justified by the Frank-Condon principle, which states that the nuclear motion is slow compared to the electronic reorganization [54][36]. As a consequence, the bath remains in the equilibrium state when the electronic excitation occurs at time $t = 0$, and we can write $\tilde{R}(0) = \tilde{\rho}_S(0) \otimes \rho_B$. Here, $\rho_B = e^{-\beta H_B}/Z$ is the canonical density matrix of the bath, and $Z = \text{Tr}_B[e^{-\beta H}]$ the corresponding partition function. Then:

$$\begin{aligned} \tilde{\rho}_S(t) &= \mathcal{T} \text{Tr}_B \left[\exp \left(-\frac{i}{\hbar} \int_0^t dt_1 \tilde{\mathcal{L}}_{SB}(t_1) \right) \rho_B \right] \tilde{\rho}_S(0) \\ &= \mathcal{T} \left\langle \exp \left(-\frac{i}{\hbar} \int_0^t dt_1 \tilde{\mathcal{L}}_{SB}(t_1) \right) \right\rangle \tilde{\rho}_S(0), \end{aligned} \quad (4.73)$$

where we used $\langle \tilde{O}(t) \rangle = \text{Tr}_B[\tilde{O}(t)\rho_B]$. The bath is described by an infinite set of harmonic oscillators, where the position and momenta are stochastic variables and follow a Gaussian distribution [55]. Since the linear combination of variables with a Gaussian distribution yields again a Gaussian distribution, we can infer that the interaction Liouvillian $\mathcal{L}_{SB}$ also has a Gaussian distribution. Hence, in a cumulant expansion for a Gaussian process, only the second moment survives [56] and we obtain:

$$\begin{aligned} \tilde{\rho}_S(t) &= \mathcal{T} \left\langle \exp \left(-\frac{i}{\hbar} \int_0^t dt_1 \tilde{\mathcal{L}}_{SB}(t_1) \right) \right\rangle \tilde{\rho}_S(0) \\ &= \mathcal{T} \exp \left(-\frac{1}{2\hbar^2} \int_0^t dt_1 \int_0^t dt_2 \langle \tilde{\mathcal{L}}_{SB}(t_1) \tilde{\mathcal{L}}_{SB}(t_2) \rangle \right) \tilde{\rho}_S(0). \end{aligned} \quad (4.74)$$

4.3.1 The Bath Average

To calculate $\langle \tilde{\mathcal{L}}_{SB}(t_1) \tilde{\mathcal{L}}_{SB}(t_2) \rangle$ we first must introduce a few conventions. If an operator A is written as $A^\times$ or as A° , it follows that $A^\times O = [A, O]$ and $A^\circ O = \{A, O\}$ where O is an operator, the square brackets denote the commutator and the curly brackets the anticommutator. Recalling that in general $\mathcal{L}\rho = -\frac{i}{\hbar}[H, \rho]$, and that the bath Hamiltonian can be written as $\tilde{H}_{SB} = \sum_j \tilde{S}_j \otimes \tilde{B}_j$, we obtain:

$$\tilde{\mathcal{L}}_{SB}(t) = \sum_{j=1}^N \left(\tilde{S}_j(t) \tilde{B}_j(t) \right)^\times = \frac{1}{2} \sum_{j=1}^N \left(\tilde{B}_j(t)^\circ \tilde{S}_j(t)^\times + \tilde{B}_j(t)^\times \tilde{S}_j(t)^\circ \right), \quad (4.75)$$

where in the second equality, we use the operator identity $(AB)^\times = \frac{1}{4}\{A^\times, B^\circ\} + \frac{1}{4}\{A^\circ, B^\times\} + \frac{1}{2}[A, B]^\times = \frac{1}{2}(B^\circ A^\times + B^\times A^\circ)$, which is valid for commuting operators. In this case, we have $[\tilde{S}_j(t), \tilde{B}_j(t)] = 0$, since $\tilde{S}_j(t)$ and $\tilde{B}_j(t)$ act on different Hilbert spaces. The tensor product is omitted for brevity. Thus, we obtain:

$$\begin{aligned} \langle \tilde{\mathcal{L}}_{SB}(t_1) \tilde{\mathcal{L}}_{SB}(t_2) \rangle &= \frac{1}{2} \sum_{j=1}^N \left(\tilde{S}_j(t_1)^\times \left\langle \tilde{B}_j(t_1)^\circ \tilde{B}_j(t_2)^\circ \right\rangle \tilde{S}_j(t_2)^\times \right. \\ &\quad + \tilde{S}_j(t_1)^\times \left\langle \tilde{B}_j(t_1)^\circ \tilde{B}_j(t_2)^\times \right\rangle \tilde{S}_j(t_2)^\circ \\ &\quad + \tilde{S}_j(t_1)^\circ \left\langle \tilde{B}_j(t_1)^\times \tilde{B}_j(t_2)^\circ \right\rangle \tilde{S}_j(t_2)^\times \\ &\quad \left. + \tilde{S}_j(t_1)^\circ \left\langle \tilde{B}_j(t_1)^\times \tilde{B}_j(t_2)^\times \right\rangle \tilde{S}_j(t_2)^\circ \right). \end{aligned} \quad (4.76)$$

To calculate the bath averages of the four terms, we use the cyclic property of the trace. For the last two terms, we find $\left\langle \tilde{B}_j(t_1)^\times \tilde{B}_j(t_2)^\circ \right\rangle = 0$ and $\left\langle \tilde{B}_j(t_1)^\times \tilde{B}_j(t_2)^\times \right\rangle = 0$ and, hence, only the first two remain. They can be calculated as:

$$\begin{aligned} \langle \tilde{B}_j(t_1)^\circ \tilde{B}_j(t_2)^\circ \rangle &= \text{Tr}_B [\tilde{B}_j(t_1)^\circ \tilde{B}_j(t_2)^\times \rho_B] = \text{Tr}_B [\{\tilde{B}_j(t_1), \{\tilde{B}_j(t_2), \rho_B\}\}] \\ &= \text{Tr}_B [\tilde{B}_j(t_1) \tilde{B}_j(t_2) \rho_B + \tilde{B}_j(t_2) \rho_B \tilde{B}_j(t_1) + \tilde{B}_j(t_1) \rho_B \tilde{B}_j(t_2) + \rho_B \tilde{B}_j(t_2) \tilde{B}_j(t_1)] \\ &= \text{Tr}_B [\tilde{B}_j(t_1) \tilde{B}_j(t_2) \rho_B + \tilde{B}_j(t_1) \tilde{B}_j(t_2) \rho_B + \tilde{B}_j(t_2) \tilde{B}_j(t_1) \rho_B + \tilde{B}_j(t_2) \tilde{B}_j(t_1) \rho_B] \\ &= 2 \text{Tr}_B [\{\tilde{B}_j(t_1), \tilde{B}_j(t_2)\} \rho_B] = 2 \langle \{\tilde{B}_j(t_1), \tilde{B}_j(t_2)\} \rangle \end{aligned} \quad (4.77)$$

and

$$\begin{aligned} \langle \tilde{B}_j(t_1)^\circ \tilde{B}_j(t_2)^\times \rangle &= \text{Tr}_B [\tilde{B}_j(t_1)^\circ \tilde{B}_j(t_2)^\times \rho_B] = \text{Tr}_B [\{\tilde{B}_j(t_1), [\tilde{B}_j(t_2), \rho_B]\}] \\ &= \text{Tr}_B [\tilde{B}_j(t_1) \tilde{B}_j(t_2) \rho_B + \tilde{B}_j(t_2) \rho_B \tilde{B}_j(t_1) - \tilde{B}_j(t_1) \rho_B \tilde{B}_j(t_2) - \rho_B \tilde{B}_j(t_2) \tilde{B}_j(t_1)] \\ &= \text{Tr}_B [\tilde{B}_j(t_1) \tilde{B}_j(t_2) \rho_B + \tilde{B}_j(t_1) \tilde{B}_j(t_2) \rho_B - \tilde{B}_j(t_2) \tilde{B}_j(t_1) \rho_B - \tilde{B}_j(t_2) \tilde{B}_j(t_1) \rho_B] \\ &= 2 \text{Tr}_B [[\tilde{B}_j(t_1), \tilde{B}_j(t_2)] \rho_B] = 2 \langle [\tilde{B}_j(t_1), \tilde{B}_j(t_2)] \rangle. \end{aligned} \quad (4.78)$$

In the next step, we introduce a complex function $\alpha_j(t) = \alpha_{r,j}(t) + i\alpha_{i,j}(t)$, and define its real part as $\alpha_{r,j}(t_1 - t_2) \equiv \frac{1}{2} \langle \{\tilde{B}_j(t_1), \tilde{B}_j(t_2)\} \rangle$, and its imaginary part as $\alpha_{i,j}(t_1 - t_2) \equiv \frac{1}{2i} \langle [\tilde{B}_j(t_1), \tilde{B}_j(t_2)] \rangle$. The function $\alpha_j(t)$ contains the bath averages, and hence serves as a bath correlation function. Inserting $\alpha_{r,j}(t_1 - t_2)$ and $\alpha_{i,j}(t_1 - t_2)$ into Equation 4.76, we find:

$$\left\langle \tilde{\mathcal{L}}_{SB}(t_1) \tilde{\mathcal{L}}_{SB}(t_2) \right\rangle = 2 \sum_{j=1}^N \tilde{S}_j(t)^\times \left(\alpha_{r,j}(t_1 - t_2) \tilde{S}_j(t_2)^\times + i \alpha_{i,j}(t_1 - t_2) \tilde{S}_j(t_2)^\circ \right). \quad (4.79)$$

When substituting Equation 4.79 back into Equation 4.74, the sum in the exponent can be written as a product over exponentials, and thus yields:

$$\tilde{\rho}_S(t) = \mathcal{T} \prod_{j=1}^N \exp \left(-\frac{1}{\hbar^2} \int_0^{t_1} dt_1 \int_0^{t_1} dt_2 \tilde{S}_j(t_1)^\times [\alpha_{r,j}(t_1 - t_2) \tilde{S}_j(t_2)^\times + i \alpha_{i,j}(t_1 - t_2) \tilde{S}_j(t_2)^\circ] \right) \tilde{\rho}_S(0). \quad (4.80)$$

Since the bath oscillators are described with stochastic variables for the position and momenta, Equation 4.80 expresses the time propagation of the reduced density matrix in terms of statistical quantities of the environment. The operator acting on $\tilde{\rho}_S(0)$ is called the influence operator, and in the position basis, represents the influence functional of Feynman and Vernon [57]. The bath correlation function $\alpha_j(t)$ can be written in terms of the Drude-Lorentz spectral density, which is discussed in appendix B.1. For the bath correlation function at site j , we then have:

$$\alpha_j(t) = \int_0^\infty J_j(\omega) \left[\coth \left(\frac{\beta \hbar \omega}{2} \right) \cos(\omega t) - i \sin(\omega t) \right] d\omega. \quad (4.81)$$

For a detailed derivation of Equation 4.81, I refer to Reference [55]. As a result of the interaction between the system and the bath, an excitation in the system will influence the phonons of the bath, while the bath, in turn, exerts an influence on the system that decays over time. This effect corresponds to the reorganization of the nuclei, described in Section 2.2.2, and is characterized by the temporal bath correlation function $\alpha_j(t)$ and the influence functional. The evaluation of the integral in Equation 4.81 results in a series, which is called the Matsubara series. However, the Matsubara series converges very slowly to an exact result, and thus, a much faster converging series, the so-called Padé series, is more frequently used. Both series are derived in the appendix B.1. Using the series representation for the bath correlation function, we obtain:

$$\alpha_j(t) = p_{j0} \delta_+(t) + \sum_{k=1}^K p_{jk} e^{-\nu_{jk} t}, \quad (4.82)$$

where $p_{jk} = a_{jk} + i b_{jk}$ with real numbers a_{jk} and b_{jk} . Depending on which series representation is used, the ν_{jk} are the Matsubara or Padé frequencies associated with site j . The one-sided delta function $\delta_+(t)$ limits the integration to $(0, \infty)$, and in general, the bath correlation function is approximated to consist of K exponential terms. With the revised definition of the bath correlation function, we can find an equation of motion for the system density matrix:

$$\begin{aligned} \tilde{\rho}_S(t) = \mathcal{T} \prod_j \exp \left(\frac{1}{\hbar^2} \int_0^t dt_1 \tilde{\Phi}_j(t_1) \tilde{\Theta}_{j0}(t_1) \right) \cdot \\ \prod_k \exp \left(\frac{1}{\hbar^2} \int_0^t dt_1 \int_0^{t_1} dt_2 \tilde{\Phi}_j(t_1) e^{-\nu_{jk}(t_1 - t_2)} \tilde{\Theta}_{jk}(t_2) \right) \tilde{\rho}_S(0), \end{aligned} \quad (4.83)$$

where we defined the abbreviation $\tilde{\Phi}_j(t) = i \tilde{S}_j(t)^\times$ and $\tilde{\Theta}_{jk}(t) = i a_{jk} \tilde{S}_j(t)^\times - b_{jk} \tilde{S}_j(t)^\circ$.

4.3.2 Auxiliary Density Operators and the Hierarchy

We now introduce a set of so-called auxiliary density operators $\sigma_{\mathbf{n}}(t)$, which allow us to find an equation of motion with a nice hierarchical structure. Here, $\mathbf{n}$ is a matrix with matrix elements n_{jk} . We then find a similar equation to Equation 4.83, for the auxiliary density operators:

$$\begin{aligned} \tilde{\sigma}_{\mathbf{n}}(t) = & \mathcal{T} \prod_j \exp \left(\frac{1}{\hbar^2} \int_0^t dt_1 \tilde{\Phi}_j(t_1) \tilde{\Theta}_{j0}(t_1) \right) \cdot \\ & \prod_k \left(\frac{1}{\hbar^2} \int_0^t dt_1 e^{-\nu_{jk}(t-t_1)} \tilde{\Theta}_{jk}(t_1) \right)^{n_{jk}} \cdot \\ & \exp \left(\frac{1}{\hbar^2} \int_0^t dt_1 \int_0^{t_1} dt_2 \tilde{\Phi}_j(t_1) e^{-\nu_{jk}(t_1-t_2)} \tilde{\Theta}_{jk}(t_2) \right) \tilde{\rho}_S(0). \end{aligned} \quad (4.84)$$

Notice that if we set $\mathbf{n} = 0$, we recover Equation 4.83 for $\tilde{\rho}_S(t)$. Hence, the operator $\sigma_{\mathbf{n}}$ with $\mathbf{n} = 0$ is called the reduced density operator and represents the system density matrix $\tilde{\rho}_S(t)$. The other $\sigma_{\mathbf{n}}$ with $\mathbf{n} \neq 0$ are auxiliary density operators and encode the degrees of freedom of the bath. Taking the derivative of Equation 4.84, we obtain a set of differential equations that form a hierarchy:

$$\begin{aligned} \frac{d}{dt} \tilde{\sigma}_{\mathbf{n}}(t) = & \sum_{jk} \frac{1}{\hbar^2} \left(\tilde{\Phi}_j(t) \tilde{\Theta}_{j0}(t) - \hbar^2 n_{jk} \nu_{jk} \right) \tilde{\sigma}_{\mathbf{n}}(t) \\ & + \sum_{jk} \frac{1}{\hbar^2} \left(\tilde{\Phi}_j(t) \tilde{\sigma}_{\mathbf{n}_{jk+}}(t) + n_{jk} \tilde{\Theta}_{jk}(t) \tilde{\sigma}_{\mathbf{n}_{jk-}}(t) \right). \end{aligned} \quad (4.85)$$

Using $\frac{d\tilde{O}(t)}{dt} = e^{\frac{i}{\hbar}\mathcal{L}_0 t} \frac{d\hat{O}}{dt} e^{-\frac{i}{\hbar}\mathcal{L}_0 t}$ with $\mathcal{L}_0 = \mathcal{L}_S + \mathcal{L}_B$, a transformation to the Schrödinger picture yields:

$$\begin{aligned} \frac{d}{dt} \sigma_{\mathbf{n}}(t) = & -\frac{i}{\hbar} \mathcal{L}_S \sigma_{\mathbf{n}}(t) - \frac{1}{\hbar^2} \sum_{jk} \left(a_{j0} S_j^\times S_j^\times - i b_{j0} S_j^\times S_j^\circ + \hbar^2 n_{jk} \nu_{jk} \right) \sigma_{\mathbf{n}}(t) \\ & + \frac{1}{\hbar^2} \sum_{jk} \left(i S_j^\times \hat{\sigma}_{\mathbf{n}_{jk+}}(t) + i n_{jk} p_{jk} S_j \hat{\sigma}_{\mathbf{n}_{jk-}}(t) + i n_{jk} p_{jk}^* \hat{\sigma}_{\mathbf{n}_{jk-}}(t) S_j \right). \end{aligned} \quad (4.86)$$

Summing all values n_{jk} , we obtain $\mathcal{M}_{\mathbf{n}} = \sum_j^N \sum_k^K n_{jk}$, which indicates the level of the hierarchy. Here, N is the number of sites, or BChl molecules, and K is the number of exponentials considered in the bath correlation function in Equation 4.82. The hierarchy can be read off the auxiliary density operators, where for each level, determined by $\mathbf{n}$, contributions arise from the next level, indicated by $\mathbf{n}_{jk+}$, and the previous level, indicated by $\mathbf{n}_{jk-}$. So each auxiliary density operator is coupled to one of the previous level and one of the next level. The notation $\mathbf{n}_{jk\pm}$ implies that the element of $\mathbf{n}$, indexed by jk , is replaced by $jk \pm 1$. In theory, Equation 4.86 can be evaluated for arbitrarily high levels, and must thus be truncated at level $\mathcal{N}$ when increasing $\mathcal{N}$ has a negligible effect on the dynamics.

The number of auxiliary density operators increases rapidly with each level. If we denote the truncation level of the hierarchy with $\mathcal{N}$, then $\frac{(\mathcal{N}+(NK))!}{\mathcal{N}!(NK)!}$ auxiliary density operators are needed to calculate the dynamics of $\tilde{\rho}_S(t)$ until level $\mathcal{N}$. To further illustrate this with an example, we choose $N = 2$ and $K = 2$ and thus $\mathbf{n}$ is a two-by-two matrix. For the truncation level, we select $\mathcal{N} = 2$, and therefore obtain auxiliary density operators for the levels $\mathcal{M}_{\mathbf{n}} = 0$, $\mathcal{M}_{\mathbf{n}} = 1$, and $\mathcal{M}_{\mathbf{n}} = 2$. For $\mathcal{M}_{\mathbf{n}} = 0$, there is only one possibility, namely that all four $n_{jk} = 0$, which corresponds to the system density matrix. For $\mathcal{M}_{\mathbf{n}} = 1$, one of the four n_{jk}

must be one and the others zero, which yields four auxiliary density operators. For $\mathcal{M}_{\mathbf{n}} = 2$ either two of the four n_{jk} are one or one n_{jk} is two which yields 10 auxiliary density operators. Thus, for truncation level $\mathcal{N} = 2$ of the hierarchy we sum to obtain $15 = \frac{(\mathcal{N} + (NK))!}{\mathcal{N}!(NK)!}$ auxiliary density operators. This shows that if the truncation level cannot be kept small, a substantial number of differential equations must be solved for the auxiliary density operators. Hence, HEOM is potentially a very expensive method, and a compromise between computational cost and accuracy must be found. To this end, several approaches for the efficient implementation of HEOM are proposed in References [\[58\]](#)[\[59\]](#)[\[60\]](#).

The HEOM method can also be viewed in terms of annihilation and creation operators. To this end, we interpret each element n_{jk} of $\mathbf{n}$ as a bath mode labeled by jk , indicating that a certain auxiliary density operator $\sigma_{\mathbf{n}}$ contains n_{jk} quanta in bath mode jk . Since each level in the hierarchy corresponds to a certain number of quanta, auxiliary density operators from adjacent levels can be created by annihilating or creating one quanta.

The non-Markovian HEOM method allows for the calculation of the time evolution of the system density matrix, while the effect of the bath degrees of freedom is encoded in the auxiliary density operators. In the derivation of HEOM, no approximations were used, which makes it a numerically exact method. In practice, the accuracy is only limited by the finite truncation depth $\mathcal{N}$.

5 Quantum Computation and Quantum Circuits

Classical computers store and process data as individual bits and are limited by the fact that classical systems can occupy only one state at a time. In contrast, quantum computers are based on quantum mechanics and benefit from the ability of quantum systems to be in multiple states simultaneously, which is called superposition. Quantum computation aims to harness the computational power arising from the superposition principle [7].

Implementing quantum algorithms requires a fundamentally different circuit model. While classical circuits manipulate information using logic gates, quantum circuits use unitary gates, which perform linear transformations on the quantum counterparts of classical bits, the qubits [61].

In this chapter, we introduce the principal concepts of quantum computation, quantum information, and quantum circuits. Most of the basic principles are well described in the book by Nielsen and Chuang [7]. Hence, if not stated otherwise, this shall be considered as the main source, together with lecture notes on quantum computation by Ronald de Wolf [61].

5.1 Quantum Information

In classical computation, information is stored in bits, whereas in quantum computation, the unit of information is a quantum bit or qubit. Classical bits can take on a value of either 0 or 1. In contrast, qubits cannot only exist in the states $|0\rangle$ and $|1\rangle$, but also in a superposition of these states and thus live in the vector space $\mathbb{C}^2$ with the computational basis states $|0\rangle$ and $|1\rangle$. Thus, the state $|\psi\rangle$ of a qubit is given as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle. \quad (5.1)$$

Here, α and β are complex numbers that determine how similar the state of the qubit is to $|0\rangle$ or $|1\rangle$. To gain information about $|\psi\rangle$, we must measure the qubit, which yields either 0 with probability $|\alpha|^2$ or 1 with probability $|\beta|^2$, and implies the condition $|\alpha|^2 + |\beta|^2 = 1$. Consequently, with a single measurement, it is impossible to directly obtain α and β , and therefore the quantum state $|\psi\rangle$ of the qubit. To learn more about the quantum state of the qubit, repeated measurements are necessary.

Similarly, we can think of a system with two qubits. Each qubit lives in its own Hilbert space, indicating that the Hilbert space of the composite two-qubit system can be calculated as a tensor product of the two one-qubit Hilbert spaces. This yields four possible basis states, since after a measurement each qubit can be found in the basis state $|0\rangle$ or $|1\rangle$. If we invoke the superposition principle, the two-qubit state vector can be written as:

$$|\psi\rangle = \alpha_{00}|0\rangle \otimes |0\rangle + \alpha_{01}|0\rangle \otimes |1\rangle + \alpha_{10}|1\rangle \otimes |0\rangle + \alpha_{11}|1\rangle \otimes |1\rangle. \quad (5.2)$$

A measurement of $|\psi\rangle$ yields $|00\rangle$, $|01\rangle$, $|10\rangle$, or $|11\rangle$, where we have omitted the tensor product for brevity. The respective probabilities are $|\alpha_{00}|^2$, $|\alpha_{01}|^2$, $|\alpha_{10}|^2$, and $|\alpha_{11}|^2$. The conservation of probabilities again demands that $\sum_{x \in \{0,1\}} |\alpha_x|^2 = 1$, which implies that the qubit's state must be normalized.

A two-qubit system already suffices to explain an important quantum property, entanglement. Suppose we have the following quantum state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad (5.3)$$

which is one of the four Bell basis states. We claim that, in the state $|\Phi^+\rangle$, the two qubits are somehow correlated. And indeed, if we measure the first qubit as 0, the state collapses to $|00\rangle$, and we immediately know that the second qubit will take on a value of 0 as well. This works analogously if the measurement of the first qubit yields 1. Thus, measuring the first qubit determines the state of the second qubit. This correlation between the qubits is called entanglement. In general, a bipartite state $|\psi\rangle$ is said to be entangled when it cannot be expressed as a tensor product of $|\psi_A\rangle$ and $|\psi_B\rangle$, where $|\psi_A\rangle$ and $|\psi_B\rangle$ belong to two different Hilbert spaces. We also say that $|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$ does not factorize.

Everything discussed so far can be neatly extended to a system of n qubits with 2^n basis states. Suppose we have a system of 500 qubits, yielding 2^{500} basis states of the form $|x_1x_2\dots x_{500}\rangle$ with 2^{500} coefficients, where $x_1, \dots, x_{500} \in \{0, 1\}$. This number is larger than the estimated number of particles in the universe, which demonstrates the enormous potential power of quantum computing. However, the difficulty in harnessing this power lies in the measurement. To extract information from such enormous quantum states, a plethora of measurements would be needed. Therefore, not many algorithms exist where a quantum computer brings a significant advantage over a classical computer. To avoid this problem, quantum circuits for implementing the non-unitary evolution of open quantum systems are kept rather small, since finding an efficient algorithm is challenging enough.

5.2 The Bloch Sphere

So far, the only constraint for a qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. Thus, we can write $\alpha = \cos \frac{\theta}{2}$ and $\beta = e^{i\varphi} \sin \frac{\theta}{2}$ to obtain:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad (5.4)$$

which still fulfills $|\alpha|^2 + |\beta|^2 = 1$. The numbers θ and φ can be interpreted as angles on a three-dimensional sphere with radius 1. This sphere is usually called the Bloch sphere and can be seen in Figure 5.1.

The Bloch sphere is a nice tool to visualize the action of single-qubit gates, which will be introduced in the next section. It can also distinguish between pure and mixed states, since pure states lie on the surface of the sphere and mixed states lie beneath the surface. In terms of quantum computation, we are mostly concerned with pure states. The Bloch sphere shows the state $|0\rangle$ on its North Pole and the state $|1\rangle$ on its South Pole. States in a superposition between $|0\rangle$ and $|1\rangle$ are somewhere in between, but still on the surface, depending on their coefficients. As powerful as the Bloch sphere is, its use case is limited to single qubits and cannot be extended to visualize multi-qubit gates.

5.3 Elementary Gates

In classical computing, logic gates are used to manipulate information. For example, the classical NOT gate transforms a 0 into a 1 and vice versa. Similarly, in quantum computing, matrices act on several qubits, which transforms their joint state. Amazingly, these matrices have to fulfill only one property. The conservation of probability demands that a quantum state $|\psi\rangle$ must be normalized at any time. Thus, the

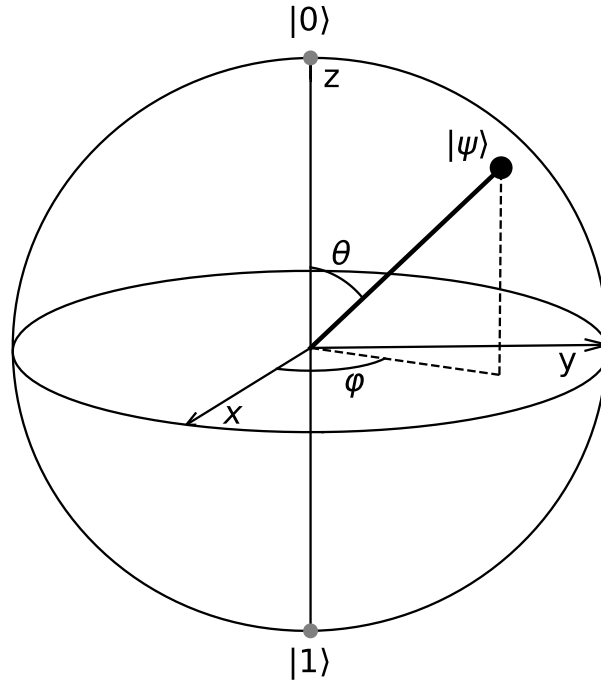


Figure 5.1: Pure single-qubit states correspond to a point on the Bloch sphere. The position of this point depends on the coefficients α and β , which can be parametrized by the angles θ and φ . The polar angle θ determines the amplitude ratio between $|0\rangle$ and $|1\rangle$, where $\theta = 0$ corresponds to $|0\rangle$ and $\theta = \pi$ corresponds to $|1\rangle$. The azimuth angle φ denotes the relative phase between $|0\rangle$ and $|1\rangle$.

operations acting on the state $|\psi\rangle$ must preserve its norm, which is only fulfilled if the operation is unitary. Therefore, a matrix U which acts on a quantum state must be unitary, i.e., $UU^\dagger = U^\dagger U = I$, where I describes the identity operation. Since unitary operations describe rotations, their action on a quantum state can be described as a rotation of that state on the Bloch sphere.

Unitary matrices that act on n qubits are called n -qubit gates. The dimension of such matrices must be exactly $2^n \times 2^n$. However, if we wish to implement a matrix with a different dimension, we need to embed it into a sufficiently large unit matrix of dimension $2^n \times 2^n$, ensuring that it is still unitary.

We will continue our investigation of elementary gates by starting with single-qubit gates. Single-qubit gates act on only one qubit, which results in a dimension of $2^1 \times 2^1$. Examples for single-qubit gates are the NOT gate X or the Hadamard gate H :

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (5.5)$$

which act as follows on the basis state $|0\rangle$:

$$X|0\rangle = |1\rangle, \quad H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \quad (5.6)$$

The NOT gate flips the state $|0\rangle$ to $|1\rangle$ and, in general, transforms a qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ into $|\psi'\rangle = \beta|0\rangle + \alpha|1\rangle$. The Hadamard gate produces an equal superposition between $|0\rangle$ and $|1\rangle$, with $\alpha = \beta = \frac{1}{\sqrt{2}}$. On the Bloch sphere, the X gate performs a rotation of π around the x -axis. Analogously,

single-qubit gates Y and Z exist, which rotate around the y - and z -axis by the same angle. As for the NOT gate, their matrix representation is equivalent to the Pauli matrices σ_y and σ_z , which reminds us that unitary operations on general quantum states are essentially rotations in Hilbert space. Also, the Hadamard gate can be visualized on the Bloch sphere. It corresponds to a rotation of $\frac{\pi}{2}$ around the y -axis followed by a rotation of π around the x -axis. Another important gate is the phase gate R_Φ :

$$R_\Phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Phi} \end{pmatrix}, \quad (5.7)$$

which rotates the phase of a qubit by an angle Φ around the z -axis. It acts as $R_\Phi|0\rangle = |0\rangle$ and $R_\Phi|1\rangle = e^{i\Phi}|1\rangle$, and thus $R_\Phi(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle + e^{i\Phi}\beta|1\rangle$.

In general, infinitely many single-qubit gates exist, since one can find infinitely many ways to construct unitary matrices. Therefore, it is all the more useful that an arbitrary single-qubit gate can be built using a finite set of single-qubit gates. In fact, an arbitrary unitary 2×2 matrix U can be decomposed as [7]:

$$U = e^{i\alpha} \begin{pmatrix} e^{-i\beta/2} & 0 \\ 0 & e^{i\beta/2} \end{pmatrix} \begin{pmatrix} \cos \frac{\gamma}{2} & -\sin \frac{\gamma}{2} \\ \sin \frac{\gamma}{2} & \cos \frac{\gamma}{2} \end{pmatrix} \begin{pmatrix} e^{-i\delta/2} & 0 \\ 0 & e^{i\delta/2} \end{pmatrix}, \quad (5.8)$$

using 4 real-valued angles α , β , γ , and δ . Such decompositions are very powerful tools, since they allow us to build any matrix using only a few basis gates. In the following, we will encounter this concept repeatedly, since quantum computers only use a specific set of elementary gates to build arbitrary operations.

In principle, most of what has been discussed so far also applies to multiple-qubit gates. We will now introduce a two-qubit gate that, together with a few single-qubit gates, enables us to build a complete set of elementary gates. This two-qubit gate is the controlled NOT gate, or simply the CNOT gate. The CNOT gate acts on two qubits, known as the control and target qubits. The action of the CNOT is as follows: if the control qubit is $|1\rangle$, the target qubit is flipped, and if the control qubit is $|0\rangle$, the target qubit remains unchanged. With the matrix representation of the CNOT gate:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (5.9)$$

it becomes already clear how it acts on the four basis states of a two-qubit system:

$$\text{CNOT}|00\rangle = |00\rangle; \quad \text{CNOT}|01\rangle = |01\rangle; \quad \text{CNOT}|10\rangle = |11\rangle; \quad \text{CNOT}|11\rangle = |10\rangle. \quad (5.10)$$

Here, the control is on the first qubit, and thus the state does not change if the first qubit is $|0\rangle$. However, if the first qubit is $|1\rangle$, the second qubit is flipped. By writing $|00\rangle = (1, 0, 0, 0)^T$, and similarly for the remaining three basis states, one can verify the action of the CNOT gate with a simple matrix-vector multiplication. Like all quantum gates, the CNOT gate must be unitary, which can be verified by multiplying it by its adjoint. As already mentioned, the CNOT gate, together with a single-qubit gate, forms a complete set, which means that an arbitrary operation acting on an arbitrary number of qubits can be built from CNOT gates and single-qubit gates. To prove this central theorem, we first must introduce quantum circuits and learn how to decompose an arbitrary unitary gate into so-called two-level unitary gates.

5.4 Quantum Circuits

A quantum circuit is the quantum analog of the classical circuit. Classical circuits consist of wires and logic gates for the transport and manipulation of information. In a quantum circuit, the logic gates are replaced by a set of elementary unitary gates that perform linear operations on a set of qubits, as introduced in the last section. In contrast to a classical circuit, a quantum circuit does not contain physical wires. Thus, the flow of information is simply indicated by black lines, representing a passage of time or the movement of a physical particle, like a photon, through space. An exemplary quantum circuit, consisting of two qubits $|q_0\rangle$ and $|q_1\rangle$, is shown in Figure 5.2.

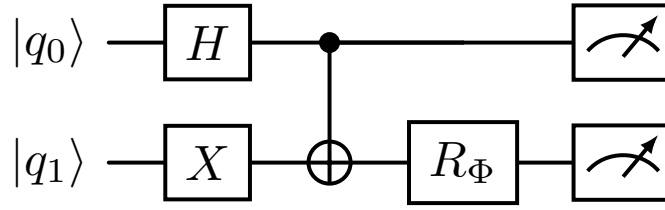


Figure 5.2: Illustration of an exemplary quantum circuit. The input state is formed by $|q_0\rangle \otimes |q_1\rangle$, and the elementary gates H , X , CNOT, and R_Φ are applied sequentially from the left to the right. The output state can be obtained by measuring both qubits at the end of the circuit, which is indicated by the two “meter” symbols.

Quantum circuits are read from the left to the right. Thus, we first consider the input state or initial state, which is given as a tensor product of the input qubits, i.e., in this case $|q_0\rangle \otimes |q_1\rangle$. When not explicitly stated otherwise, the ground state $|0\dots 0\rangle$ is chosen as the input state, since in practice it can be created most reliably [62]. It is also worth mentioning that the ground state can be transformed into an arbitrary new initial state using the appropriate gate sequence [63]. Thus, the quantum circuit can always safely be initialized to $|0\dots 0\rangle$. Proceeding further in the circuit, the leftmost gate acts first on the input state and performs a linear operation, which is described by a matrix-vector multiplication. Then, the remaining gates of the circuit are applied from left to right, and the output state can be measured at the end of the circuit. To this end, each qubit is measured, which yields either 0 or 1, and multiple measurements must be conducted to find the probabilities for the possible output states. To get more familiar with quantum circuits, we will calculate the action of the quantum circuit in Figure 5.2 on the input state $|00\rangle$ and indicate the applied gates with an arrow:

$$\begin{aligned}
 |0\rangle \otimes |0\rangle &\xrightarrow{H,X} H|0\rangle \otimes X|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\
 &\xrightarrow{R_\Phi} \frac{1}{\sqrt{2}}(|0\rangle \otimes R_\Phi|1\rangle + |1\rangle \otimes R_\Phi|0\rangle) = \frac{1}{\sqrt{2}}(|0\rangle \otimes e^{i\Phi}|1\rangle + |1\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}}(e^{i\Phi}|01\rangle + |10\rangle).
 \end{aligned} \tag{5.11}$$

The input qubits $|q_0\rangle$ and $|q_1\rangle$ are both initialized to $|0\rangle$, which yields the input state $|0\rangle \otimes |0\rangle$. Then, a Hadamard gate H is applied to $|q_0\rangle$, and a NOT gate X to $|q_1\rangle$. The action of the Hadamard gate can be easily calculated with its matrix representation and yields $H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$,

whereas the NOT gate just flips the qubit and thus yields $X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$. Next, a CNOT gate is applied, which flips the second qubit if the first qubit is in $|1\rangle$, since the first qubit is the control qubit and the second the target qubit. Finally, the rotation gate R_Φ acts on the second qubit yielding $R_\Phi|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Phi} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$ and $R_\Phi|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Phi} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e^{i\Phi}|1\rangle$.

The output state of the circuit can be obtained by repeated measurements of both qubits, which is indicated by the two “meter” symbols at the end of the circuit. Each measurement will either yield 0 for the first and 1 for the second qubit with a probability of $|\frac{1}{\sqrt{2}}e^{i\Phi}|^2 = \frac{1}{2}$, or 1 for the first and 0 for the second qubit with a probability of $|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$. This implies that sufficient statistics will show both outcomes with the same frequency. We further note that the complex phase $e^{i\Phi}$ does not contribute to the probabilities, since its absolute square is 1. Obviously, such a simple measurement is unable to determine the whole quantum state, since its phase cannot be captured. However, ways to also extract the phase by measuring repeatedly in different bases exist. This is called quantum tomography and will be introduced in Section 5.8.

It remains to be noted that more advanced quantum circuits are usually not illustrated solely with elementary gates. Mostly, more complex matrices are used as gates, which are just illustrated by appending them to the correct number of qubits, as will become clear in Section 6. However, it is understood that every unitary matrix that is appended to a quantum circuit must ultimately be decomposed into elementary gates, since quantum computers only use a specific set of elementary gates. Examples for such sets are given in Reference [64] for different IBM quantum processors.

5.5 Decomposing Arbitrary Gates into Two-Level Matrices

In this section, we introduce an algorithm to decompose an arbitrary unitary gate into two-level unitary matrices. A two-level matrix is a unitary matrix that only acts non-trivially on two or fewer vector components. Thus, it has at most four symmetrically arranged entries that are distinct from 0 and 1. To outline the algorithm, we consider a 3×3 unitary matrix U of the form:

$$U = \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & j \end{pmatrix}. \quad (5.12)$$

We show that it is possible to find unitary two-level matrices $U_1, \dots, U_3$ that satisfy $U_3 U_2 U_1 U = I$. Using the unitarity, this can be inverted to express U as:

$$U = U_1^\dagger U_2^\dagger U_3^\dagger. \quad (5.13)$$

It is easy to verify that the adjoint of a two-level matrix is also a two-level matrix. In the first step of the algorithm, we aim to transform the entries under the main diagonal, i.e., b , c , and f , to 0. Thus, if we already have $b = 0$, we set $U_1 = I$, where I is the identity. However, if $b \neq 0$, we set:

$$U_1 = \begin{pmatrix} \frac{a^*}{\sqrt{|a|^2+|b|^2}} & \frac{b^*}{\sqrt{|a|^2+|b|^2}} & 0 \\ \frac{b}{\sqrt{|a|^2+|b|^2}} & \frac{-a}{\sqrt{|a|^2+|b|^2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (5.14)$$

This ensures that the result of the multiplication $U_1 U$ contains $b = 0$, since:

$$U_1 U = \begin{pmatrix} a' & d' & g' \\ 0 & e' & h' \\ c' & f' & j' \end{pmatrix}. \quad (5.15)$$

Note that U_1 was indeed chosen as a unitary two-level matrix. To obtain $c' = 0$, we proceed in a similar manner. If $c' = 0$ is already satisfied, we set:

$$U_2 = \begin{pmatrix} a'^* & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (5.16)$$

where the left upper element was set to a'^* , and the unitarity enforces $a' a'^* = 1$. If $c \neq 0$, we set:

$$U_2 = \begin{pmatrix} \frac{a'^*}{\sqrt{|a'|^2 + |c'|^2}} & 0 & \frac{c'^*}{\sqrt{|a'|^2 + |c'|^2}} \\ 0 & 1 & 0 \\ \frac{c'}{\sqrt{|a'|^2 + |c'|^2}} & 0 & \frac{-a'}{\sqrt{|a'|^2 + |c'|^2}} \end{pmatrix}, \quad (5.17)$$

which yields:

$$U_2 U_1 U = \begin{pmatrix} 1 & d'' & g'' \\ 0 & e'' & h'' \\ 0 & f'' & j'' \end{pmatrix}. \quad (5.18)$$

We can now use that $d'' = g'' = 0$, which is enforced by the unitarity of $U_2 U_1 U$, since U , U_1 , and U_2 are all unitary. This ensures that the first row of $U_2 U_1 U$ has a norm of 1. Thus, we finally find:

$$U_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e''^* & f''^* \\ 0 & h''^* & j''^* \end{pmatrix}, \quad (5.19)$$

where, again, the unitarity of the two-level matrix U_3 enforces that $e'' e''^* + f'' f''^* = 1$, $e''^* h'' + f''^* j'' = 0$, $h'' h''^* + j'' j''^* = 1$, and $h''^* e'' + j''^* f'' = 0$, which then leads to $U_3 U_2 U_1 U = I$. By solving for U and using the unitarity once again, we obtain:

$$U = U_1^\dagger U_2^\dagger U_3^\dagger, \quad (5.20)$$

where $U_1^\dagger$, $U_2^\dagger$, and $U_3^\dagger$ are unitary two-level matrices, thus completing the algorithm. It is not a surprise that this algorithm can be extended for the decomposition of arbitrary unitary matrices of dimension d . Thus, for a $d \times d$ unitary matrix U , unitary two-level matrices V_k can be found such that:

$$U = V_1 \dots V_k, \quad (5.21)$$

where $k \leq (d-1) + (d-2) + \dots + 1 = d(d-1)/2$.

5.6 Decomposing Arbitrary Gates into Single-Qubit and CNOT Gates

The previous section illustrated a procedure to decompose any unitary matrix into two-level matrices. In this section, it is shown that each two-level matrix can be realized using single-qubit gates and CNOTs, thereby expressing any unitary matrix in terms of elementary gates. To this end, suppose we wish to implement the two-level matrix:

$$U = \begin{pmatrix} a & 0 & 0 & 0 & 0 & 0 & 0 & c \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ b & 0 & 0 & 0 & 0 & 0 & 0 & d \end{pmatrix}, \quad (5.22)$$

where we define $\tilde{U} \equiv \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ as a single-qubit matrix containing the non-trivial elements of U . We aim to implement U on a quantum circuit using only the single-qubit gate $\tilde{U}$ and CNOTs. We note that U acts only non-trivially on the basis states $|000\rangle$ and $|111\rangle$. Thus, the circuit should apply $\tilde{U}$ only to these states, and act as identity on the other basis states. This can be achieved if the circuit first transforms $|000\rangle$ such that it differs in only one bit from $|111\rangle$. Then, $\tilde{U}$ must be applied to the bit that still differs to ensure that the circuit does not have the same effect on both basis states $|000\rangle$ and $|111\rangle$. The corresponding circuit is shown in Figure 5.3. For a more in-depth explanation, I refer to chapter 4.5 of Nielsen and Chuang [7].

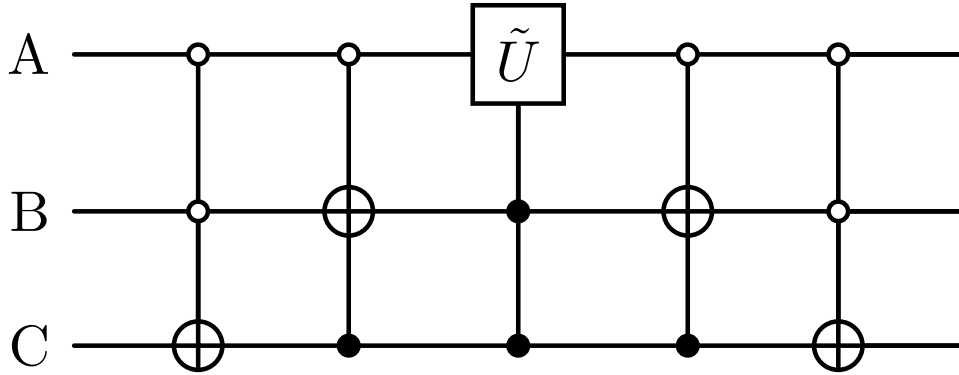


Figure 5.3: Quantum circuit for the implementation of U . It consists of a single-qubit gate $\tilde{U}$ and CNOTs, and thus shows an example of how to decompose an arbitrary unitary matrix into a set of elementary gates. The CNOTs are arranged such that $\tilde{U}$ acts differently at the input states $|000\rangle$ and $|111\rangle$, and does not act on any other basis states.

In Figure 5.3, we first use two CNOTs with multiple controls to transform $|000\rangle$ to $|011\rangle$, whereas $|111\rangle$ is not affected. If the control is depicted by an open circle, the target qubit is negated if the control qubit is 0, and if it is depicted by a closed circle, the target qubit is negated if the control qubit is 1. The single qubit gate $\tilde{U}$ is controlled by the qubits B and C and hence only acts on qubit A if B and C are 1. Thus,

$\tilde{U}$ is only applied if the input states are $|000\rangle$ or $|111\rangle$, since no other states exist where the last two bits are transformed to 1. Furthermore, $\tilde{U}$ is applied to qubit A , which is $|0\rangle$ if the input state is $|000\rangle$, and $|1\rangle$ if the input state is $|111\rangle$, ensuring that $\tilde{U}$ acts differently on these two states, while leaving the others unaffected. In the next step, the transformation applied by the first two CNOTs must be reversed, which is achieved by the last two CNOTs. In addition, this guarantees that the circuit acts as identity on the other basis states, since any transformation by the first two CNOTs is simply reversed.

To get a “feeling” for quantum circuits, we calculate the action of the circuit on the basis states $|000\rangle$ and $|111\rangle$, since only these two are affected by the circuit. Starting with $|000\rangle$, and indicating each step in the circuit with an arrow and the corresponding gate yields:

$$|000\rangle \xrightarrow{\text{CNOT}} |001\rangle \xrightarrow{\text{CNOT}} |011\rangle \xrightarrow{\tilde{U}} \tilde{U}|011\rangle = a|011\rangle + b|111\rangle \xrightarrow{\text{CNOT}} a|001\rangle + b|111\rangle \xrightarrow{\text{CNOT}} a|000\rangle + b|111\rangle, \quad (5.23)$$

where we used that $\tilde{U}|011\rangle = \tilde{U}|0\rangle \otimes |1\rangle \otimes |1\rangle = (a, b)^T \otimes |1\rangle \otimes |1\rangle = (a|0\rangle + b|1\rangle) \otimes |1\rangle \otimes |1\rangle = a|011\rangle + b|111\rangle$. Similarly, for $|111\rangle$ we get:

$$|111\rangle \xrightarrow{\text{CNOT}} |111\rangle \xrightarrow{\text{CNOT}} |111\rangle \xrightarrow{\tilde{U}} \tilde{U}|111\rangle = c|011\rangle + d|111\rangle \xrightarrow{\text{CNOT}} c|001\rangle + d|111\rangle \xrightarrow{\text{CNOT}} c|000\rangle + d|111\rangle, \quad (5.24)$$

where we used that $\tilde{U}|111\rangle = \tilde{U}|1\rangle \otimes |1\rangle \otimes |1\rangle = (c, d)^T \otimes |1\rangle \otimes |1\rangle = (c|0\rangle + d|1\rangle) \otimes |1\rangle \otimes |1\rangle = c|011\rangle + d|111\rangle$. Comparing the results obtained with the circuit and with the matrix multiplications $\tilde{U}|000\rangle$ and $\tilde{U}|111\rangle$, it is obvious that both yield the same result:

$$U|000\rangle = a|000\rangle + b|111\rangle, \quad U|111\rangle = c|000\rangle + d|111\rangle, \quad (5.25)$$

which shows that the circuit indeed implements U . Thus, we have presented an algorithm that implements an arbitrary unitary matrix U using a set of elementary gates, containing a single-qubit gate and CNOTs.

5.7 Decomposing Diagonal Unitary Matrices

In the last section, we discussed how to decompose an arbitrary unitary matrix into elementary gates. However, later in this work, we will encounter diagonal unitary matrices, which can be implemented more efficiently using so-called Walsh operators. To this end, we follow Reference [16] and [65] to introduce the corresponding algorithm.

In the following, we aim to decompose an arbitrary diagonal unitary matrix U_Σ of dimensions $2^n \times 2^n$ into elementary gates. To this end, we first express the integer $j, k \in \{0, 2^n - 1\}$ in their binary and dyadic expansion:

$$j = \sum_{i=1}^n j_i 2^{i-1}, \quad (5.26)$$

$$k = \sum_{i=1}^n k_i 2^{n-i}. \quad (5.27)$$

Next, we introduce the discrete Walsh functions and the vector components f_k as:

$$W_{jk} = (-1)^{\sum_{i=1}^n j_i k_i}, \quad f_k = (\ln U_\Sigma)_k, \quad (5.28)$$

where W_{jk} is a matrix, and the vector components of $\vec{f}$ contain the logarithm of the diagonal elements of U_Σ . The so-called Walsh coefficient a_j is obtained by a Walsh-Fourier transform of f_k :

$$a_j = \frac{1}{2^n} \sum_{k=1}^{2^n} W_{jk} f_k. \quad (5.29)$$

We can then write the unitary diagonal U_Σ as:

$$U_\Sigma = \prod_{j=1}^{2^n} e^{ia_j Q_j}. \quad (5.30)$$

The operators Q_j are called Walsh operators, and are given as:

$$Q_j = (Z)^{j_1} \otimes (Z)^{j_2} \otimes \dots \otimes (Z)^{j_n}, \quad (5.31)$$

where $j_1, \dots, j_n \in \{0, 1\}$, since they arise from the binary expansion of j . It is important to note that the Q_j are formed by a tensor product of either Z or I matrices, which are both diagonal. Hence, also Q_j , and of course U_Σ , are diagonal. We have thus limited the implementation of U_Σ to the implementation of Z and I matrices with a specific Walsh coefficient a_j .

Each factor comprising U_Σ is of the form $U_{\Sigma_j} = e^{-i \frac{\theta_j}{2} \otimes_i (Z_i)^{j_i}}$, where we defined $\theta_j = -2a_j$ and inserted the expansion of Q_j . Thus, the circuit for U_{Σ_j} is identical to that for Q_j if the Z gate is replaced with a rotation around the z -axis by an angle of $\theta_j = -2a_j$. The gate describing the rotation is given as $R_z(\theta) = e^{-i \frac{\theta}{2} Z}$. The quantum circuit for U_Σ is constructed by successively implementing the U_{Σ_j} .

To this end, we consider the binary expansion of j and place a rotation gate $R_z(-2a_j)$ on the most significant non-zero bit of j . Then, CNOTs are applied to the left and right of the rotation gate. The targets are on the same bit as the rotation gate, and the controls are on the other non-zero bits in the expansion of j . If the most significant non-zero bit is the only non-zero bit, no CNOTs are applied.

To further illustrate the algorithm, Figure 5.4 shows the implementation of the unitary diagonal matrix U_{Σ_7} , where $j = 7$. Expressing 7 in its binary expansion yields a bit string of $(j_2, j_1, j_0) = (111)$. The rightmost bit corresponds to $|q_0\rangle$, the middle bit to $|q_1\rangle$, and the leftmost bit to $|q_2\rangle$. The rotation gate and the CNOT gates must be applied accordingly to correctly implement U_{Σ_7} .

First, the most significant bit in the bit string is located. In the case of $(j_2, j_1, j_0) = (111)$, the most significant bit is the leftmost bit, which corresponds to j_2 . Thus, we append a rotation gate on $|q_2\rangle$. The angle of the rotation is given by the Walsh coefficient a_7 . Next, the CNOT gates are applied, which are targeted at the most significant non-zero bit. The control is applied to $|q_0\rangle$ and $|q_1\rangle$, since the other non-zero bits correspond to j_0 and j_1 . To implement U_Σ , this procedure must be repeated until all factors of U_{Σ_7} are

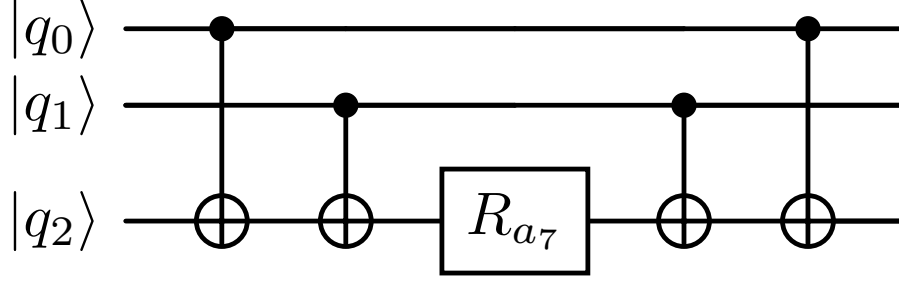


Figure 5.4: Quantum circuit for the decomposition of the unitary diagonal matrix U_{Σ_7} . First, the corresponding bit string $(j_2, j_1, j_0) = (111)$ is constructed. Then, the rotation gate and the target of the CNOTs are applied to the most significant non-zero bit. The control for the CNOTs is on the remaining non-zero bits.

appended to the circuit. The decomposition is not unique, and equivalent circuits for the implementation exist. For further detail, I refer to Reference [65].

The introduced procedure can be used to efficiently implement diagonal unitary matrices. In Sections 6, we will introduce algorithms to calculate the time evolution of the system density matrix of an open quantum system. The largest matrices in these algorithms are diagonal unitary matrices. Thus, it will prove useful to implement them efficiently.

5.8 Quantum Tomography

Quantum tomography provides a method to experimentally reconstruct quantum states. Suppose we have a quantum circuit with an output state $|\psi\rangle$, which we want to determine experimentally. To this end, we need to measure many copies of $|\psi\rangle$ to get sufficient statistics. Unfortunately, a measurement in a single basis only yields the probabilities for each component and does not capture its phase. Hence, we must perform measurements in multiple bases [7][66]. We will start our investigation for the case of a single qubit, which is then generalized to multiple qubits. This will be followed by two different methods for quantum tomography, the linear least-squares inversion tomography and the Gaussian linear least-squares tomography.

5.8.1 General Procedure for Single Qubits

The density matrix of a single qubit can be written as a linear combination of the Pauli matrices and the identity matrix, since they form a basis for the space of Hermitian 2×2 matrices [67]. We will denote the Pauli matrices as $\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\} \equiv \{I, \sigma_x, \sigma_y, \sigma_z\} \equiv \{I, X, Y, Z\}$. We then obtain:

$$\rho = \frac{1}{2} (I + \vec{r} \cdot \vec{\sigma}) = \frac{1}{2} \left(I + \sum_{j=x,y,z} r_j \sigma_j \right), \quad (5.32)$$

where the r_j are the coefficients in the linear combination. Since they indicate the position of the state on

the Bloch sphere, the vector $\vec{r}$ is also called the Bloch vector. To fully determine ρ , it is thus sufficient to determine $\vec{r}$. An expression for $\vec{r}$ can be obtained by employing the trace:

$$\begin{aligned}\langle \sigma_i \rangle &= \text{Tr}[\rho \sigma_i] = \text{Tr} \left[\frac{1}{2} \left(I + \sum_j r_j \sigma_j \right) \sigma_i \right] = \frac{1}{2} \text{Tr} \left[I \sigma_i + \sum_j r_j \sigma_j \sigma_i \right] \\ &= \frac{1}{2} \left[\text{Tr}[\sigma_i] + \sum_j r_j \text{Tr}[\sigma_j \sigma_i] \right] = \frac{1}{2} \left[0 + \sum_j r_j \cdot 2\delta_{ij} \right] = \frac{1}{2} \cdot 2r_i = r_i,\end{aligned}\tag{5.33}$$

where we used the properties of the Pauli matrices, $\text{Tr}[\sigma_i] = 0$, and $\text{Tr}[\sigma_j \sigma_i] = 2\delta_{ij}$. This then leads to:

$$\rho = \frac{1}{2} \left(I + \sum_{j=x,y,z} \langle \sigma_j \rangle \sigma_j \right),\tag{5.34}$$

where the Pauli matrices can be considered as basis operators. To determine the expectation values, we introduce the measurement operators E_k , which are enumerated by the subscript k . They correspond to projectors onto the eigenstates of the Pauli operators. For X , Y , and Z , these projectors are [68]:

$$P_{\pm}^{(X)} = |\pm\rangle\langle\pm|, \quad P_{\pm}^{(Y)} = |\pm i\rangle\langle\pm i|, \quad P_0^{(Z)} = |0\rangle\langle 0|, \quad P_1^{(Z)} = |1\rangle\langle 1|,\tag{5.35}$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$, and $|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$. The expectation value can then be calculated as the product of the eigenvalues and the probability of finding the state in the corresponding eigenstate after the measurement. Using the spectral decomposition, for X we thus have:

$$\begin{aligned}\langle X \rangle &= \text{Tr}[\rho X] = \text{Tr}[\rho \lambda_+^X |+\rangle\langle +| + \rho \lambda_-^X |-\rangle\langle -|] = \lambda_+^X \text{Tr}[\rho |+\rangle\langle +|] + \lambda_-^X \text{Tr}[\rho |-\rangle\langle -|] \\ &= \lambda_+^X \langle + | \rho | + \rangle + \lambda_-^X \langle - | \rho | - \rangle = \lambda_+^X p_+ + \lambda_-^X p_-, \end{aligned}\tag{5.36}$$

where p_+ and p_- are the probabilities that the state ρ collapses into the eigenstates $|+\rangle$ and $|-\rangle$ after the measurement. Since the measurement results correspond to the eigenvalues of X , the eigenstate into which the system collapses can always be determined. If the measurement is repeated on many copies of ρ , the absolute frequency for $|+\rangle$ and $|-\rangle$, and thus p_+ and p_- , can be determined.

To realize the measurement on a quantum circuit, we must consider that, usually, the Z basis is used for measurements. Suppose we have a quantum circuit that produces the output state $|\psi\rangle$. Then, the components of $|\psi\rangle$ are measured as $p_0 = |\langle 0 | \psi \rangle|^2$ and $p_1 = |\langle 1 | \psi \rangle|^2$. To obtain $p_{\pm} = |\langle \pm | \psi \rangle|^2$, a Hadamard gate must be appended at the end of the circuit, which transforms $|\langle 0 | H | \psi \rangle|^2 = |\langle 0 | H^\dagger | \psi \rangle|^2 = |\langle + | \psi \rangle|^2 = p_+$.

The expectation values $\langle Y \rangle$ and $\langle Z \rangle$ can be determined similarly. Inserting them into Equation 5.34 allows us to reconstruct the density matrix ρ , provided sufficient measurements are conducted. It also shows that the $\langle \sigma_i \rangle$ can be interpreted as coordinates on the Bloch sphere, where the basis is given by the σ_i .

5.8.2 General Procedure for Multiple Qubits

The formalism can be extended from a single qubit to multiple qubits. To this end, we write the n -qubit quantum state ρ as a linear combination of tensor products of n Pauli matrices [69]:

$$\rho = \frac{1}{2^n} \sum_{v_1, \dots, v_n \in \{0, x, y, z\}} r_{v_1 \dots v_n} \sigma_{v_1} \otimes \dots \otimes \sigma_{v_n}, \quad (5.37)$$

where the coefficients are given by expectation values $r_{v_1 \dots v_n} = \text{Tr}(\rho(\sigma_{v_1} \otimes \dots \otimes \sigma_{v_n})) = \langle \sigma_{v_1} \otimes \dots \otimes \sigma_{v_n} \rangle$, and the basis operators M_j are defined as $M_{v_1 \dots v_n} = \sigma_{v_1} \otimes \dots \otimes \sigma_{v_n}$ [67]. In order to measure a coefficient $\langle \sigma_{v_1} \otimes \dots \otimes \sigma_{v_n} \rangle$, a projective measurement in the eigenbasis of the Pauli operators must be performed. The projectors for each qubit are given in Equation 5.35, and can be written in a more compact form as $P_{\pm}^{(v)} = \frac{1}{2}(I \pm \sigma_v)$, where $v \in \{0, x, y, z\}$. A multi-qubit measurement operator can then be constructed as a tensor product of the single-qubit projectors:

$$E_{s_1, \dots, s_n}^{(v_1, \dots, v_n)} = P_{s_1}^{(v_1)} \otimes \dots \otimes P_{s_n}^{(v_n)}, \quad (5.38)$$

where $s_1, \dots, s_n \in \{+, -\}$. As in Equation 5.36, the measurement probability can be calculated from the measurement operators:

$$p_{s_1, \dots, s_n}^{(v_1, \dots, v_n)} = \text{Tr} \left[E_{s_1, \dots, s_n}^{(v_1, \dots, v_n)} \rho \right] = \langle P_{s_1}^{(v_1)} \otimes \dots \otimes P_{s_n}^{(v_n)} \rangle. \quad (5.39)$$

The corresponding coefficient is computed as:

$$\langle \sigma_{v_1} \otimes \dots \otimes \sigma_{v_n} \rangle = \sum_{s_1, \dots, s_n \in \{+, -\}} (s_1 1 \cdot \dots \cdot s_n 1) \cdot p_{s_1, \dots, s_n}^{(v_1, \dots, v_n)}, \quad (5.40)$$

where -1 and $+1$ are the eigenvalues of the Pauli operators [70]. This procedure must be repeated for all coefficients $\langle \sigma_{v_1} \otimes \dots \otimes \sigma_{v_n} \rangle$, to obtain ρ by using Equation 5.37.

For further illustration, we compute the coefficient $\langle X \otimes Z \rangle$ for a two-qubit system. Then, the projectors on the eigenstates of X and Z are given as:

$$P_{\pm}^{(X)} = \frac{1}{2}(I \pm X), \quad P_{\pm}^{(Z)} = \frac{1}{2}(I \pm Z), \quad (5.41)$$

where $P_{\pm}^{(X)} = |\pm\rangle \langle \pm|$, $P_{+}^{(Z)} = |0\rangle \langle 0|$, and $P_{-}^{(Z)} = |1\rangle \langle 1|$. The four measurement operators for the composite system can be formed by a tensor product of the single-qubit projectors:

$$E_{s_1, s_2}^{(X, Z)} = P_{s_1}^{(X)} \otimes P_{s_2}^{(Z)}, \quad (5.42)$$

where $s_1, s_2 \in \{+, -\}$. This yields the four probabilities:

$$p_{s_1 s_2} = \text{Tr}[\rho E_{s_1, s_2}^{(X, Z)}]. \quad (5.43)$$

If $s_1 = +$ and $s_2 = -$, we obtain $p_{+-} = \text{Tr}[\rho E_{+, -}^{(X, Z)}] = \text{Tr}[\rho (P_{+}^{(X)} \otimes P_{-}^{(Z)})] = \text{Tr}[\rho \cdot (|+\rangle \langle +| \otimes |1\rangle \langle 1|)]$, where we used $P_{+, -} = |+\rangle \langle +| \otimes |1\rangle \langle 1|$. Repeating the calculation for p_{++} , p_{-+} , and p_{--} , we can finally find:

$$\begin{aligned} \langle X \otimes Z \rangle &= (+1)(+1) \cdot p_{++} + (+1)(-1) \cdot p_{+-} + (-1)(+1) \cdot p_{-+} + (-1)(-1) \cdot p_{--} \\ &= p_{++} - p_{+-} - p_{-+} + p_{--}. \end{aligned} \quad (5.44)$$

Since we consider a two-qubit system, the output state of a quantum circuit will have four components. In the corresponding measurement, a Hadamard gate must be used on the first qubit to transform it into the X basis. Then, both qubits can be measured, which yields either 0 for both qubits, which corresponds to p_{++} ; or 0 for the first and 1 for the second qubit, which corresponds to p_{+-} ; or 1 for the first and 0 for the second qubit, which corresponds to p_{-+} ; or 1 for both qubits, which corresponds to p_{--} . If sufficient states are prepared and measurements conducted, the four probabilities can be determined, which, in turn, yields $\langle X \otimes Z \rangle$.

5.8.3 Linear Inversion

We now introduce the simplest version of quantum tomography, which is called linear inversion. To this end, we first define the Hilbert-Schmidt inner product of two linear operators as $A \cdot B = \text{Tr}[A^\dagger B] = \vec{A}^\dagger \vec{B}$. It is defined via the trace, since this definition guarantees linearity, positive definiteness, and invariance under unitary transformations. Next, we introduce the general, but fixed, measurement operators E_k , which can be written in a vectorized form and represent the rows of a matrix A [71]:

$$A = \begin{pmatrix} \vec{E}_1^\dagger \\ \vec{E}_2^\dagger \\ \vec{E}_3^\dagger \\ \vdots \end{pmatrix}. \quad (5.45)$$

The number of rows of A equals the number of measurement operators E_k , and the number of columns depends on their dimension. This convenient definition allows us to write [70]:

$$A\vec{\rho} = \begin{pmatrix} \vec{E}_1^\dagger \vec{\rho} \\ \vec{E}_2^\dagger \vec{\rho} \\ \vec{E}_3^\dagger \vec{\rho} \\ \vdots \end{pmatrix} = \begin{pmatrix} \text{Tr}[E_1^\dagger \rho] \\ \text{Tr}[E_2^\dagger \rho] \\ \text{Tr}[E_3^\dagger \rho] \\ \vdots \end{pmatrix} = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ \vdots \end{pmatrix} = \vec{p}, \quad (5.46)$$

where we used the definition of the Hilbert-Schmidt inner product in the second equality, and Equation 5.39 in the third equality. To obtain the vectorized density matrix $\vec{\rho}$, the matrix A must be inverted. However, A is usually not a square matrix, which means that a direct matrix inversion does not exist, and the so-called pseudoinverse must be employed. To this end, we multiply Equation 5.46 with $A^\dagger$ from the left and solve for $\vec{\rho}$:

$$\vec{\rho} = (A^\dagger A)^{-1} A^\dagger \vec{p}. \quad (5.47)$$

The matrix product $(A^\dagger A)^{-1}$ is Hermitian and positive semidefinite. It is invertible if the columns of A are linearly independent, which is usually fulfilled, since the operators E_k span the entire operator space. However, the pseudoinverse results from minimizing the least squares error, and hence does not deliver an exact solution for $\vec{\rho}$. Its origin can be understood as follows. Since the equation $A\vec{\rho} = \vec{p}$ has no exact solution, we aim to find $\vec{\rho}$ by minimizing the squared Euclidean norm $\|A\vec{\rho} - \vec{p}\|_2^2$ [72]. Thus, we need to find the minimum of $f(\vec{\rho}) = \|A\vec{\rho} - \vec{p}\|_2^2$, which yields $\nabla_{\vec{\rho}} f(\vec{\rho}) = 2A^\dagger(A\vec{\rho} - \vec{p}) = 0$, and hence $\vec{\rho} = (A^\dagger A)^{-1} A^\dagger \vec{p}$.

The pseudoinverse usually yields a good approximation for the vectorized density matrix $\vec{\rho}$, but does not guarantee physical results. Unfortunately, the calculated density matrix $\vec{\rho}$ may not be positive semidefinite, and its trace can differ from 1, since the only requirement is to minimize the error. To ensure physicality, additional constraints on the trace or positive definiteness must be enforced.

5.8.4 Gaussian Linear Least-Squares Tomography

In this section, we follow the source code of the constrained Gaussian linear least-squares tomography fitter from the Python library Qiskit [73][74]. We will only briefly and conceptually introduce the method, since a detailed description is beyond the scope of this work.

In general, the method is similar to linear inversion, and hence approximates the density matrix by minimizing the squared Euclidean norm $\|W(A\vec{\rho} - \vec{p})\|_2^2$, where the variable W is a Gaussian weight. These weights can be calculated using:

$$W_i(s) = \frac{1}{\sqrt{\text{Var}[p_i(s)]}}, \quad (5.48)$$

where $p_i(s)$ is the expected probability for the result $s \in \{0, 1\}$ and tomography basis index i . Probabilities with small variances indicate small uncertainties and are therefore given greater weights. The variances are computed by:

$$\text{Var}[p_i(s)] = \frac{p_i(s)(1 - p_i(s))}{\bar{\alpha}_i + N_i + 1}, \quad (5.49)$$

where

$$p_i(s) = \frac{\alpha_i(s) + f_i(s)}{\bar{\alpha}_i + N_i} \quad (5.50)$$

and $\bar{\alpha}_i = \alpha_i(0) + \alpha_i(1)$. The number of measurements is denoted by N_i , and the absolute frequency of outcome s by $f_i(s)$. The $\alpha(i)$ are a priori probabilities, and correspond to a Dirichlet distribution. This Dirichlet prior contains the assumed prior knowledge about the measurement probabilities, which can stem from previous experiments or physical reasoning. For details on the Dirichlet distribution, I refer to Reference [75].

The Constrained Gaussian linear least-squares tomography fitter is designed to deliver more accurate results than a plain linear inversion tomography fitter. However, this accuracy comes at the cost of computation time, making the fitter unsuitable for systems larger than five or six qubits [76].

6 Non-Unitary Dynamics on a Quantum Computer

Non-unitary dynamics are described by non-unitary operations. However, a quantum computer itself is ideally a closed system and thus only supports unitary operations that preserve the norm of the qubits [15]. Hence, implementing non-unitary operations on a quantum circuit is challenging, since such circuits are restricted to unitary transformations. To address this limitation, two methods will be introduced. Both dilate a non-unitary matrix into a unitary one, which can be implemented on a quantum computer [16][17]. The term “dilate” refers to embedding a non-unitary matrix into a larger unitary matrix, which acts on a higher-dimensional Hilbert space. This artificial enlargement of the operations describing the system also leads to an enlarged output state vector in the quantum circuit. To undo the rescaling of the Hilbert space and obtain a state vector with the appropriate dimension, we only take a part of the output state vector and discard the rest.

We will start by introducing the implementation of Markovian non-unitary dynamics on a quantum circuit. Here, we follow publications from Seneviratne et al. [16] from 2024, and Hu et al. [17] from 2022. Then, we discuss non-Markovian dynamics as presented in Seneviratne et al. from 2024 and Dan et al. from 2025 [18]. After introducing the algorithms, their results will be reproduced, and their limitations analyzed. Based on these insights, an improved algorithm developed in the context of this work will be presented.

6.1 Markovian Dynamics with Arbitrary Step Size

We aim to construct a quantum circuit to calculate the Markovian dynamics for open quantum systems on a quantum computer. To this end, we use the Lindblad equation and the operator sum representation introduced in Sections 4.1 and 4.2. In the algorithm, we calculate the Lindblad superoperator by enlarging the Lindblad equation to a higher-dimensional Hilbert space, which allows for an arbitrarily large step size in the time evolution. The Kraus operators are obtained by a Choi decomposition, and a singular value decomposition yields the correct operators for the implementation on a quantum circuit. Here, we must dilate the Kraus operator to a unitary matrix, since a quantum circuit only supports unitary operations.

6.1.1 Evaluation of the Kraus Operators

The Lindblad equation describes the Markovian dynamics of the density matrix of an open quantum system and was derived in Section 4.1. We thus have:

$$\frac{d}{dt}\rho_S(t) = -\frac{i}{\hbar}[H, \rho_S(t)] + \sum_{\alpha} \gamma_{\alpha} \left(L_{\alpha} \rho_S(t) L_{\alpha}^{\dagger} - \frac{1}{2} \{ L_{\alpha}^{\dagger} L_{\alpha}, \rho_S(t) \} \right). \quad (6.1)$$

In general, the Hamiltonian, the system density matrix, the Lindblad operators, and the decay rates can depend on time. However, in this work, only the system density matrix is time-dependent. To solve the Lindblad equation, we make a vectorization Ansatz, where we use the identities:

$$\text{vec}(ABC) = (I_n \otimes AB) \text{vec}(C) = (C^T B^T \otimes I_k) \text{vec}(A) = (C^T \otimes A) \text{vec}(B), \quad (6.2)$$

$$\text{vec}(AB) = (I_m \otimes A) \text{vec}(B) = (B^T \otimes I_k) \text{vec}(A), \quad (6.3)$$

where A is a $k \times l$ matrix, B is an $l \times m$ matrix, and C is an $m \times n$ matrix. The dimension of the unit

matrix I is given by the corresponding subscript. Applying this to each term of the Lindblad equation, we can omit the subscript for I , since the Hamiltonian, the Lindblad operators, and the density matrix have the same dimension:

$$\text{vec}(H\rho_S) = (I \otimes H)\text{vec}(\rho_S), \quad (6.4)$$

$$\text{vec}(\rho_S H) = (H^T \otimes I)\text{vec}(\rho_S), \quad (6.5)$$

$$\text{vec}(L_\alpha \rho_S L_\alpha^\dagger) = (L_\alpha^* \otimes L_\alpha)\text{vec}(\rho_S). \quad (6.6)$$

$$\text{vec}(L_\alpha^\dagger L_\alpha \rho_S) = (I \otimes L_\alpha^\dagger L_\alpha)\text{vec}(\rho_S), \quad (6.7)$$

$$\text{vec}(\rho_S L_\alpha^\dagger L_\alpha) = (L_\alpha^T L_\alpha^* \otimes I)\text{vec}(\rho_S), \quad (6.8)$$

Notice that, in each case, the identities were chosen such that we obtain the vectorized density matrix as the last term on the right-hand side. By adding all terms, and pulling out $\text{vec}(\rho_S)$ to the right, we arrive at the vectorized Lindblad equation:

$$\begin{aligned} \frac{d}{dt} \text{vec}(\rho_S) = & \left[-\frac{i}{\hbar}(I \otimes H) - \frac{i}{\hbar}(H^T \otimes I) + \sum_{\alpha} \gamma_{\alpha} \left((L_{\alpha}^* \otimes L_{\alpha}) \right. \right. \\ & \left. \left. - \frac{1}{2}(I \otimes L_{\alpha}^{\dagger} L_{\alpha}) - \frac{1}{2}(L_{\alpha}^T L_{\alpha}^* \otimes I) \right) \right] \text{vec}(\rho_S) = \mathbb{L} \text{vec}(\rho_S). \end{aligned} \quad (6.9)$$

The differential Equation 6.9 can be solved with an exponential Ansatz, and yields an equation for the time-dependent vectorized density matrix:

$$\text{vec}(\rho_S(t)) = e^{\mathbb{L}t} \text{vec}(\rho_S(0)) = \mathcal{L}(t) \text{vec}(\rho_S(0)). \quad (6.10)$$

Here, $\mathbb{L}$ does not depend on time, since in this work only $\rho_S(t)$ is time-dependent. Hence, the time step t for the time evolution can be arbitrarily large, as is expressed in the exponent.

In Section 4.2.6, we derived that the solution of the Lindblad equation $\rho_S(t) = \mathcal{L}(t)\rho_S(0)$ is a quantum map, and in Section 4.2.7 we showed that this map is completely positive and trace-preserving, and thus has an operator sum representation. Hence, it seems reasonable that an operator sum representation can also be found in terms of the vectorized Lindblad Equation 6.10. And indeed, it is possible to find the corresponding Kraus operators using a so-called Choi decomposition. To this end, we first construct the Choi matrix using the involution operation acting on $\mathcal{L}(t)$:

$$C(t) := \sum_{i,j=1}^N (E_{i,j} \otimes I) \mathcal{L}(t) (I \otimes E_{i,j}). \quad (6.11)$$

Here, I is an $N \times N$ matrix, where N equals the number of rows or columns of H . Also, $E_{i,j}$ is an $N \times N$ matrix with a 1 in the $(ij)^{\text{th}}$ component, and 0 elsewhere. For details on the Choi decomposition, I refer to Reference [77] and [78].

Since $\mathcal{L}(t)$ is a completely positive trace-preserving map, the Choi matrix is guaranteed to be positive and Hermitian, which ensures that it has positive eigenvalues [16]. To find the Kraus operators corresponding to $\mathcal{L}(t)$, we perform an eigen decomposition of the Choi matrix:

$$C(t) = \sum_{k=1}^{N^2} \sigma_k(t) u_k(t) u_k^\dagger(t), \quad (6.12)$$

where $\sigma_k(t)$ are the eigenvalues, and $u_k(t)$ are the eigenvectors of the Choi matrix at time t . These can be used to find the vectorized Kraus operators $M_k(t)$ [16]:

$$\text{vec}(M_k(t)) = \sqrt{\sigma_k(t)} u_k(t). \quad (6.13)$$

To obtain a Kraus operator of the correct dimensions, $\text{vec}(M_k(t))$ must be reshaped into matrix form again. According to formula 4.33, the time evolution of the density matrix can then be calculated as:

$$\rho_S(t) = \sum_k M_k(t) \rho_S(0) M_k^\dagger(t), \quad (6.14)$$

where a single index k is used for the Kraus operators. The number of Kraus operators equals the number of eigenvectors of the Choi matrix. Note that, the Kraus operators can be used to evolve to an arbitrary time step t , as indicated by the time dependence of Equation 6.10. Hence, small time steps are not required to keep the error small.

6.1.2 Singular Value Decomposition and Dilation

Quantum circuits are limited to unitary operations. Since the Kraus operators in Equation 6.14 represent the non-unitary evolution of an open quantum system, they are inherently non-unitary. Therefore, it is necessary to dilate the Kraus operators to unitary operators. To this end, we first employ a singular value decomposition (SVD) [79], which exploits that every real or complex matrix can be factorized into a rotation, a rescaling, and a second rotation. Performing the singular value decomposition on the Kraus operator $M_k(t)$ yields:

$$M_k(t) = U \Sigma V^\dagger. \quad (6.15)$$

Here, U , V , and Σ have the same dimensions as $M_k(t)$ and, in principle, depend on time, since they differ for each $M_k(t)$. However, we will omit the time argument for brevity. The operators U and V correspond to the rotations, and hence are already unitary. The non-unitary part of $M_k(t)$ is encoded into Σ , which corresponds to a rescaling, and is therefore a non-unitary diagonal matrix. Thus, to implement Σ , we must dilate it into a unitary matrix U_Σ :

$$U_\Sigma = \begin{pmatrix} \Sigma_+ & 0 \\ 0 & \Sigma_- \end{pmatrix}, \quad (6.16)$$

where $\Sigma_{\pm jj} = \sigma_{jj} \pm i\sqrt{1 - \sigma_{jj}^2}$, with σ_{jj} being the diagonal elements of Σ . We construct Σ_{+jj} and Σ_{-jj} for each diagonal element σ_{jj} of Σ , and hence the resulting unitary U_Σ is twice as large as Σ , $M_k(t)$, U , and $V^\dagger$. Since $\mathcal{L}(t)$ is a completely positive, trace-preserving map, $|\sigma_{jj}|$ is always smaller than 1, preventing negative expressions in the square root.

6.1.3 Construction of the Circuit

The operators $V^\dagger$, U_Σ , and U , obtained from the singular value decomposition, can be used to construct a quantum circuit, as shown in Figure 6.1. The aim for the circuit is to apply the Kraus operator $M_k(t)$ to the input state $|\psi\rangle$, which would give $M_k(t)|\psi\rangle$ as a result. Taking the outer product with itself yields $M_k(t)|\psi\rangle\langle\psi|M_k(t) = M_k(t)\rho_S M_k(t)$, which corresponds to one summand in the operator sum representation of Equation 6.14. Thus, it is expected that the circuit shares similarities with the singular value decomposition $M_k(t) = U\Sigma V^\dagger$, as it aims to calculate $M_k(t)|\psi\rangle$ in a unitary manner.

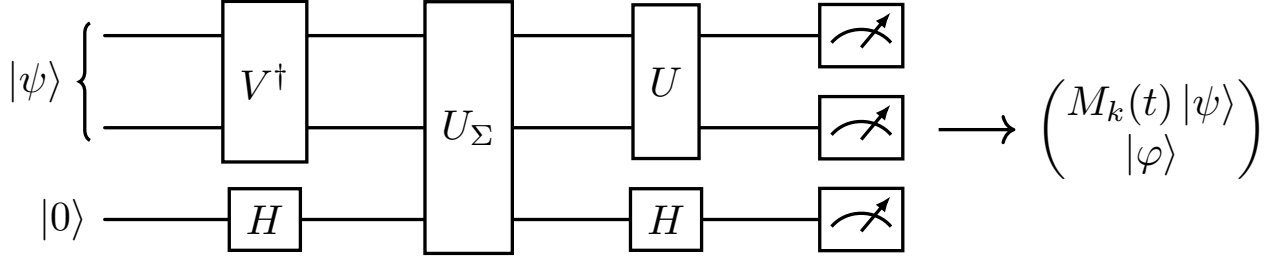


Figure 6.1: Quantum circuit for the calculation of the time dependence of the system density matrix. The operators $V^\dagger$, U , and U_Σ stem from the singular value decomposition. Applying them to the circuits allows us to perform a non-unitary evolution on a unitary quantum circuit. The result of the circuit, given an input state $|\psi\rangle$, is an output state vector whose first half yields the desired outcome $M_k(t)|\psi\rangle$. Repeating the calculation for all Kraus operators $M_k(t)$ and summing the results yields the correct time evolution of the system density matrix.

In general, the circuit consists of $n + 1$ qubits, where the Kraus operator, and thus also the system Hamiltonian and system density matrix, are of dimension $2^n \times 2^n$. The first n qubits implement the initial state $|\psi\rangle$ of the system, and thus correspond to the size of the system Hilbert space. Here, two system qubits were chosen, since the algorithm will be tested on a four-by-four Hamiltonian of the FMO complex.

The $(n + 1)^{\text{th}}$ qubit is a so-called ancilla qubit. Ancilla qubits can be viewed as “helper qubits” that are artificially introduced to make certain calculations feasible. Here, the ancilla qubit arises, since Σ is dilated to a unitary U_Σ in a higher-dimensional Hilbert space. In contrast to the two system qubits, it must always be initialized to $|0\rangle$. Besides U_Σ , we also append two Hadamard gates on the ancilla qubit. These prepare the system into a superposition of the basis state, as will be clear shortly.

To prove that the quantum circuit introduced in Figure 6.1 implements $M_k(t)|\psi\rangle$, we can calculate the action of the circuit on the input state $|\psi\rangle$. We start with the two states $|\psi\rangle$ and $|0\rangle$, which together form the initial state of the quantum circuit as:

$$|\psi\rangle \otimes |0\rangle. \quad (6.17)$$

Following the circuit, we must first apply $V^\dagger$ to $|\psi\rangle$ and the Hadamard gate H to $|0\rangle$. This leads to:

$$V^\dagger|\psi\rangle \otimes H|0\rangle = V^\dagger|\psi\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}(V^\dagger|\psi\rangle \otimes |0\rangle + V^\dagger|\psi\rangle \otimes |1\rangle), \quad (6.18)$$

where we calculated the action of the Hadamard gate on the ancilla qubit $|0\rangle$. Next, we apply U_Σ , which

acts on all 3 qubits and can be written as $U_\Sigma = \Sigma_+ \otimes |0\rangle\langle 0| + \Sigma_- \otimes |1\rangle\langle 1|$. Then:

$$\begin{aligned}
U_\Sigma \frac{1}{\sqrt{2}} \left(V^\dagger |\psi\rangle \otimes |0\rangle + V^\dagger |\psi\rangle \otimes |1\rangle \right) &= \frac{1}{\sqrt{2}} \left[U_\Sigma \left(V^\dagger |\psi\rangle \otimes |0\rangle \right) + U_\Sigma \left(V^\dagger |\psi\rangle \otimes |1\rangle \right) \right] \\
&= \frac{1}{\sqrt{2}} \left[\left(\Sigma_+ \otimes |0\rangle\langle 0| \right) \left(V^\dagger |\psi\rangle \otimes |0\rangle \right) + \left(\Sigma_- \otimes |1\rangle\langle 1| \right) \left(V^\dagger |\psi\rangle \otimes |1\rangle \right) \right] \\
&= \frac{1}{\sqrt{2}} \left[\Sigma_+ V^\dagger |\psi\rangle \otimes (|0\rangle\langle 0|0\rangle) + \Sigma_- V^\dagger |\psi\rangle \otimes (|1\rangle\langle 1|1\rangle) \right] \\
&= \frac{1}{\sqrt{2}} \left[\Sigma_+ V^\dagger |\psi\rangle \otimes |0\rangle + \Sigma_- V^\dagger |\psi\rangle \otimes |1\rangle \right],
\end{aligned} \tag{6.19}$$

where we used the orthonormality of the computational basis states $|0\rangle$ and $|1\rangle$. Finally, we must apply H to $|0\rangle$ and $|1\rangle$, and U to $|\psi\rangle$. This results in:

$$\begin{aligned}
&\frac{1}{\sqrt{2}} \left(U \Sigma_+ V^\dagger |\psi\rangle \otimes H|0\rangle + U \Sigma_- V^\dagger |\psi\rangle \otimes H|1\rangle \right) \\
&= \frac{1}{\sqrt{2}} (U \Sigma_+ V^\dagger |\psi\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) + \frac{1}{\sqrt{2}} (U \Sigma_- V^\dagger |\psi\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \\
&= \frac{1}{2} \left((U \Sigma_+ V^\dagger + U \Sigma_- V^\dagger) |\psi\rangle \otimes |0\rangle + (U \Sigma_+ V^\dagger - U \Sigma_- V^\dagger) |\psi\rangle \otimes |1\rangle \right) \\
&= \frac{1}{2} \begin{pmatrix} U(\Sigma_+ + \Sigma_-) V^\dagger |\psi\rangle \\ U(\Sigma_+ - \Sigma_-) V^\dagger |\psi\rangle \end{pmatrix} = \begin{pmatrix} M_k(t) |\psi\rangle \\ |\varphi\rangle \end{pmatrix},
\end{aligned}$$

since $\Sigma_+ + \Sigma_- = 2\Sigma$, and thus $\frac{1}{2}U(\Sigma_+ + \Sigma_-)V^\dagger = M_k(t)$ as desired. The first half of the output vector is given by $M_k(t) |\psi\rangle$ and can be used to obtain $M_k(t) |\psi\rangle \langle \psi| M_k(t)$ by taking the outer product with itself. The circuit must be repeated for each Kraus operator $M_k(t)$ at time t . Summing the results, we obtain:

$$\rho_S(t) = \sum_k M_k(t) |\psi(0)\rangle \langle \psi(0)| M_k^\dagger(t) = \sum_k M_k(t) \rho_S(0) M_k^\dagger(t). \tag{6.20}$$

The second half of the output vector is given by $|\varphi\rangle$, and can be safely discarded. It is just a residual of the expansion of Σ to a higher dimension, and thus has no physical meaning. The proof works for an arbitrary number of qubits, which must be adjusted to the dimensions of the system Hilbert space. If the system Hilbert space is of dimension $2^n \times 2^n$, then $|\psi\rangle$ has a shape of $2^n \times 1$, $V^\dagger$ and U a shape of $2^n \times 2^n$, and U_Σ a shape of $2^{n+1} \times 2^{n+1}$. The Hadamard gate H and the computational basis states $|0\rangle$ and $|1\rangle$ are, of course, always two-dimensional by definition.

Having completed all steps of the algorithm, a graphical summary is shown in Figure 6.2. In principle, with the developed method, we can calculate the time evolution of a multitude of open quantum systems. Here, the system of choice is the FMO complex. In Seneviratne et al., the time evolution was calculated for a four-site FMO complex with a four-by-four Hamiltonian. Thus, three qubits, two for the system and one ancilla qubit, are needed.

The following sections contain the results for the time evolution of the FMO complex. First, the results obtained in Seneviratne et al. are reproduced for the time evolution of the four-site FMO complex, then the algorithm is extended to seven sites, and finally, an enhanced version of the algorithm is introduced.

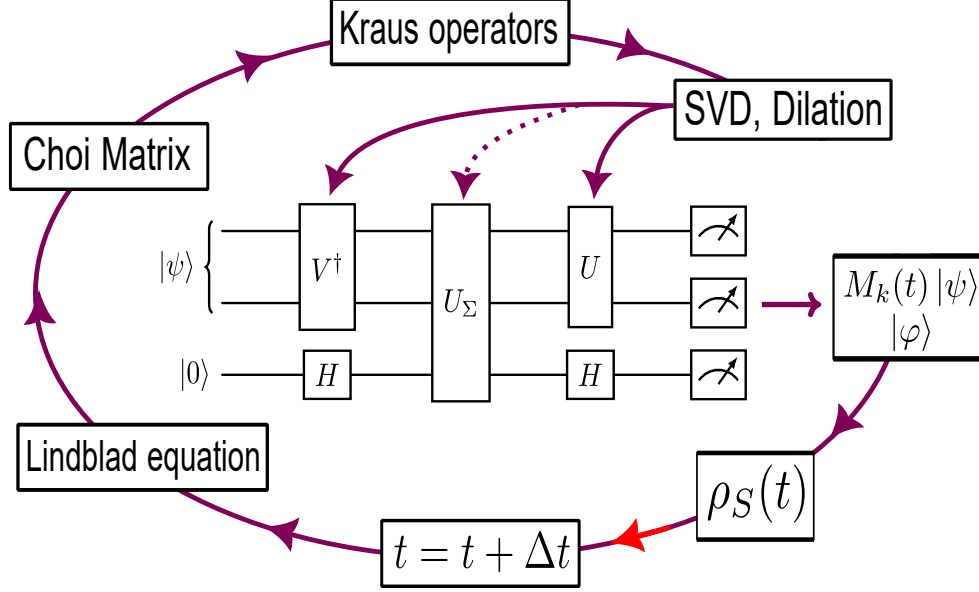


Figure 6.2: Illustration of the algorithm proposed by Seneviratne et al. to compute the dynamics of $\rho_S(t)$. For different time steps, the same procedure must be performed repeatedly, which is depicted by the purple ellipse. The algorithm starts at a specific time t and runs through all the steps until $\rho_S(t)$ is obtained. To advance in time, it updates t to the next step and repeats the entire cycle, as indicated by the red arrow.

6.1.4 Reproduction of the Results for the FMO Complex

In this section, we reproduce the dynamics of the FMO complex as presented by Seneviratne et al. [16] in 2024. As described in Section 6.1.1, the time evolution of the system density matrix is calculated using the Lindblad equation. To arrive at the Lindblad equation, we apply the Markovian approximation, which neglects the backflow of information from the bath into the system [80].

Calculating the Markovian dynamics via the Lindblad equation requires the initial system density matrix, the system Hamiltonian, the dissipation rates, and the corresponding Lindblad operators. The initial system density matrix is chosen as:

$$\rho_S(0) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (6.21)$$

It is expressed in the site basis, and thus corresponds to an initial excitation at site 1, since the corresponding diagonal element shows a probability of 1. For dynamics starting with an excitation at site 2, the 1 in Equation 6.21 would appear in the second diagonal element instead of the first. Initial excitations at other sites are obtained analogously.

The four-site Hamiltonian for the FMO complex is taken from Read et al. [81] and is given in units of inverse centimeters:

$$H = \begin{pmatrix} 12375 & -87.7 & 5.5 & -5.9 \\ -87.7 & 12495 & 30.8 & 8.2 \\ 5.5 & 30.8 & 12175 & -53.5 \\ -5.9 & 8.2 & -53.5 & 12285 \end{pmatrix}. \quad (6.22)$$

The Hamiltonian is written in the site basis, and hence the diagonal elements describe the site energy, and the off-diagonal elements the coupling between the sites. Each site is coupled to its local bath, and all baths are described with the same Drude-Lorentz spectral density, which is introduced in the Equation B.2:

$$J(\omega) = \frac{2\lambda\nu\omega}{\omega^2 + \nu^2}. \quad (6.23)$$

The reorganization energy is set to $\lambda = 35\text{cm}^{-1}$, the phonon relaxation rate to $\nu = 106.18\text{cm}^{-1}$, and for the temperature, a value of 300 K is used. For details on the spectral density, I refer to appendix B.1.

The Lindblad operators are calculated with Equation 4.10, derived in Section 4.1. Their number depends on the number of sites. For each site, one Lindblad operator is obtained for every transition frequency corresponding to the energy difference between the sites $E_n - E_m$. In a four-site system, this yields 12 non-zero combinations of $E_n - E_m$ per site, giving a total of 48 Lindblad operators.

The dissipation rates are calculated using Equation 4.27. It is assumed that the rates are identical for each site, with the same Drude-Lorentz spectral density and the same parameters for the reorganization energy, phonon relaxation rate, and temperature as above.

Using the above-described objects for the Lindblad equation, the Kraus operators are constructed as explained in Section 6.1.1. After performing the singular value decomposition, the circuit is built as illustrated in Figure 6.1. By executing the circuit for all Kraus operators at a given time t , the system density matrix $\rho_S(t)$ is obtained, as described in Equation 6.20. The results of the simulation are shown in Figure 6.3.

The vertical axis represents the population, and the horizontal axis the time in femtoseconds. The population at site m at time t corresponds to the m^{th} diagonal element of $\rho_S(t)$, and is a measure for the probability to find the excitation at site m . Since the initial excitation is on site 1, its population starts at 1, corresponding to 100% probability. As time progresses, the excitation delocalizes over the four sites until it eventually reaches a steady state. The simulation considers only the four sites of the FMO complex and excludes the reaction center. Thus, the populations do not decay to 0 and stay in the complex indefinitely, since no transition to the reaction center is included. Furthermore, the model assumes that at a given time only 1 excitation is in the system, neglecting the cases where multiple sites are excited simultaneously.

The solid lines represent classically performed calculations without involving quantum circuits. To obtain the corresponding dynamics, the Lindblad equation was used, where the colors correspond to different sites. The circles refer to the results obtained with the quantum computer simulator from the Python library Qiskit [82]. This simulator is not a real quantum computer, and therefore shows no quantum noise, but only statistical errors. The experiment was repeated 10000 times to obtain sufficient statistics.

Figure 6.3 shows that the quantum simulator results closely match the classical results, with slight deviations due to insufficient statistics. If the experiment were repeated a sufficiently large number of times, the two results would agree with arbitrary precision.

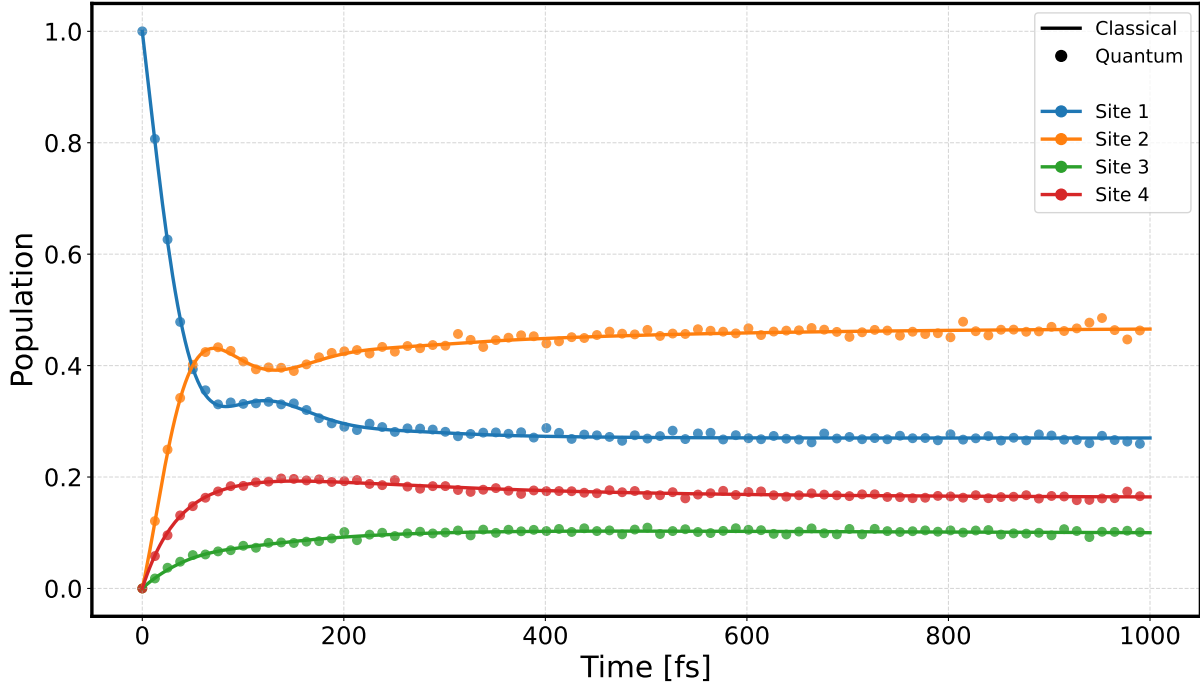


Figure 6.3: The populations are shown as a function of time for the four sites of the FMO complex, which are indicated in different colors. The initial excitation is at site 1, and the exciton delocalizes over the sites as time progresses. The solid lines denote the classical result obtained with the Lindblad equation without involving quantum circuits, whereas the circles refer to the results calculated with the quantum computer simulator from the Python library Qiskit [82].

6.1.5 Extension to a Larger Hilbert Space

In Seneviratne et al. [16], the algorithm was applied only to a four-site FMO complex, with a Hamiltonian given in Equation 6.22. However, it is possible to extend the algorithm to seven sites. Thus, the system Hamiltonian [18] and the density matrix are 7×7 matrices:

$$H = \begin{pmatrix} 310.0 & -97.9 & 5.5 & -5.8 & 6.7 & -12.1 & -10.3 \\ -97.9 & 230.0 & 30.1 & 7.3 & 2.0 & 11.5 & 4.8 \\ 5.5 & 30.1 & 0.0 & -58.8 & -1.5 & -9.6 & 4.7 \\ -5.8 & 7.3 & -58.8 & 180.0 & -64.9 & -17.4 & -64.4 \\ 6.7 & 2.0 & -1.5 & -64.9 & 405.0 & 89.0 & -6.4 \\ -12.1 & 11.5 & -9.6 & -17.4 & 89.0 & 320.0 & 31.7 \\ -10.3 & 4.8 & 4.7 & -64.4 & -6.4 & 31.7 & 270.0 \end{pmatrix}, \quad \rho_S(0) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (6.24)$$

where the initial excitation is again on site 1. Since only matrices of dimension $2^n \times 2^n$ can be applied to a quantum circuit, they must be enlarged to 8×8 matrices, which corresponds to $n = 3$ qubits for the system. The enlargement should not change the outcome of the calculation, and the enlarged matrix must still be unitary. Thus, we define an 8×8 unit matrix and assign the system Hamiltonian or system density matrix to the top left 7×7 block of the unit matrix. Since the Kraus operator has the same dimensions as

the system Hamiltonian and the system density matrix, the operators $V^\dagger$ and U are also 7×7 matrices, and thus must be embedded into an 8×8 unit matrix. The operator U_Σ , however, has a dimension of 14×14 , and hence must be embedded into a 16×16 unit matrix. As proved above, the quantum circuit can be extended to larger systems by increasing the number of system qubits, while keeping one ancilla qubit. The circuit for the seven-site FMO complex is shown in Figure 6.4:

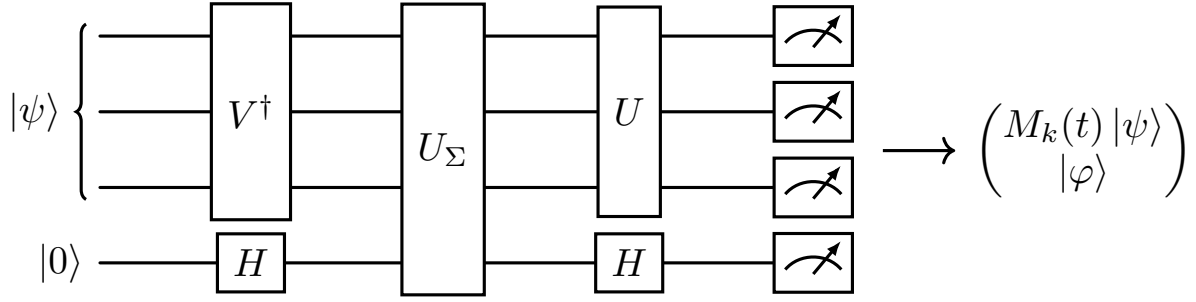


Figure 6.4: Extended quantum circuit for the calculation of the time evolution for a seven-site FMO complex. The matrices $V^\dagger$ and U have dimensions of 7×7 , and are thus embedded into an 8×8 unit matrix, which requires three system qubits. The matrix U_Σ is twice as large as $V^\dagger$ and U , and hence needs four qubits for the implementation. Apart from different dimensions, the circuit works as described above.

Using the circuit, together with the system Hamiltonian and density matrix from Equation 6.24, the time evolution of the seven-site FMO complex was calculated. The results are shown in Figure 6.5:

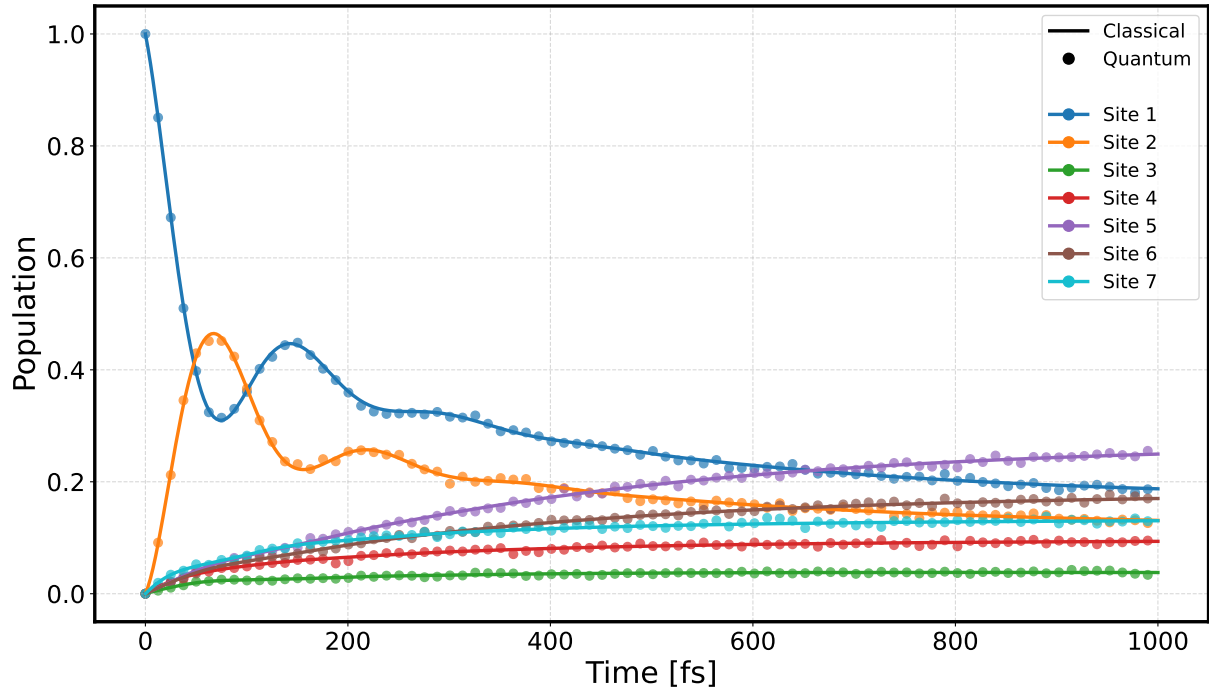


Figure 6.5: The populations are depicted as a function of time for the seven sites of the FMO complex, which are indicated in different colors. The solid lines denote the classical results without involving quantum circuits, while the circles refer to the results obtained using the quantum computer simulator from Qiskit.

The vertical axis shows the population of each site, and the horizontal axis the time from 0 to 1000 fs. The seven sites are indicated in different colors, and the initial excitation is on site 1. To conserve the total probability, the population of the other sites starts at 0. The solid lines denote the result of the classical computation using the Lindblad equation, whereas the circles refer to the results obtained with the quantum computer simulator from Qiskit. The experiment was performed 10000 times to obtain sufficient statistics. As shown in Figure 6.5, the quantum simulator results closely match those obtained with the classical computation. The slight deviations are statistical, since the simulator does not model quantum noise.

So far, we have reproduced the results of an algorithm that aims to compute the dynamics of open quantum systems on a quantum computer. However, as illustrated in Figure 6.2, a lot of work must still be performed by the classical routine. To propagate from time 0 to time t , the Kraus operators $M_k(t)$ and the corresponding circuit must be recalculated for each time step. This process includes the construction and eigendecomposition of the Choi matrix, as well as the singular value decomposition and the dilation of Σ to a unitary U_Σ . The classical processing concludes only after these steps, and the quantum computer takes over. This implies that the majority of the computational effort is carried out using classical methods, and only when the Kraus operators are calculated and the problem is essentially solved, the quantum circuits are employed. Consequently, this algorithm, proposed by Seneviratne et al., is only a proof of concept, since no quantum computer is required to solve the problem. Hence, we aim to enhance the algorithm such that the entire time evolution can be calculated using quantum circuits, and only minimal classical pre-processing must be performed.

6.1.6 Enhanced Algorithm

Ideally, the algorithm presented in the previous sections would be executed primarily on a quantum computer. However, a significant portion of the computational effort is performed classically, since the Kraus operators must be computed repeatedly for each time step. For each t , the superoperator $\mathcal{L}(t)$ is evaluated to construct the corresponding Choi matrix, and derive the Kraus operators. The vast majority of the calculation time for the classical algorithm could be spared if the Kraus operators only had to be calculated once for an arbitrary large time step Δt . The idea is to repeatedly apply the same circuit with a Kraus operator that propagates $\rho_S(t)$ to $\rho_S(t + \Delta t)$. Thus, to propagate from 0 to 1000 femtoseconds with a time step of 10 femtoseconds, the same circuit must be applied $n_{\text{steps}} = 100$ times. Consequently, the computation executed by the classical routine must be performed only once, rather than n_{steps} times, which is quite a substantial improvement.

To develop the algorithm, we aim to reuse the state from the previous time step for the next time step. Instead of repeatedly starting at $\rho_S(0)$, and directly propagating to $\rho_S(t)$, we propagate from $\rho_S(0)$ to $\rho_S(\Delta t)$, and continue iteratively until we reach $\rho_S(t = n_{\text{steps}}\Delta t)$. Thus, the operator sum representation reads:

$$\rho_S(t + \Delta t) = \sum_k M_k(\Delta t) \rho_S(t) M_k^\dagger(\Delta t). \quad (6.25)$$

To realize the quantum circuit, we must find the correct state to initialize it for the next time step. As explained in section 6.1.3, the relevant output state of the circuit is $M_k |\psi\rangle$, and the density matrix is

obtained by taking the outer product of the output vector with itself and summing the results, which is expressed as $\rho_S(t + \Delta t) = \sum_k M_k(\Delta t) |\psi(t)\rangle \langle \psi(t)| M_k(\Delta t)$. To initialize the quantum circuit for the next time step, we require a vector $|\psi'\rangle$ that corresponds to the density matrix $\rho_S(t)$, and thus fulfills $\rho_S(t) = |\psi'\rangle \langle \psi'|$. This condition implies that $\rho_S(t)$ must be pure. However, the interaction between the system and the environment leads to decoherence, which means that the initially pure state $\rho_S(0)$ becomes mixed. Thus, it is, in general, not possible to find a state vector $|\psi'\rangle$ that fulfills $\rho_S(t) = |\psi'\rangle \langle \psi'|$.

Since the correct state vector for initializing the next time step cannot be determined, an alternative approach is required. To this end, we note that all information about the quantum mechanical state of the FMO complex at time t is stored in the system density matrix, which can be correctly calculated for each time step. Therefore, we aim to use the system density matrix instead of the state vector to initialize the circuit. Since a quantum circuit can only be initialized with a vector, the system density matrix must be vectorized:

$$\text{vec}(\rho_S(0)) = (1, 0, \dots, 0)^T, \quad (6.26)$$

where the length of $\text{vec}(\rho_S(0))$ depends on the dimension of the system. In the following, we will implement this algorithm for the seven-site FMO complex, and thus $\text{vec}(\rho_S(0))$ has $8 \cdot 8 = 64 = 2^6$ entries. These considerations already indicate that the quantum circuit must be expanded from four to seven qubits. Since we now aim to implement $\text{vec}(\rho_S(0))$ instead of $\rho_S(0)$, the operator sum representation must be vectorized:

$$\text{vec}(\rho_S(\Delta t)) = \text{vec}\left(\sum_k M_k(\Delta t) \rho_S(0) M_k^\dagger(\Delta t)\right) = \sum_k \left(M_k^*(\Delta t) \otimes M_k(\Delta t)\right) \text{vec}(\rho_S(0)), \quad (6.27)$$

where we used Equation 6.2, and $(M_k^\dagger)^T = M_k^*$. The correct density matrix is obtained after each time step Δt , and can consequently be used to initialize the circuit for the next time step. Reviewing the circuit in Figure 6.4, we see that it is initialized with $|\psi\rangle$, and produces the relevant output $M_k(t) |\psi\rangle$, where the gates are obtained by a singular value decomposition of $M_k(t)$. By initializing the circuit with $\text{vec}(\rho_S)$, we aim to obtain the relevant output $(M_k^*(\Delta t) \otimes M_k(\Delta t)) |\text{vec}(\rho_S)\rangle$. Accordingly, it appears promising to perform a singular value decomposition of $M_k^*(\Delta t) \otimes M_k(\Delta t)$ for each M_k , and sum the results to resemble Equation 6.27. We thus have:

$$M_k^*(\Delta t) \otimes M_k(\Delta t) = U \Sigma V^\dagger, \quad (6.28)$$

where U and $V^\dagger$ are already unitary, and Σ is non-unitary and diagonal. In terms of the seven-site FMO complex, $M_k(t)$ is a 7×7 matrix, which means that $M_k^*(\Delta t) \otimes M_k(\Delta t)$, and thus also U , Σ , and $V^\dagger$, are 49×49 matrices, and U_Σ is a 98×98 matrix. To apply the matrices on a quantum circuit, they must be embedded into a unit matrix of shape $2^n \times 2^n$, and $2^{n+1} \times 2^{n+1}$, respectively, where $n = 6$. Together with the ancilla qubit for U_Σ , and the Hadamards, the new circuit is built of seven qubits, three more than in Figure 6.4. The corresponding circuit is shown in Figure 6.6 and a graphical summary of the algorithm can be seen in Figure 6.10.

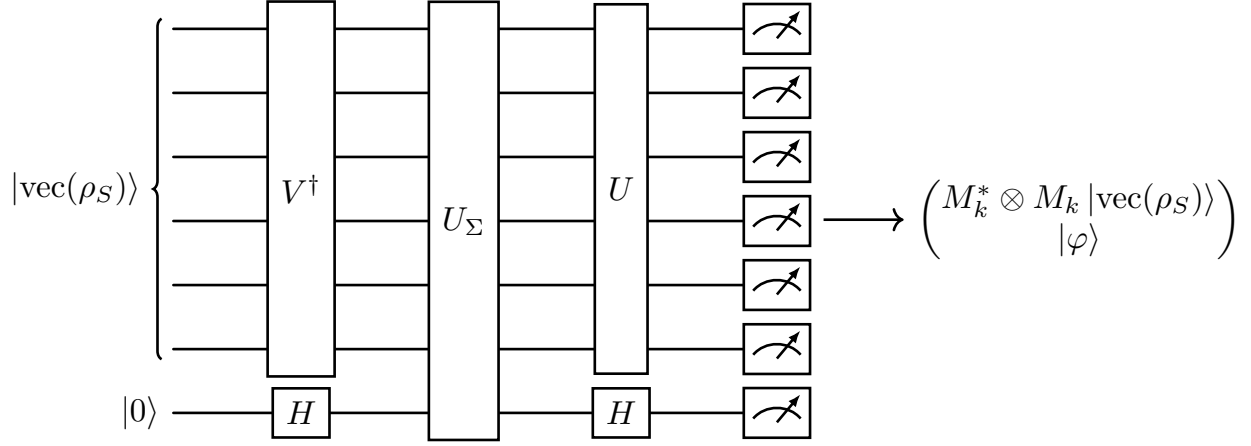


Figure 6.6: Quantum circuit for the implementation of the enhanced algorithm. To evolve the vectorized density matrix in time, a singular value decomposition of $M_k^*(\Delta t) \otimes M_k(\Delta t)$ is performed. Hence, compared to the algorithm presented by Seneviratne et al., more qubits are required. Using this circuit, the Kraus operators must be calculated only once with the classical method, whereas the entire time evolution can be calculated on the quantum computer simulator.

To verify the algorithm, the time evolution of the system density matrix was calculated in Figure 6.7 for the seven-site FMO complex, using the Hamiltonian and the initial density matrix from Equation 6.24:

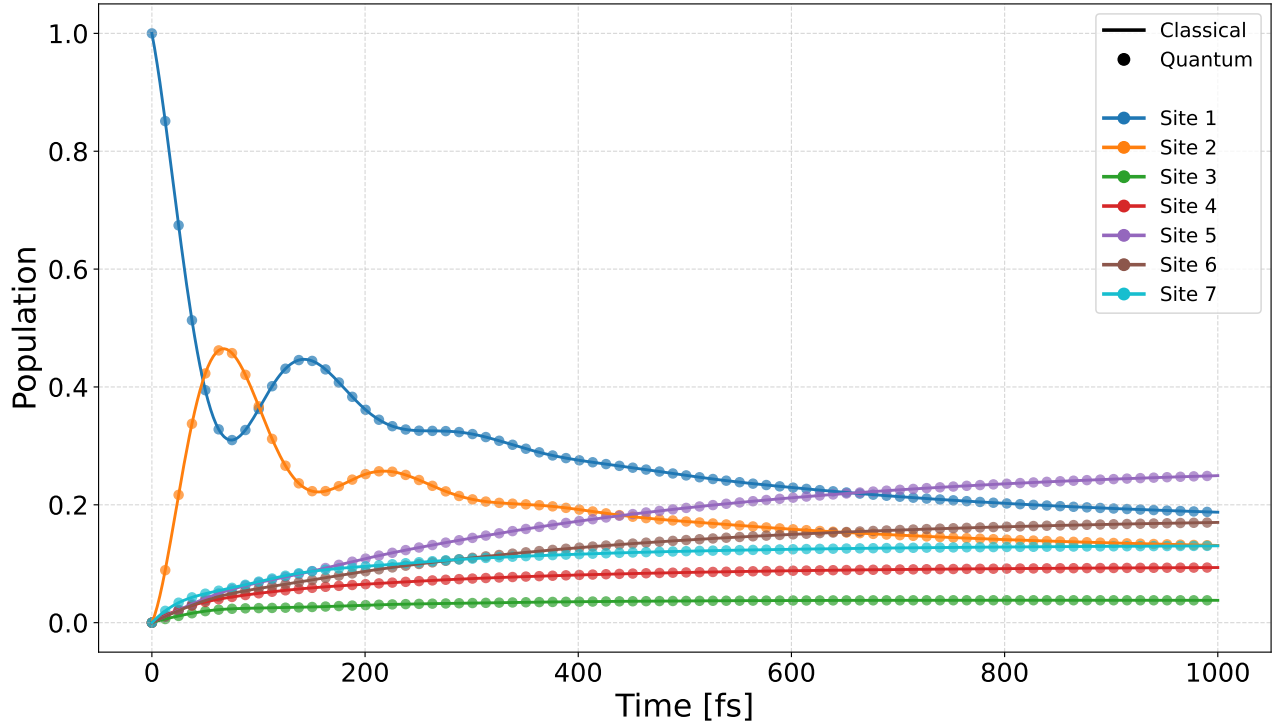


Figure 6.7: Populations as a function of time, calculated with the enhanced algorithm. The solid lines and the circles exactly match, since, instead of measuring counts in a statistical measurement, a function from Qiskit was used to return the quantum state after each measurement. Such a function is only a theoretical tool, and in practice, considerable effort is required to retrieve the full quantum state via tomography.

The initial excitation is again on site 1, and thus the circuit is initialized with $\text{vec}(\rho_S(0))$, where $\rho_S(0)$ is taken from Equation 6.24. The results of the calculation can be seen in Figures 6.7, 6.8 and 6.9. In Figure 6.7, the populations are depicted as a function of time, where the solid lines are computed with the classical routine without involving quantum circuits, and the circles with the quantum computer simulator of the Python library Qiskit. In this case, the circles show no deviation from the classical results, since no statistical simulation was conducted. Instead of repeatedly measuring the qubits of the quantum circuit, a function from Qiskit was used that simply returns the full quantum state after each time step, enabling the calculation of the populations. However, in practice, it is challenging to obtain the full quantum state, and hence, a procedure called quantum tomography must be employed, as discussed in Sections 5.8 and 6.4 and the [outlook](#).

Figure 6.8 shows the real and imaginary parts of the off-diagonal elements of the system density matrix. In the excitonic basis, these terms represent quantum coherences that quantify the extent to which the system exists in a linear superposition of different basis states. This superposition leads to time-dependent oscillations at frequencies corresponding to the energy differences between basis states. The coherences are subject to rapid dephasing due to interactions with the protein environment [83].

The coherences in the FMO complex provide evidence for a wavelike energy transfer that enhances the efficiency of excitonic transport at low temperatures [84]. Experiments have shown that these off-diagonal elements can persist for hundreds of femtoseconds at physiological temperatures, even though coupling to the protein environment typically causes electronic dephasing on sub-picosecond timescales [83].

In Figure 6.8, the coherences are shown as a function of time. Since the system density matrix is Hermitian, the off-diagonal elements above and below the main diagonal are equal. For a seven-site system, this results in 21 distinct off-diagonal terms. Each of the 21 graphs displays both the real and imaginary parts of the coherences. The real part is shown as a solid line in a slightly darker color, while the imaginary part is represented by a dashed line in a lighter shade. In addition, markers are used to indicate the data obtained with the quantum algorithm. Circular markers represent the real part and triangular markers the imaginary part. The solid and dashed lines correspond to the classical results obtained with the Lindblad equation without involving the circuit in Figure 6.6.

To better visualize the magnitude of the coherences, their absolute value is illustrated as a heatmap in Figure 6.9. The coherence between sites 1 and 2 initially has the largest value, around 0.25, as indicated by the red color. As time progresses, this value decays, and the color turns greener, corresponding to smaller values. The other coherences are generally smaller, corresponding to bluer colors. For instance, the off-diagonal elements (5,1), (6,1), and (7,1) rise steeply to higher values, indicated by a lighter blue, but decay over time as their color becomes bluer again. In contrast, elements such as (7,4) or (6,4) initially have low values, depicted in blue, but their values rise over time, as seen by the transition to a lighter blue.

While the procedure introduced in this section works theoretically, it is also important to consider the practical implementation of the quantum algorithm. To this end, we first note that one could in principle construct a large quantum circuit by sequentially appending the circuit shown in Figure 6.6. Then, the large circuit would consist of the basic circuit block, followed by the same block again, repeated n_{steps} times. In this way, the system would be propagated to $t = n_{\text{steps}} \Delta t$ without the need to measure the full quantum

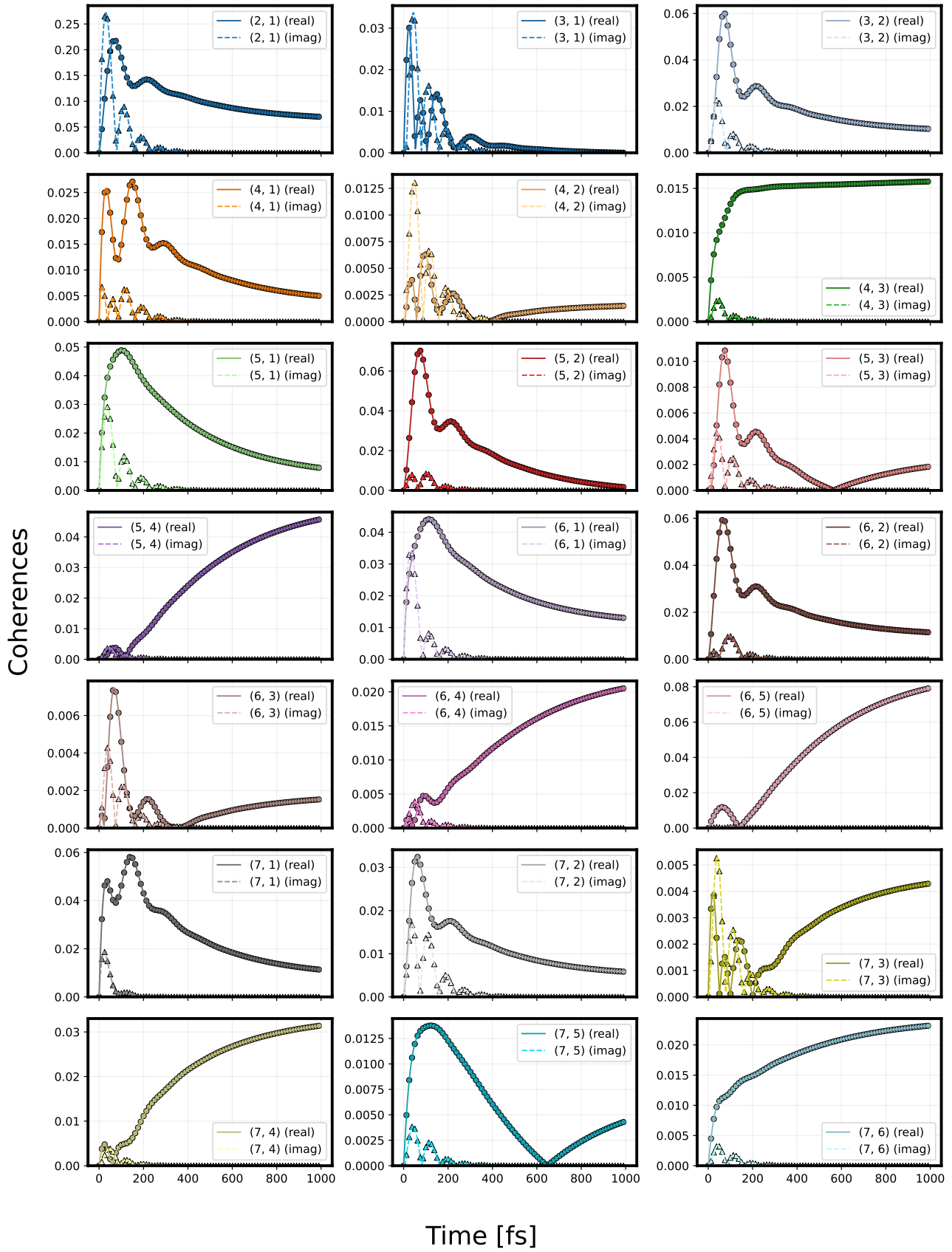


Figure 6.8: Time evolution of the real and imaginary parts of the off-diagonal elements of the system density matrix. The vertical axis represents the magnitude of the coherences, while the horizontal axis shows the time from 0 to 1000 femtoseconds. The numbers in braces in the legend of each plot refer to the index of the off-diagonal element. The solid lines correspond to the results obtained from a classical simulation using the Lindblad equation. In contrast, the triangular and circular markers represent the data generated by a quantum computer simulator provided by Qiskit. The results are in good agreement, further confirming the validity of the enhanced algorithm.

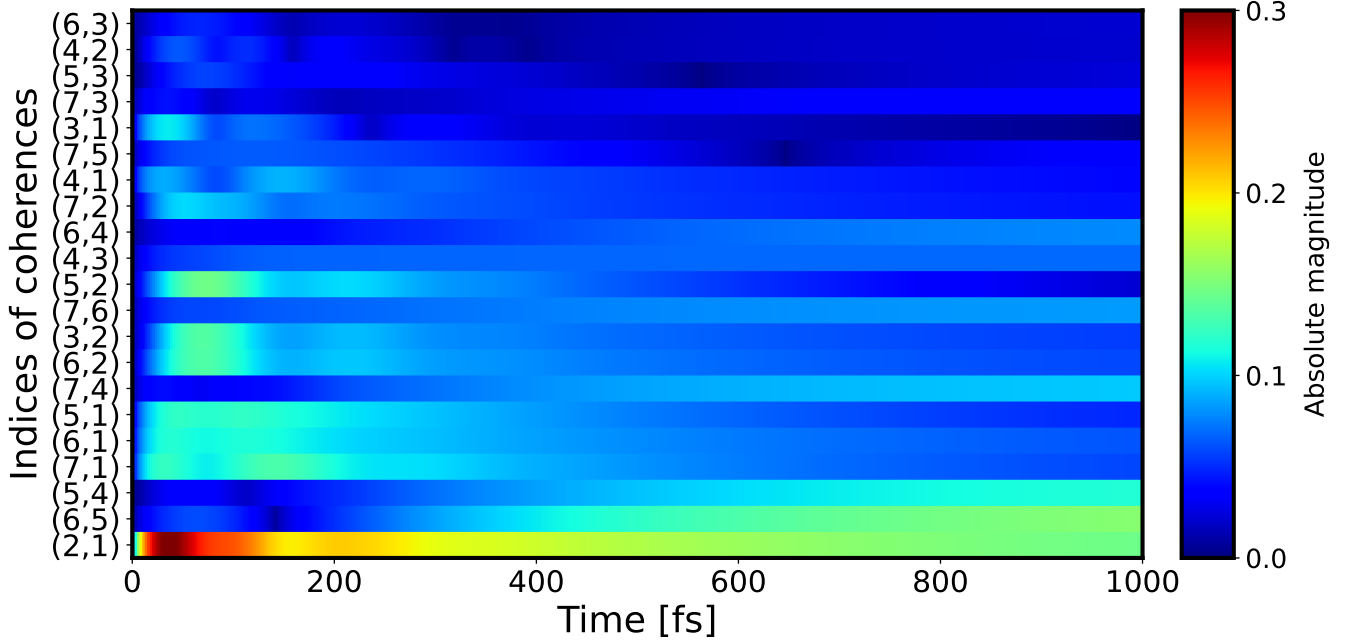


Figure 6.9: The absolute value of the coherences is depicted as a heatmap. The vertical axis represents the indices of the off-diagonal elements, while the horizontal axis shows the time in femtoseconds. The color bar on the right maps values to colors. Bluish and greenish tones correspond to small coherence values, whereas yellow and red tones indicate large values.

state after each intermediate step. However, this approach is not feasible since the results from each individual circuit must be summed, which cannot be done reliably on a quantum computer. In addition, due to the dilation process, only a portion of each output vector from the quantum circuits is required, which complicates the procedure even further. Therefore, the full quantum state must be obtained after each time step Δt in order to use it as the input for the subsequent step, which poses a practical challenge despite the circuit functioning as intended in theory. Simple projective measurements yield only the probabilities, so the absolute squares of the coefficients in each basis state, without revealing the phases. Since the phase information is essential to reuse the quantum state, projective measurements alone are insufficient. To address this limitation, a procedure known as quantum tomography is employed, which performs measurements in different bases to reconstruct the complete quantum state as described in Section 5.8. The results for the dynamics of $\rho_S(t)$ obtained via quantum tomography are presented in Section 6.4.

In this section, we have outlined an improved algorithm that shifts most of the computational workload from classical to quantum routines. This is achieved by reusing the state from the previous time step as the initial state of the following time step. Consequently, instead of calculating the Kraus operators $M_k(t)$

for each time step, $M_k(\Delta t)$ must only be computed once, and the same circuit can be applied iteratively. The algorithm is graphically summarized in Figure 6.10.

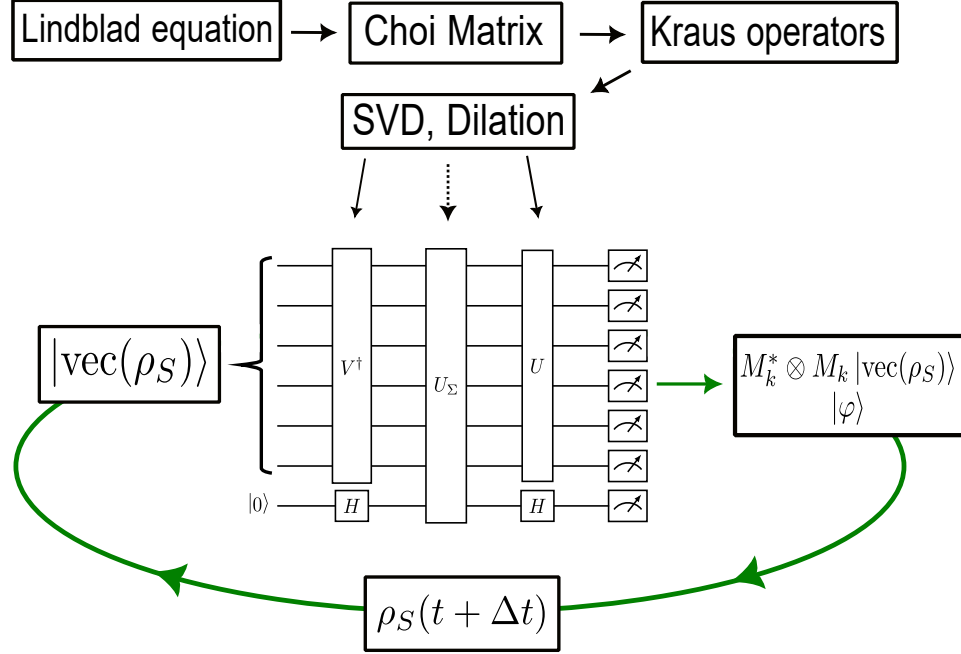


Figure 6.10: Graphical illustration of the enhanced algorithm. Instead of repeatedly calculating the Kraus operators, the circuit is only constructed once. To obtain $\rho_S(t + \Delta t)$, the vectorized ρ_S , computed in the previous time step, is used as the initial state for the next time step, as indicated by the green ellipse.

6.2 Markovian Dynamics with Small-Step Evolution

This section introduces a second algorithm for the calculation of the Markovian dynamics of open quantum systems. The algorithm was published in a paper by Hu et al. [17] in 2022. It employs the Kraus operator sum representation to propagate the system density matrix over small time steps dt . Since the Kraus operators describe the dynamics of open quantum systems, they are non-unitary and must be dilated to unitary matrices to be implemented on a quantum circuit. However, instead of a singular value decomposition, the paper uses a simpler, so-called Sz.-Nagy dilation. The system of interest is again the FMO complex, but for simplicity, only three sites are considered. In the following, we will first reproduce the results of the paper, and then apply a similar algorithm as presented in Section 6.1.6 to achieve an exponential saving in computational cost.

6.2.1 Kraus Operators and the Sz.-Nagy Dilation

To calculate the dynamics of the system density matrix of an open quantum system, we use the operator sum representation, which was introduced in Section 4.2.1:

$$\rho_S(t) = \sum_k M_k(t) \rho_S(0) M_k^\dagger(t). \quad (6.29)$$

As described in Section 4.2.5, the operator sum representation has a map Φ , such that $\rho_S(t+dt) = \Phi[\rho_S(t)]$. The idea is to apply the map Φ iteratively to obtain:

$$\rho_S(t) = \Phi[\dots\Phi[\rho_S(0)]], \quad (6.30)$$

where Φ propagates n_{steps} times by a small time step dt . To obtain the Kraus operators, we recall that the Lindblad equation has an operator sum representation, and thus, a connection between the Lindblad operators and Kraus operators exists. Using the concepts of Section 4.2.6, we find:

$$\begin{aligned} \rho_S(t+dt) &= M_0(dt)\rho_S(t)M_0^\dagger(dt) + \sum_{k>0} M_k(dt)\rho_S(t)M_k^\dagger(dt) \\ &= \rho_S(t) - \frac{1}{2} \sum_{k>0} \gamma_k \left(L_k \rho_S(t) L_k^\dagger + \{L_k^\dagger L_k, \rho_S(t)\} \right) dt + \mathcal{O}(dt^2). \end{aligned} \quad (6.31)$$

The Kraus operators for $k > 0$ are defined as $M_k = \sqrt{\gamma_k dt} L_k$, where the L_k are the Lindblad operators, the γ_k the dissipation rates, and dt is a small time step. In addition, we must introduce $M_0 = I - \frac{1}{2} dt \sum_{k>0} \gamma_k L_k^\dagger L_k$ to obtain:

$$\begin{aligned} \sum_k M_k^\dagger M_k &= M_0^\dagger M_0 + \sum_{k>0} M_k^\dagger M_k \\ &= I - dt \sum_{k>0} \gamma_k L_k^\dagger L_k + dt \sum_{k>0} \gamma_k L_k^\dagger L_k + \mathcal{O}(dt^2) = I + \mathcal{O}(dt^2), \end{aligned} \quad (6.32)$$

and thus ensure the conservation of probability. The time argument for the Kraus operators was omitted for brevity. To verify the equations, one can easily plug in the expressions for the Kraus operators. For a detailed derivation, I refer to Section 4.2.6. Equation 6.31 shows that the operator sum representation evolves the dissipator of the Lindblad equation with the Euler method, with an error on the order of $\mathcal{O}(dt^2)$. Thus, to keep the error small, the approach is limited to small time steps dt . Since Equation 6.31 accounts only for the dissipative part of the Lindblad equation, we must multiply the diagonal Kraus operators with $U = e^{-\frac{i}{\hbar} H dt}$ to also consider the unitary part of the evolution. The Kraus operators are stated in Equation 6.35, from which the diagonal operators can be easily identified.

Using the above-discussed formulas, the first three time steps are computed as:

$$\begin{aligned} \rho_S(1) &= \sum_k M_k \rho_S(0) M_k^\dagger, \\ \rho_S(2) &= \sum_k M_k \rho_S(1) M_k^\dagger = \sum_j \sum_k M_j M_k \rho_S(0) M_k^\dagger M_j^\dagger, \\ \rho_S(3) &= \sum_k M_k \rho_S(2) M_k^\dagger = \sum_i \sum_j \sum_k M_i M_j M_k \rho_S(0) M_k^\dagger M_j^\dagger M_i^\dagger. \end{aligned} \quad (6.33)$$

Each Kraus operator M_k propagates by a small time step dt , however, the time argument was omitted for brevity. Hence, the notation $\rho(n_{steps} = N)$ corresponds to time $t = Ndt$. Accordingly, ρ_S will be denoted either as $\rho_S(N)$ or as $\rho_S(t)$, depending on whether the emphasis is on the number of discrete steps or on the corresponding physical time. Since, for each time step, the time evolution starts at $\rho_S(0)$, the number of summands rises exponentially. To keep the computation time manageable when considering a larger number of time steps, only three sites of the FMO complex are taken into account.

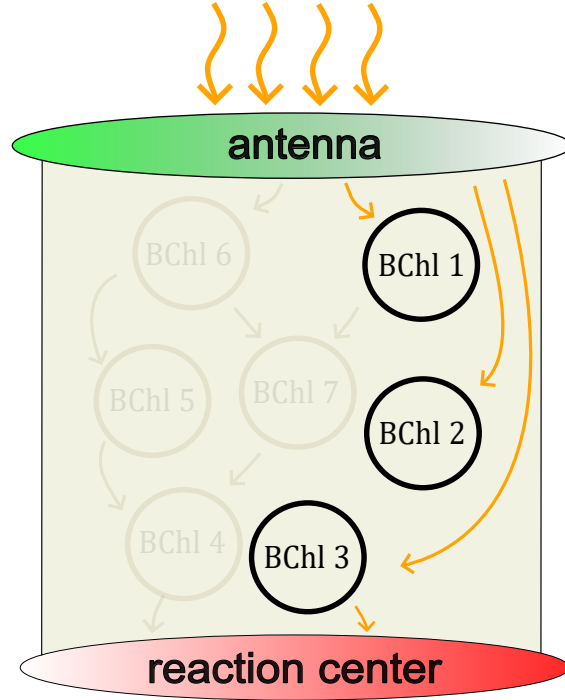


Figure 6.11: One monomer of the FMO complex, where only three sites are considered in the simulations. The arrows indicate the transfer of the excitation from the antenna through the complex to the reaction center. Each arrow corresponds to one transition, described by the Kraus operators in Equation 6.35.

The Hamiltonian of the system [17] is given in Equation 6.34, and a schematic illustration of the FMO complex can be seen in Figure 6.11.

$$H = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0.0267 & -0.0129 & 0.000632 & 0 \\ 0 & -0.0129 & 0.0273 & 0.00404 & 0 \\ 0 & 0.000632 & 0.00404 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (6.34)$$

The system Hamiltonian is a 5×5 matrix, since it contains, besides the three sites, a sink and a ground state. The sink corresponds to the reaction center, and the ground to the antenna. The three sites are considered by the non-zero 3×3 block in the center of the Hamiltonian. The ground and sink show no couplings with the other sites, implying that the excitation cannot jump back from a site to the antenna, or from the reaction center to a site. This is expressed by the zeros in the first and last row and column of the Hamiltonian. The diagonal elements H_{00} and H_{44} do not affect the populations on sites 1 to 3, and can thus be set to 0. Moreover, the site energies are shifted such that the lowest-lying site, site three, has a site energy of 0, which yields $H_{33} = 0$.

In Figure 6.11, the three BChl molecules considered in the simulation are depicted in stronger colors. The yellow arrows indicate the transfer of the excitation through the complex to the reaction center. Each arrow represents a possible jump of the excitation from one site to another. Jumps can occur from the antenna

to sites 1, 2, and 3, and from site 3 to the reaction center. Accordingly, the system is described by three dephasing operators $L_{deph}(m) = \sqrt{\alpha}|m\rangle\langle m|$, three dissipation operators $L_{diss}(m) = \sqrt{\beta}|0\rangle\langle m|$, and one sink operator $L_{sink} = \sqrt{\gamma}|4\rangle\langle 3|$, where $m = 1, \dots, 3$. The dephasing operators model the loss of coherence at site m , and the dissipation operators and the sink operator describe the change in population on the three sites due to the interaction with the antenna and the sink. Since sites 1, 2, and 3 are not coherently coupled, a part of the non-unitary evolution is omitted. This implies that the simulations will yield a result more similar to a unitary evolution.

Using the expressions for the Lindblad operators, the Kraus operators can be calculated as $M_k = \sqrt{\gamma_k dt} L_k$ for $k > 0$, and $M_0 = I - \frac{1}{2} dt \sum_{k>0} \gamma_k L_k^\dagger L_k$ for $k = 0$. This yields the following 8 Kraus operators:

$$\begin{aligned} M_1 &= \sqrt{\alpha dt} |1\rangle\langle 1|, & M_2 &= \sqrt{\alpha dt} |2\rangle\langle 2|, \\ M_3 &= \sqrt{\alpha dt} |3\rangle\langle 3|, & M_4 &= \sqrt{\beta dt} |0\rangle\langle 1|, \\ M_5 &= \sqrt{\beta dt} |0\rangle\langle 2|, & M_6 &= \sqrt{\beta dt} |0\rangle\langle 3|, \\ M_7 &= \sqrt{\gamma dt} |4\rangle\langle 3|, & M_0 &= \sqrt{I - \sum_{k>0} M_k^\dagger M_k}. \end{aligned} \quad (6.35)$$

The parameters α , β , and γ are the dephasing rates, dissipation rates, and sink rates, respectively, and are given as [17]:

$$\begin{aligned} \alpha &= 3.00 \times 10^{-3} \text{ fs}^{-1}, \\ \beta &= 5.00 \times 10^{-7} \text{ fs}^{-1}, \\ \gamma &= 6.28 \times 10^{-3} \text{ fs}^{-1}. \end{aligned} \quad (6.36)$$

With Equations 6.33 to 6.36, the procedure to calculate the time evolution of $\rho_S(t)$ is straightforward. The remaining question is how to implement this approach on a quantum circuit. To this end, we review Equation 6.33 and notice that the system density matrix at time $n_{steps} dt$ can be calculated by summing $8^{n_{steps}}$ terms, since 8 Kraus operators were introduced. For $n_{steps} = N$, these terms are all of the form $\mathcal{M}_\alpha(dt) \rho_S(0) \mathcal{M}_\alpha^\dagger(dt)$, where $\mathcal{M}_\alpha(dt) = M_{i_1}(dt) \cdots M_{i_N}(dt)$ with $i_1, \dots, i_N \in \{1, \dots, 8\}$ is a product of N Kraus operators, and $\alpha \in \{1, \dots, 8^N\}$. Thus, we can rewrite Equation 6.33 as:

$$\rho_S(N) = \sum_{\alpha} \mathcal{M}_\alpha(dt) \rho_S(0) \mathcal{M}_\alpha^\dagger(dt). \quad (6.37)$$

We aim to implement Equation 6.37 on a quantum circuit. However, a quantum circuit only allows unitary operations, and hence the non-unitary operator $\mathcal{M}_\alpha$ must be dilated to a unitary one. To this end, we introduce a so-called Sz.-Nagy dilation to construct a unitary operator $U_{\mathcal{M}_\alpha}$:

$$U_{\mathcal{M}_\alpha} = \begin{pmatrix} \mathcal{M}_\alpha & D_{\mathcal{M}_\alpha}^\dagger \\ D_{\mathcal{M}_\alpha} & -\mathcal{M}_\alpha^\dagger \end{pmatrix}, \quad (6.38)$$

where $D_{\mathcal{M}_\alpha} = \sqrt{I - \mathcal{M}_\alpha^\dagger \mathcal{M}_\alpha}$ is called the defect operator of $\mathcal{M}_\alpha$, and the time argument was omitted for brevity. Due to the block matrix structure, it is easy to verify that $U_{\mathcal{M}_\alpha}$ is twice as large as $\mathcal{M}_\alpha$. Thus, an additional ancilla qubit must be introduced to account for the larger Hilbert space. The ancilla qubit is always set to $|0\rangle$, which will become clearer shortly.

Moreover, for the Sz.-Nagy dilation to be applicable, the Kraus operators must be contractions, which means that they shrink or preserve the norm of a vector. The proof that each Kraus operator is indeed a contraction can be found in appendix A.3. For further reading, I refer to Reference [85].

6.2.2 Construction of the Circuit

The operator $U_{\mathcal{M}_\alpha}$ can be applied to a simple quantum circuit, which can be seen in Figure 6.12. Of course, the unitary gates $U_{\mathcal{M}_\alpha}$ must be decomposed into elementary gates to allow an implementation on a quantum computer.

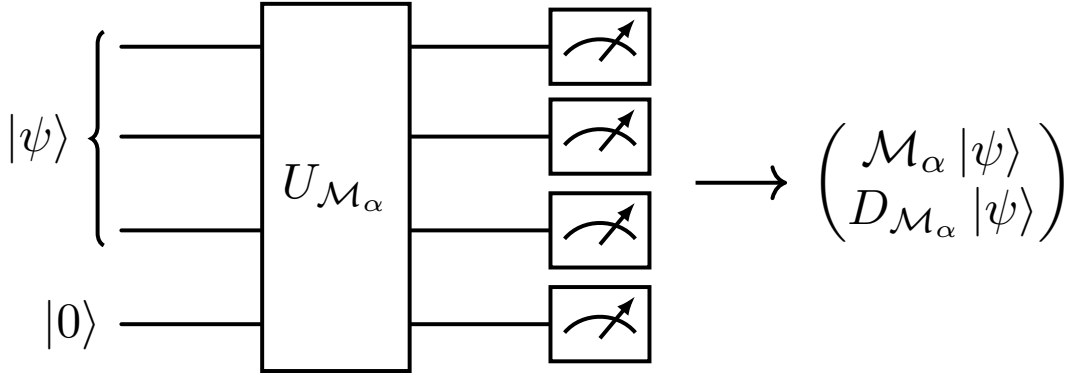


Figure 6.12: Quantum circuit for the calculation of the time evolution of $\rho_S(t)$. The operator $U_{\mathcal{M}_\alpha}$ is applied to the input vector $|\psi\rangle \otimes |0\rangle$. The first half of the output vector yields the desired outcome. To compute $\rho_S(t)$, the circuit must be applied to each $U_{\mathcal{M}_\alpha}$ and the resulting contributions must be summed.

To better understand the circuit, we can calculate its action on an input state $|\psi\rangle \otimes |0\rangle$. The calculation is rather trivial and yields:

$$U_{\mathcal{M}_\alpha}(|\psi\rangle \otimes |0\rangle) = \begin{pmatrix} \mathcal{M}_\alpha |\psi\rangle \\ D_{\mathcal{M}_\alpha} |\psi\rangle \end{pmatrix}. \quad (6.39)$$

Here, we used that $|\psi\rangle \otimes |0\rangle = (\psi, 0, \dots, 0)^T$, where the number of zeros equals the length of $|\psi\rangle$. Since $|\psi\rangle$, $\mathcal{M}_\alpha$, and $D_{\mathcal{M}_\alpha}^\dagger$ have the same dimension, $D_{\mathcal{M}_\alpha}^\dagger$ and $-\mathcal{M}_\alpha^\dagger$ are effectively multiplied by a zero-vector and thus vanish. Hence, the outcome is a state vector which contains $\mathcal{M}_\alpha |\psi\rangle$ in the first half and $\mathcal{D}_{\mathcal{M}_\alpha} |\psi\rangle$ in the second half. The second half does not have any physical meaning and can thus be safely discarded. It is only a remainder of the dilation to a larger subspace. In contrast, the first half of the output vector yields the desired result. Taking the outer product of $\mathcal{M}_\alpha |\psi\rangle$ with itself, we obtain $\mathcal{M}_\alpha |\psi\rangle \langle \psi| \mathcal{M}_\alpha$. Repeating the circuit for all Kraus operators $\mathcal{M}_\alpha$ and summing the results, the system density matrix $\rho_S(N)$ at time step N can be calculated by Equation 6.37. The process must be repeated for each time step dt . Thus, to calculate $\rho_S(N)$ corresponding to $n_{steps} = N$ and time $t = Ndt$, the operator $\mathcal{M}_\alpha$ is obtained by a multiplication of N Kraus operators, and the sum in Equation 6.33 contains 8^N terms. The algorithm is graphically illustrated in Figure 6.13.

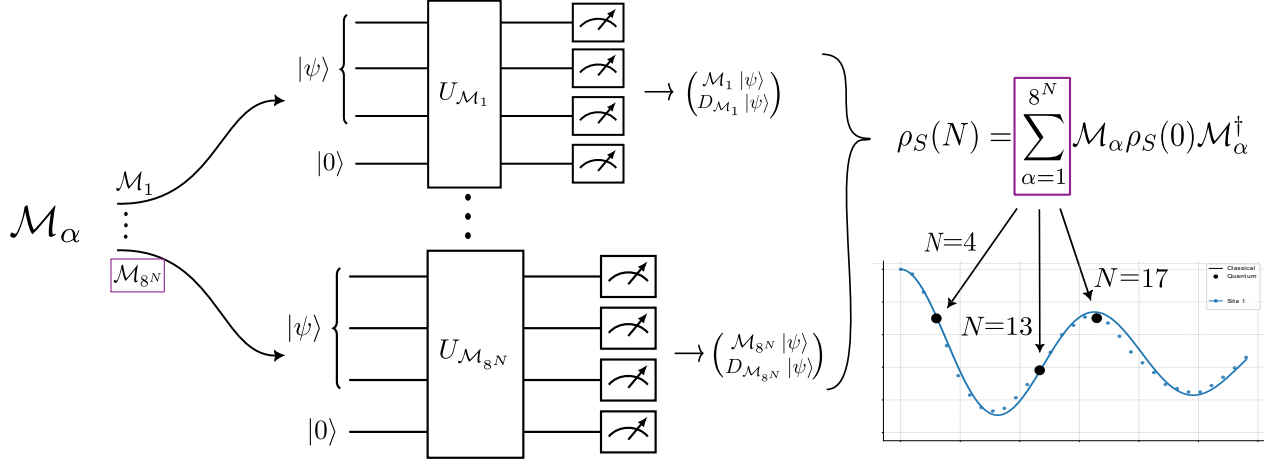


Figure 6.13: Illustration of the algorithm to compute $\rho_S(N)$, where each Kraus operator $\mathcal{M}_\alpha$ propagates by a small time step dt , i.e, in this case until time $t = Ndt$. Each $\mathcal{M}_\alpha$ is dilated to a unitary $U_{\mathcal{M}_\alpha}$ and subsequently appended onto the circuit. Hence, 8^N circuits must be evaluated to get ρ at time step N . The outcomes of the circuit are summed to obtain $\rho_S(N)$, and the results are plotted, as is exemplarily illustrated by the black dots for $N = 4$, $N = 13$, and $N = 17$. The purple box indicates that a significant amount of summands is required, which makes the algorithm computationally very expensive.

6.2.3 Reproduction of the Results for the FMO Complex

To keep the number of terms manageable, the time evolution of $\rho_S(t)$ was computed in 5 iterations. Iteration n_{it} propagates $\rho_S(0)$ by $n_{it} \cdot 9.68$ femtoseconds, where $n_{it} \in \{0, \dots, 4\}$. Then, starting from $\rho_S(n_{it} \cdot 9.68)$, Kraus operators that propagate by 48.4 femtoseconds are applied up to 6 times to obtain $\rho_S(48.4 + n_{it} \cdot 9.68)$ to $\rho_S(6 \cdot 48.4 + n_{it} \cdot 9.68)$. Thus, to calculate $\rho_S(6 \cdot 48.4 + n_{it} \cdot 9.68)$, the operator $\mathcal{M}_\alpha$ consists of a multiplication of 6 Kraus operators, and the sum contains 8^6 terms. Due to the phenomenological choice for the Lindblad operators, most of the 8^6 terms are close to 0 and can thus be discarded. The number of relevant terms is on the order of 10^3 .

The results reproduced from Hu et al. [17] are shown in Figure 6.14. The vertical axis represents the populations, while the horizontal axis corresponds to the time, ranging from 0 to 300 femtoseconds. Since the initial excitation was at site 1, the population on site 1 starts at 1. As time progresses, the excitation delocalizes over the three sites. In addition, it is assumed that only one excitation at a time is in the system. The solid lines represent calculations performed by the classical routine using the Lindblad equation. The circles refer to the results obtained for the calculation using the quantum computer simulator from the Python library Qiskit. This simulator is not a real quantum computer and therefore shows no quantum noise, but only statistical errors. The experiment was repeated 10000 times to obtain sufficient statistics. Figure 6.14 shows that the results obtained with the quantum computer simulator deviate from the classically calculated results using the Lindblad equation. The reason is not only the statistical error, but primarily that the time step dt could not be chosen small enough without requiring too much computation time.

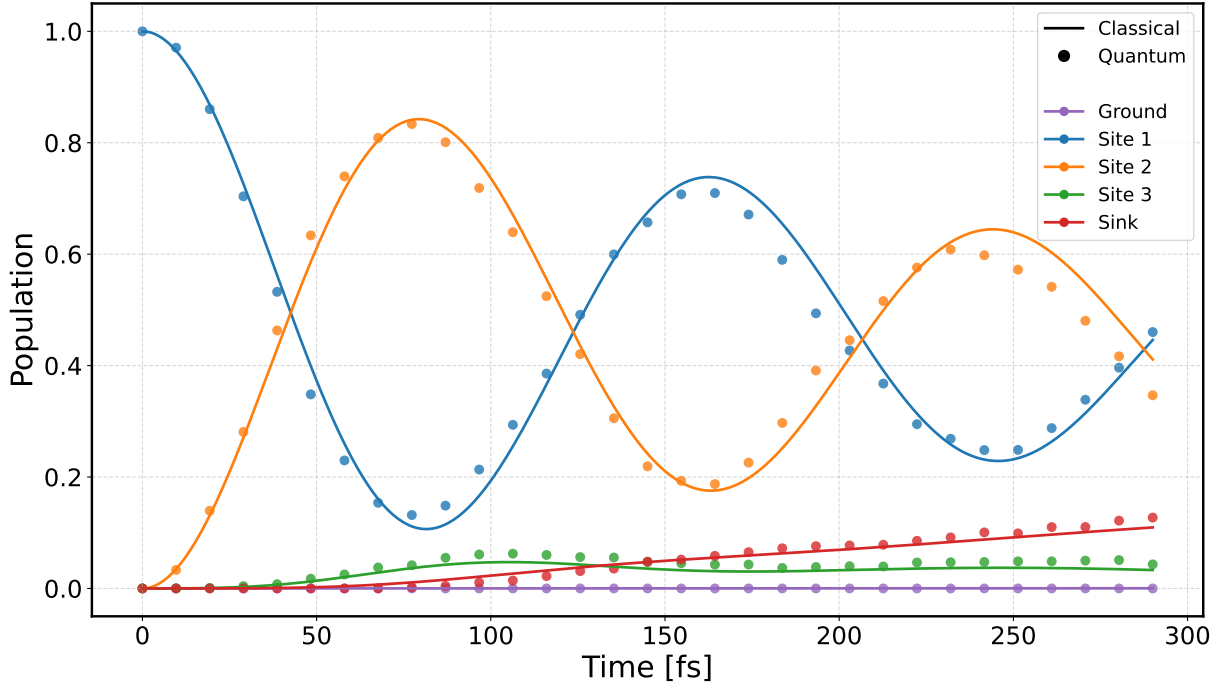


Figure 6.14: The populations are depicted as a function of time for the three sites, the ground and the sink state. The solid lines denote the result of the classical computation, whereas the circles refer to the results obtained by the quantum computer simulator from the Python library Qiskit. Since the time steps could not be chosen small enough, the results differ slightly due to the error on the order of $\mathcal{O}(dt^2)$.

In the following, we will present an enhanced algorithm that leads to an exponential saving in the computational cost. Similarly to Section 6.1.6, we reuse the output state from the previous time step to initialize the circuit for the next time step. This enables us to use a sufficiently small time step, so that the discrepancies vanish.

6.2.4 Enhanced Algorithm

Briefly reviewing the algorithm from the previous sections shows that the computational complexity rises exponentially for each iteration. To compute $\rho_S(N)$, Equation 6.37 contains 8^N summands, which implies that 8^N multiplications of N matrices must be performed by a classical method. For $N = 3$, the 512 summands seem manageable, but already for $N = 10$ the number of summands surpasses one billion. Thus, it is out of question that this algorithm scales very poorly and cannot be used for longer propagations. The main reason for the poor performance is that the calculations start at $\rho_S(0)$ for each time step, which means that all the products must be expanded when, for the next time step, an additional sum over Kraus operators is applied. If the state resulting from the previous time step could be reused for the next time step, the number of terms sinks significantly. Reusing the state, we can write:

$$\rho_S(N) = \sum_k M_k(dt) \rho_S(N-1) M_k^\dagger(dt). \quad (6.40)$$

Thus, to propagate from time step N to time step $N+1$, we only need to apply the 8 Kraus operators to

$\rho_S(N-1)$, which yields 8 summands. Propagating from $\rho_S(0)$ to $\rho_S(10)$ thus only contains the calculation of 80 summands, which is an enormous reduction compared to the $\sim 10^9$ terms needed before. In addition, no matrix multiplications by a classical routine are required, which means that most of the work can now be performed on a quantum computer.

To develop the algorithm, we must first find the correct state to initialize the circuit for the next time step. As explained in Section 6.2.2, the relevant output state of the circuit is $\mathcal{M}_\alpha(dt) |\psi\rangle$, and the density matrix is obtained by taking the outer product of the output vector with itself and summing all the results, which is expressed as $\rho_S(t) = \sum_\alpha \mathcal{M}_\alpha(dt) |\psi(0)\rangle \langle \psi(0)| \mathcal{M}_\alpha(dt)$. To initialize the quantum circuit for the next time step, we would need a vector $|\psi'\rangle$ that corresponds to the density matrix $\rho_S(t)$ and thus fulfills $\rho_S(t) = |\psi'\rangle \langle \psi'|$. However, as described in Section 6.1.6, such a state vector does, in general, not exist. Since the correct state vector for initializing the next time step cannot be determined, the system density matrix is used instead. However, a quantum circuit can only be initialized with a vector, and hence, the system density matrix must be vectorized. Assuming an initial excitation on site 1, this yields:

$$\text{vec}(\rho_S(0)) = (1, 0, \dots, 0)^T, \quad (6.41)$$

where the length of $\text{vec}(\rho_S(0))$ depends on the dimension of the system. Since we now aim to implement $\text{vec}(\rho_S(0))$ instead of $\rho_S(0)$, we need to vectorize the operator sum representation:

$$\text{vec}(\rho_S(dt)) = \text{vec}\left(\sum_k M_k(dt) \rho_S(0) M_k^\dagger(dt)\right) = \sum_k \left(M_k^*(dt) \otimes M_k(dt)\right) \text{vec}(\rho_S(0)). \quad (6.42)$$

Briefly reviewing the circuit in Figure 6.12 shows that it is initialized with $|\psi\rangle$ and yields the relevant output $\mathcal{M}_\alpha(dt) |\psi\rangle$. Since now the circuit is initialized with $\text{vec}(\rho_S)$, we aim to obtain a relevant output state of $(M_k^*(dt) \otimes M_k(dt)) \text{vec}(\rho_S)$. Because we reuse the density matrix of the previous time step, no product of Kraus operators must be calculated, and $\mathcal{M}_\alpha$ is replaced by a single Kraus operator M_k . Thus, it seems promising to do a Sz.-Nagy dilation of $M_k^*(dt) \otimes M_k(dt)$ for each M_k and sum the results to resemble Equation 6.40. We thus have:

$$U_{M_k^* \otimes M_k} = \begin{pmatrix} M_k^* \otimes M_k & D_{M_k^* \otimes M_k}^\dagger \\ D_{M_k^* \otimes M_k} & -(M_k^* \otimes M_k)^\dagger \end{pmatrix}, \quad (6.43)$$

and the circuit is constructed by replacing $U_{\mathcal{M}_\alpha}$ with $U_{M_k^* \otimes M_k}$. An additional qubit must be added to the circuit, since $U_{M_k^* \otimes M_k}$ is given by the tensor product of two Kraus operators and is thus twice as large as $U_{\mathcal{M}_\alpha}$. To ensure that $U_{M_k^* \otimes M_k}$ has the correct dimensions for the quantum circuit, it must be embedded into a unitary matrix of shape $2^n \times 2^n$. Since the dimension of the Kraus operators is five by five, $M_k^* \otimes M_k$ has dimensions of 25×25 , and thus $U_{M_k^* \otimes M_k}$ has a shape of 50×50 . Hence, it must be embedded into a unitary matrix of shape 64×64 , which requires $n = 6$ qubits. The corresponding circuit can be seen in Figure 6.15.

Recalculating the action of the circuit for an input state $\text{vec}(\rho_S)$ and for the operator $U_{M_k^* \otimes M_k}$, we obtain:

$$U_{M_k^* \otimes M_k}(|\text{vec}(\rho_S)\rangle \otimes |0\rangle) = \begin{pmatrix} U_{M_k^* \otimes M_k} |\text{vec}(\rho_S)\rangle \\ D_{U_{M_k^* \otimes M_k}} |\text{vec}(\rho_S)\rangle \end{pmatrix}, \quad (6.44)$$

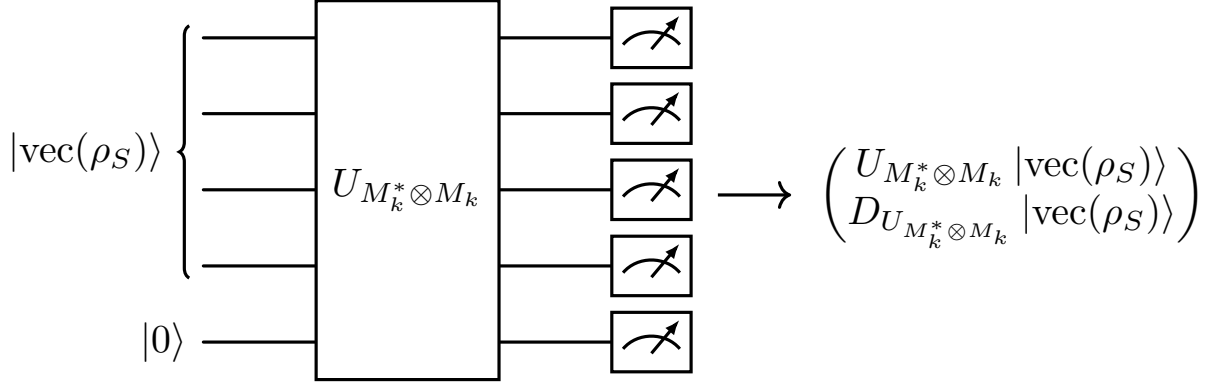


Figure 6.15: Quantum circuit for the calculation of $\rho_S(t)$. The unitary operator appended to the circuit reads $U_{M_k^* \otimes M_k}$ and allows us to reuse the state from the previous time step as the input vector for calculating the next time step. To account for the higher dimensions of $U_{M_k^* \otimes M_k}$, an additional qubit is introduced.

as desired. Thus, by repeating the circuit for all Kraus operators and summing the results, the correct time evolution of the system density matrix can be calculated according to Equation 6.42. The enhanced algorithm is schematically depicted in Figure 6.16.

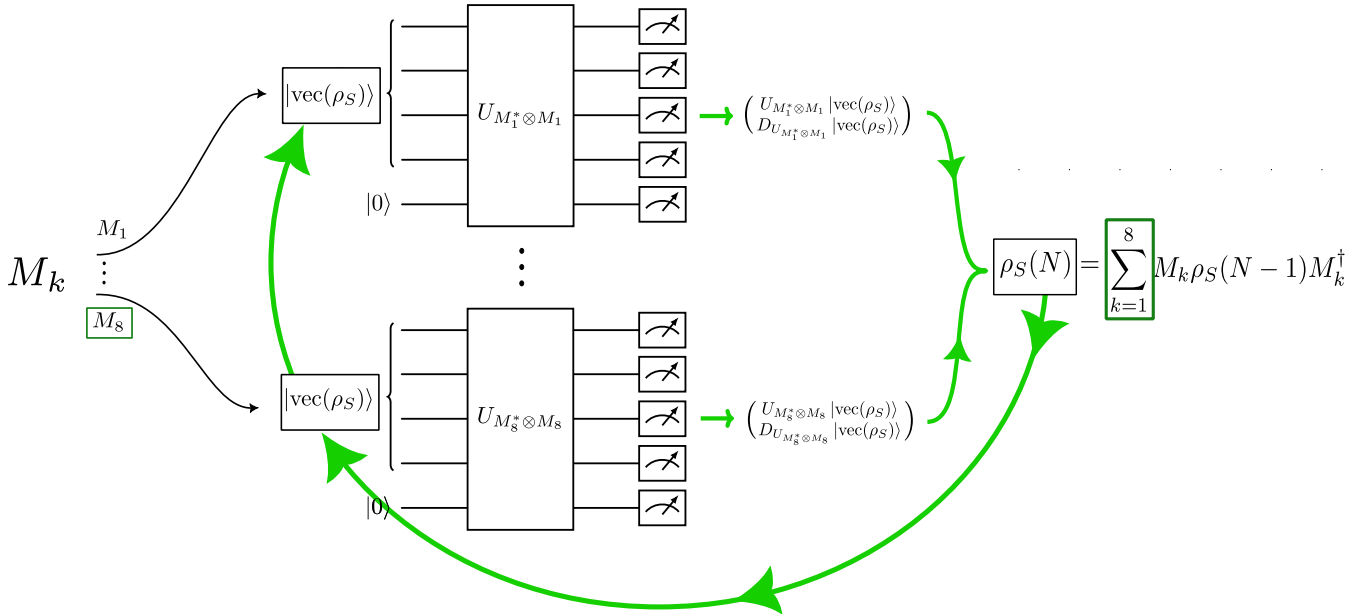


Figure 6.16: Illustration of the enhanced algorithm to compute $\rho_S(N)$. Each M_k is dilated to a unitary $U_{M_k^* \otimes M_k}$ and then appended onto the circuit. The outcomes of the circuit are summed to obtain $\rho_S(N)$, which is vectorized and reused for the next iteration. Hence the 8 circuits must only be constructed once and can be used iteratively. The green box indicates that only 8 summands are required for each time step, which is a substantial improvement compared to the 8^N summands for the N^{th} iteration alone in Figure 6.13.

The results of the calculation are shown in Figure 6.17. The time step was chosen as $dt = 1$ femtosecond, which is significantly smaller than the 9.68 and 48.4 femtoseconds used before. Instead of six, 290 iterations were computed, which corresponds to a total of $8 \cdot 290 = 2320$ summands. This is an enormous improvement compared to the $8^{290} \approx 9.07 \cdot 10^{261}$ summands previously required alone for the 290th iteration. Due to the small time step, hardly any deviation between the results obtained with the quantum computer simulator from Qiskit and the classical calculation is visible.

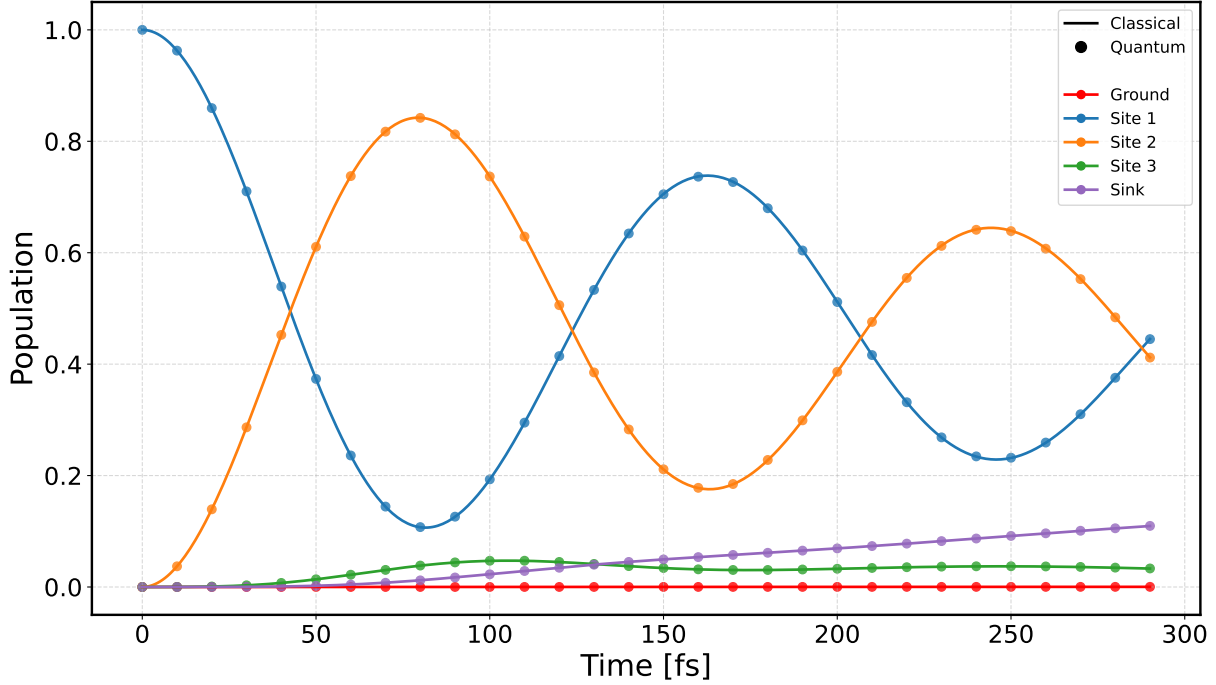


Figure 6.17: Results for the calculation of the populations using the enhanced algorithm. Since the quantum state is reused for each calculation, a time step of 1 femtosecond can be used for 290 iterations, which yields more accurate results.

To obtain the results of Figure 6.17, different settings for the Qiskit quantum computer simulator were used. Instead of running an experiment that returns the counts for each site at each time step as previously, a different instruction was used to obtain the full output state vector for each iteration of the circuit. Hence, no statistics are involved, leading to results that exactly match the ones calculated classically. Of course, this works only theoretically, and in practice, quantum tomography must be applied to retrieve the full output state.

6.3 Non-Markovian Dynamics

So far, we have discussed Markovian dynamics, where we assumed that the system changes on much larger timescales than the bath. This approximation has led to the Markovian Lindblad equation in Section 4.1. In Section 4.2.6, we proved that the Lindblad equation has an operator sum representation. However, this does not imply that the operator sum representation is limited to Markovian dynamics.

Hence, we follow Dan et al. [18] to find a non-Markovian object, called $G(t)$, that allows us to write the

non-Markovian time evolution of an open quantum system as an operator sum representation. Similar to the algorithm proposed by Seneviratne et al. [16], $G(t)$ is decomposed by a singular value decomposition and applied to a quantum circuit. It takes on the role of the Kraus operator for the non-Markovian dynamics. However, as we will see shortly, in this non-Markovian approach, the problem must be solved entirely by the classical algorithm. Consequently, no computational advantage is gained from using a quantum computer. The algorithm primarily demonstrates that non-Markovian dynamics can, in principle, be implemented on a quantum computer and does not permit the application of the enhanced algorithm.

To address this limitation, an alternative non-Markovian method [16] is introduced in Section 6.3.4, which employs a path integral formalism to construct the non-Markovian propagator. This approach generates a superoperator $\mathcal{L}(t)$ from which the Kraus operators, that capture the non-Markovian effects, can be derived. Consequently, the enhanced algorithm can be applied, extending its utility to non-Markovian dynamics.

6.3.1 The Non-Markovian Propagator

Our goal is to find a non-Markovian object $G(t)$ that acts as a propagator from time 0 to time t . Hence, applying $G(t)$ on a quantum state $|\psi\rangle$ should yield:

$$|\psi(t)\rangle = G(t) |\psi(0)\rangle. \quad (6.45)$$

The propagator $G(t)$ is a non-unitary object and thus reflects the non-unitary nature of open quantum systems. To construct $G(t)$, we employ the HEOM approach, which was introduced in Section 4.3. To this end, we calculate the system density matrix for a specific time step and extract its diagonal elements, which correspond to the populations. This must be repeated for an initial excitation on every site and for each time t . Then, the populations on site m form the m^{th} column of $G(t)$. For clarity, Equation 6.46 shows $G(t)$ for the four-site FMO complex:

$$G(t) = \left\{ \begin{pmatrix} \rho_{11}^1(t_0) & \rho_{11}^2(t_0) & \rho_{11}^3(t_0) & \rho_{11}^4(t_0) \\ \rho_{22}^1(t_0) & \rho_{22}^2(t_0) & \rho_{22}^3(t_0) & \rho_{22}^4(t_0) \\ \rho_{33}^1(t_0) & \rho_{33}^2(t_0) & \rho_{33}^3(t_0) & \rho_{33}^4(t_0) \\ \rho_{44}^1(t_0) & \rho_{44}^2(t_0) & \rho_{44}^3(t_0) & \rho_{44}^4(t_0) \end{pmatrix}, \begin{pmatrix} \rho_{11}^1(t_1) & \rho_{11}^2(t_1) & \rho_{11}^3(t_1) & \rho_{11}^4(t_1) \\ \rho_{22}^1(t_1) & \rho_{22}^2(t_1) & \rho_{22}^3(t_1) & \rho_{22}^4(t_1) \\ \rho_{33}^1(t_1) & \rho_{33}^2(t_1) & \rho_{33}^3(t_1) & \rho_{33}^4(t_1) \\ \rho_{44}^1(t_1) & \rho_{44}^2(t_1) & \rho_{44}^3(t_1) & \rho_{44}^4(t_1) \end{pmatrix}, \dots \right\}. \quad (6.46)$$

Here, $G(t)$ is a list where the first element corresponds to G at time step t_0 , the second to G at time step t_1 , and so on. The matrix elements ρ_{11} to ρ_{44} are the diagonal entries of ρ_S , and thus correspond to the population on site 1 to site 4. The superscripts in ρ^1 to ρ^4 indicate the site of the initial excitation. This implies that the first element in G at time step t_0 is a unit matrix, since the column for an initial excitation at site 1 has $\rho_{11} = 1$ and zeros elsewhere. For an initial excitation on site two, we have $\rho_{22} = 1$, and so on.

6.3.2 Construction of the Circuit

To construct a circuit for the non-Markovian dynamics, we proceed as in Section 6.1.3. To this end, we first perform a singular value decomposition on $G(t)$, which reads:

$$G(t) = U\Sigma V^\dagger, \quad (6.47)$$

where U and $V^\dagger$ are already unitary, and Σ is a non-unitary diagonal matrix. To append Σ on a quantum circuit that only allows for unitary operations, it must be dilated to a unitary matrix U_Σ :

$$U_\Sigma = \begin{pmatrix} \Sigma_+ & 0 \\ 0 & \Sigma_- \end{pmatrix}, \quad (6.48)$$

where $\Sigma_{\pm jj} = \sigma_{jj} \pm i\sqrt{1 - \sigma_{jj}^2}$ with σ_{jj} being the diagonal elements of Σ . Since we construct Σ_{+jj} and Σ_{-jj} for each diagonal element σ_{jj} of Σ , the resulting unitary U_Σ is twice as large as Σ . Applying U , $V^\dagger$, and U_Σ to the circuit yields:

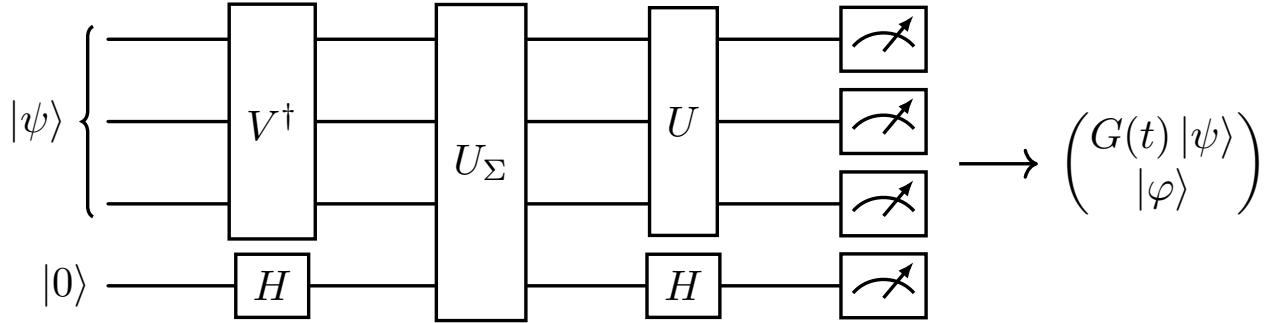


Figure 6.18: Quantum circuit for the calculation of the non-Markovian dynamics. The operators $V^\dagger$ and U stem directly from a singular value decomposition of $G(t)$. To obtain U_Σ , a dilation was needed, since Σ is not unitary, and to compute $G(t)$, the HEOM method was used.

The action of the circuit on the input state $|\psi\rangle \otimes |0\rangle$ can be calculated in the same way as in Section 6.1.3:

$$HU_\Sigma V^\dagger H \left(|\psi(0)\rangle \otimes |0\rangle \right) = \frac{1}{2} \begin{pmatrix} U(\Sigma_+ + \Sigma_-) V^\dagger |\psi(0)\rangle \\ U(\Sigma_+ - \Sigma_-) V^\dagger |\psi(0)\rangle \end{pmatrix} = \begin{pmatrix} G(t) |\psi(0)\rangle \\ |\varphi\rangle \end{pmatrix} = \begin{pmatrix} |\psi(t)\rangle \\ |\varphi\rangle \end{pmatrix}. \quad (6.49)$$

Here, H acts solely on the ancilla qubit, which is initialized as $|0\rangle$, and U , as well as $V^\dagger$, act on the three system qubits. Since U_Σ has twice the dimension of Σ , it acts on all four qubits. The output vector consists of two parts, where the first half contains the propagated wave function $|\psi(t)\rangle$, and the second half a nonphysical vector $|\varphi\rangle$, which can be discarded. The system density matrix can thus be calculated as:

$$\rho_S(t) = G(t) |\psi(0)\rangle \langle \psi(0)| G^\dagger(t) = |\psi(t)\rangle \langle \psi(t)|. \quad (6.50)$$

A summary of the procedure for the calculation of the non-Markovian dynamics for the seven-site FMO complex is schematically illustrated in Figure 6.19.

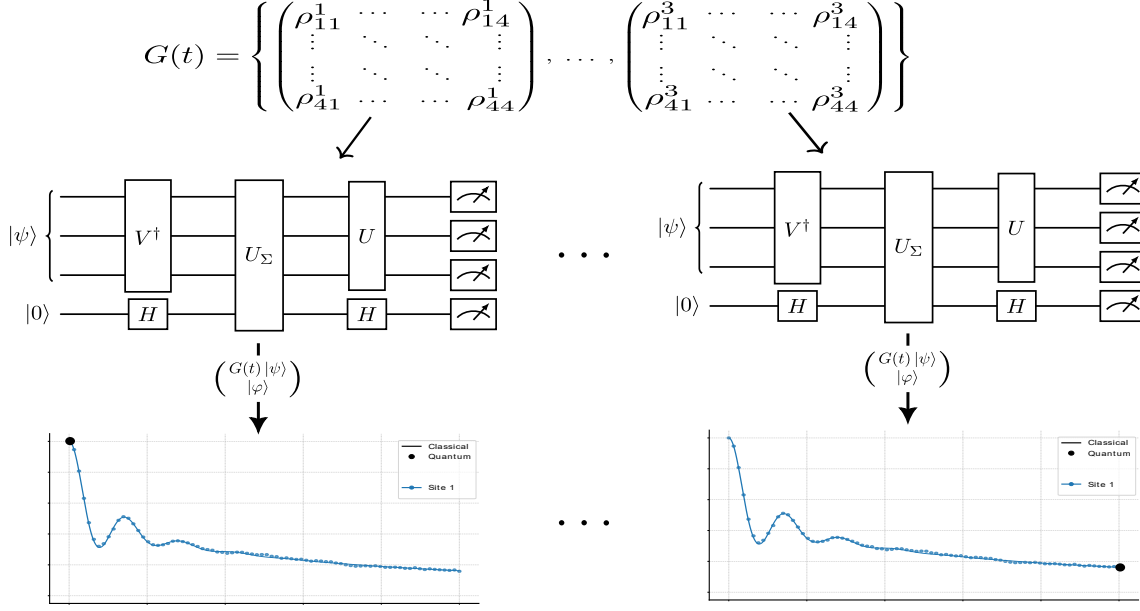


Figure 6.19: Schematic illustration of the non-Markovian algorithm. The elements of the list $G(t)$ correspond to the different time steps. For each element, a singular value decomposition is performed and the resulting matrices are appended onto the circuit. The circuit calculates the time evolution from time 0 to time t , which is shown as a function of time.

6.3.3 Reproduction of the Results for the FMO Complex

The time evolution of $\rho_S(t)$ is calculated for an initial excitation at site 1 and can be seen in Figure 6.20. The vertical axis represents the population, while the horizontal axis indicates time ranging from 0 to 1000 femtoseconds. The solid lines denote the classical result obtained by multiplying $G(t)$ with:

$$\text{vec}(\rho_S(0)) = (1, 0, 0, 0)^T. \quad (6.51)$$

Considering Equation 6.46, it is easy to see that this multiplication yields the populations for an initial excitation at site 1. The circles denote the result obtained with the quantum computer simulator from the Python library Qiskit. To arrive at these results, the circuit was applied to each element of the list $G(t)$ in Equation 6.46. To obtain sufficient statistics, the experiment was repeated 10000 times.

For all seven baths, the same Drude-Lorentz spectral density was used with parameters $\lambda = 35 \text{ cm}^{-1}$, $\nu = 106.18 \text{ cm}^{-1}$, and $T = 300 \text{ K}$. To obtain sufficient accuracy, the hierarchy of the differential equations was terminated at level four, and two terms were used in the expansion of the bath correlation function [86], which is given in appendix B.3. Comparing the non-Markovian dynamics in Figure 6.20 with the Markovian dynamics in Figure 6.5, we observe that in the non-Markovian case, the oscillations in the population between approximately 0 and 200 femtoseconds are more pronounced. The reason for this behavior is that in the Markovian approximation, the backflow of information from the bath into the system is neglected, since in Equation 4.5, the density matrix was approximated to be constant on the timescale of the bath evolution.

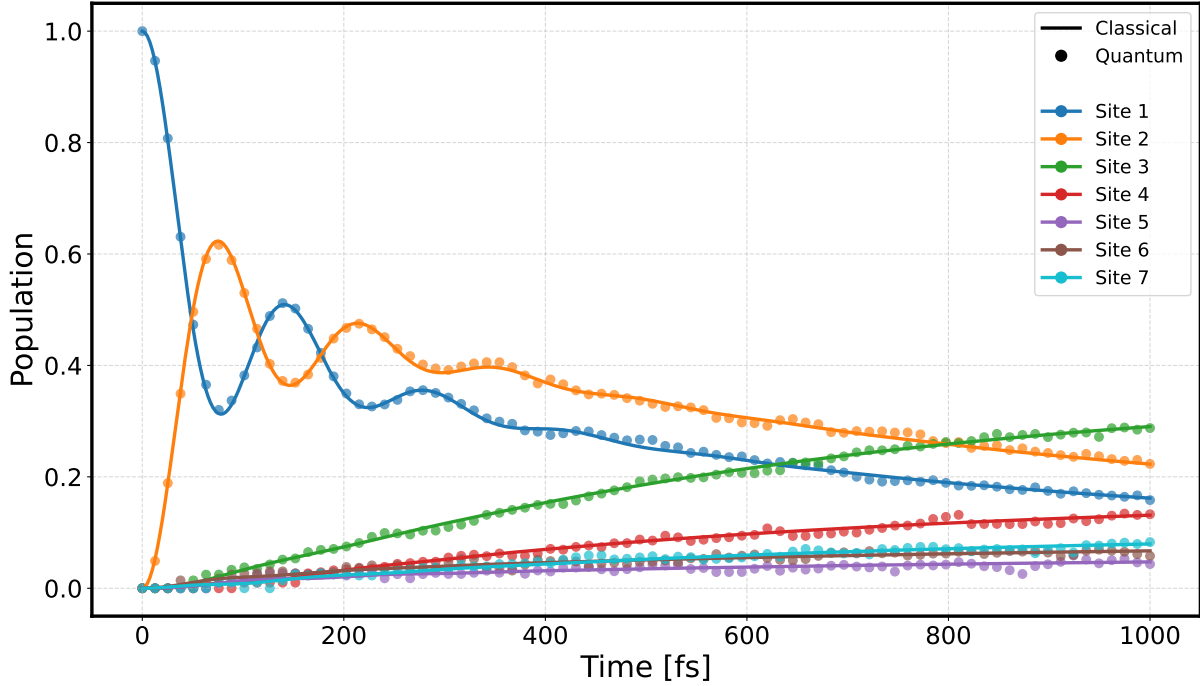


Figure 6.20: Populations as a function of time for the non-Markovian dynamics calculated with the HEOM method. The seven sites are indicated by different colors, and the solid lines depict the results obtained with the classical algorithm, whereas the results from the quantum computer simulator are illustrated with circles. The initial excitation is on site 1, and in comparison with the Markovian dynamics, the oscillations of the population are more pronounced.

The main caveat of this non-Markovian approach is that the time evolution must first be calculated using a classical algorithm. Thus, one could simply plot the elements of $G(t)$ without any need for a quantum computer. The algorithm should merely demonstrate that, in principle, the non-Markovian dynamics of an open quantum system can be calculated on a quantum computer.

6.3.4 Enhanced Algorithm

In this section, we introduce an alternative method for the calculation of the non-Markovian dynamics that permits the application of the enhanced algorithm. In a method proposed by Seneviratne et al. [16], a tensor network path-integral formalism is employed to compute the non-Markovian superoperator $\mathcal{L}(t)$ that solves Equation 6.10. We will outline the procedure only very briefly.

The time evolution of the reduced density matrix can be calculated using Feynman's path integral formalism [87] as follows:

$$\rho_s(s_t^+, s_t^-) = \int \mathcal{D}s^+ \int \mathcal{D}s^- \langle s_0^+ | \rho_0(0) | s_0^- \rangle \exp \left\{ \frac{i}{\hbar} (S[s^+] - S[s^-]) \right\} \text{IF}[s^+, s^-], \quad (6.52)$$

where

$$\text{IF}[s^+, s^-] = \exp \left\{ -\frac{1}{\hbar} \int_0^t dt' \int_0^{t'} dt'' [s^+(t') - s^-(t')] \right. \\ \left. [\alpha(t' - t'')s^+(t'') - \alpha^*(t' - t'')s^-(t'')] \right\}. \quad (6.53)$$

is the Feynman-Vernon influence functional [87][25]. The variables $s^+(t)$ and $s^-(t)$ label the ket- and bra-branches of the time contour, respectively, with the former evolving under U and the latter under $U^\dagger$. The expressions $S[s^+]$ and $S[s^-]$ are the action integrals for the free system propagation, and $\langle s_0^+ | \rho_0(0) | s_0^- \rangle$ is the initial state. The nonlocal memory kernel $\alpha(t' - t'')$ captures the non-Markovianity and can be obtained from the bath correlation function in equation 4.81. For further details on the tensor network path integral formalism I refer to Reference [88].

Equation 6.52 defines an exact linear map $\mathcal{L}(t)$ that governs the non-Markovian dynamics of the system. Several numerically efficient methods have been developed to implement this map, including quasi-adiabatic propagator path integrals [89][90], HEOM [91][92], and the tensor network path-integral formalism [93][94], which was used in Seneviratne et al. to construct the linear map, i.e., the corresponding superoperator $\mathcal{L}(t)$.

Having obtained $\mathcal{L}(t)$, the procedure follows the steps introduced in Section 6.1 and is illustrated in Figure 6.10. First, the corresponding Kraus operators, which capture the non-Markovian dynamics, are derived via the Choi decomposition. Then, the quantum circuit is constructed through a singular value decomposition and the required dilation. The enhanced algorithm can be applied using Equation 6.27 together with the circuit shown in Figure 6.6, where the Kraus operators M_k encode the non-Markovian behavior. Once the Kraus operators are obtained, the procedure is identical to the Markovian case in Figure 6.10, where the enhanced algorithm has already been validated. Consequently, the plots produced with this method would coincide with those shown in Seneviratne et al. [16], where the detailed results are presented.

Just as in the Markovian case, this approach effectively transfers the majority of the computational workload from classical to the quantum routines. As a result, the enhanced algorithm is no longer limited to Markovian dynamics and can be applied to non-Markovian processes as well.

6.4 Quantum Tomography

The basic principles of quantum tomography are introduced in Section 5.8. To illustrate its feasibility, we first reproduce the time evolution for the FMO complex, as described in Section 6.1, for the seven-site FMO complex. We then apply quantum tomography to the four-site FMO complex using the enhanced algorithm. In this procedure, the state obtained from the previous measurement is reused in subsequent steps, which leads to an accumulation of errors over time and hence increasingly inaccurate results. Both simulations were performed with Qiskit's StateTomography library using the Gaussian linear least-squares fitter [73].

In Figure 6.21 we follow the approach by Seneviratne et al. [16] described in Section 6.1, which repeatedly starts the time evolution at $\rho_S(0)$ for each step and does not reuse the quantum state. Instead of calculating

the system density matrix via the outer product of $M_k(t) |\psi\rangle$, quantum tomography was applied using 1000 shots for each of the 40 time steps. The circles are calculated with the StateTomography simulator from the Python library Qiskit, and the solid lines with the classical method, which uses the Lindblad equation for 400 time steps. As shown, the two results agree very well, demonstrating the viability of the quantum algorithm.

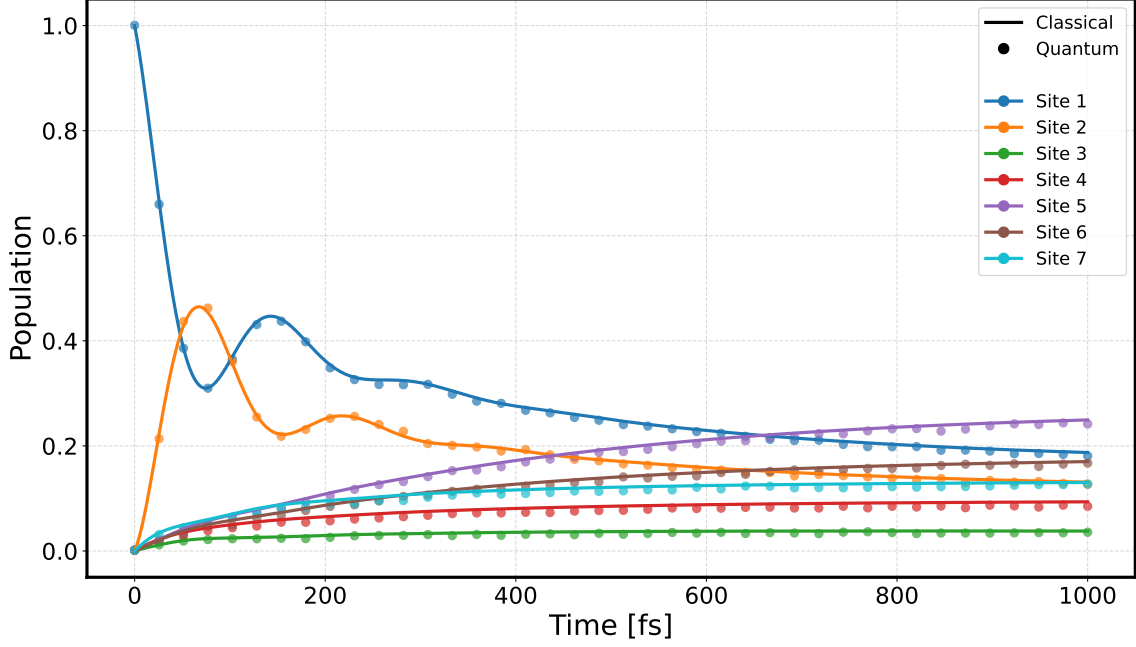


Figure 6.21: Quantum tomography applied to the seven-site FMO complex using the algorithm presented in Section 6.1. The quantum states are not reused for a subsequent calculation, and hence the results obtained with the classical routine and the quantum algorithm are in good agreement.

In Figure 6.22, quantum tomography is applied to the enhanced algorithm for the four-site FMO complex. Extending the approach to the seven-site case would require 7 qubits, which makes the procedure computationally too demanding, as the linear least-squares fitter provided by Qiskit is only suitable for systems up to 6 qubits [76]. Investigating larger systems therefore requires more advanced techniques, as discussed in the [outlook](#).

The output of the quantum circuit is the vectorized system density matrix, which must be reshaped into a quadratic matrix. Here, the challenge arises that quantum tomography yields a density matrix and not a state vector, i.e., $|\text{vec}(\rho_S)\rangle \langle \text{vec}(\rho_S)|$ instead of $|\text{vec}(\rho_S)\rangle$. Therefore, to recover $|\text{vec}(\rho_S)\rangle$, one must perform an eigendecomposition of $|\text{vec}(\rho_S)\rangle \langle \text{vec}(\rho_S)|$ and extract the eigenvector corresponding to the largest eigenvalue. If the quantum tomography was accurate, the eigenvalue is very close to 1, which implies that $|\text{vec}(\rho_S)\rangle \langle \text{vec}(\rho_S)|$ effectively describes a pure state.

However, the enhanced algorithm introduces additional errors. Since the quantum states from previous time steps are used for the subsequent steps, the error created in one time step is transferred to the next time step and thus adds up. This leads to an increasing divergence between the results produced by the quantum routine and the benchmarks obtained with the classical method. For the first time steps the

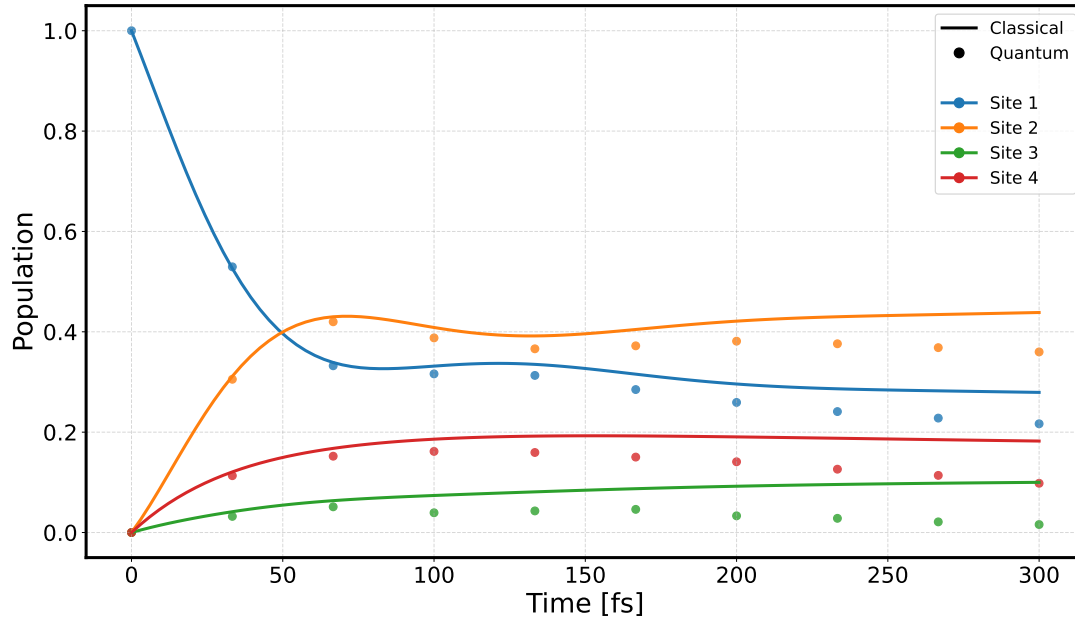


Figure 6.22: Quantum tomography applied to the four-site FMO complex using the enhanced algorithm. To keep the calculation time manageable, only four sites of the FMO complex are considered. Measurement results were collected from 1000 circuit executions to statistically reconstruct the quantum state. Since the quantum state of the previous time step is reused for the following time step, the error builds up and the results diverge.

results in Figure 6.22 align pretty well for all four sites but differ more and more for later time steps.

Quantum tomography is a very computationally expensive process since numerous measurements must be conducted in different bases. In addition the enhanced algorithm uses the vectorized density matrix as an input state and hence enlarges the quantum circuit by two qubits, which further increases the computational cost.

Nevertheless, the enhanced approach enables shifting the vast majority of the calculations from classical processing to the quantum algorithm. However, to obtain reliable results and extend the algorithm to large circuits, improvements in quantum tomography or alternative methods for full quantum state reconstruction are necessary. For further discussion on potential advances in quantum tomography, I refer to the [outlook](#).

7 Conclusion

This thesis addresses a central challenge in quantum simulations by exploring the implementation of non-unitary dynamics on hardware that is inherently limited to unitary operations. Starting from the Kraus operator sum representation and established dilation techniques, prior implementations are reproduced and, importantly, an enhanced algorithm is developed that substantially reduces classical preprocessing by reusing the quantum state produced at the previous time step. Consequently, a much larger portion of the propagation can be carried out on the quantum device itself rather than on a classical preprocessor. The shift in workload is particularly important for the simulation of large quantum systems, where the exponential growth of the state space makes classical approaches infeasible. A common example is the many-particle problem, where a state of N interacting particles requires 2^N amplitudes, making direct simulations intractable. A quantum computer avoids this scaling since the Hamiltonian directly drives the natural unitary evolution of the qubits.

The thesis investigates the implementation of both Markovian and non-Markovian descriptions of open quantum system dynamics. For the Markovian case, the existing approaches are reproduced, following Seneviratne et al. [16] and Hu et al. [17]. It is shown that these methods require considerable classical effort prior to any quantum execution, which includes the construction and eigendecomposition of the Choi matrix, a singular value decomposition, and dilation techniques. In practice, these Markovian schemes therefore act largely as a proof of principle in which classical processing performs the majority of the computational workload. To address this limitation, an enhanced algorithm is presented that computes the Kraus operators only once for a fixed time step Δt and initializes each iteration with the quantum state returned by the previous circuit execution. This reuse strategy increases efficiency, enabling the calculation of more time steps, permits repeated application of the same circuit, and shifts the dominant computational workload toward the quantum processor. Numerical implementations for the FMO complex in an idealized quantum simulator confirm that the enhanced method can, in theory, provide finer time resolution of the propagation at substantially lower classical cost. However, to realize the algorithm in practice, quantum tomography is required, which substantially increases the computational cost.

The first approach for the non-Markovian implementation uses HEOM to construct the non-Markovian propagator $G(t)$, as proposed by Dan et al. [18]. This reveals a fundamental limitation, since the non-Markovian propagator must be calculated on a classical computer. In the presented implementation, the matrix elements of $G(t)$ are computed classically via HEOM before any quantum routine is employed. Therefore, the full time evolution is already available classically, and no computational advantage is realized from the quantum circuit. To compute $G(t)$, the quantum device is not required, since the classical preprocessing produces the complete evolution in advance. Hence, the enhanced algorithm developed in this work cannot be applied. In contrast, the second approach by Seneviratne et al. [16] employs a tensor network path integral formalism to determine the non-Markovian Kraus operators. Once these operators are known, the procedure follows the same steps as in the Markovian case, where the enhanced algorithm was already validated. This shows that the method is not limited to Markovian dynamics and can also be applied to non-Markovian dynamics.

To investigate whether the enhanced scheme can be realized experimentally, quantum tomography is employed. Since the recycled state must also contain phase information and not only populations, the full quantum state has to be reconstructed after each circuit execution. When implemented as a Gaussian linear least-squares fitter, quantum tomography demonstrates high accuracy in reproducing the algorithm, yielding results that closely match the Lindblad benchmark for cases where states are not reused across time steps. However, for the enhanced algorithm, tomography introduces the practical difficulty that each reconstruction leads to a small error. Since the reconstructed state is reused in the subsequent iteration, these errors accumulate, leading to a growing deviation from the classical benchmark. Thus, while tomography enables state reuse in principle, its finite accuracy becomes the dominant source of error in iterative, state-recycling protocols.

In summary, the results offer a clear evaluation of the algorithm for simulating the dynamics of open quantum systems. With the proposed reuse strategy, Markovian as well as non-Markovian open-system dynamics can be shifted primarily onto quantum hardware, yielding significant reductions in classical computation cost and improved scalability. Finally, while feasible, quantum tomography is both costly and error-sensitive, making it a critical bottleneck in schemes that reuse full quantum states.

This thesis presents a concrete and reproducible approach for bringing more open-system simulations onto quantum devices. In addition, it identifies the main remaining challenges that primarily rely on full quantum state reconstruction. The enhanced algorithm represents a significant methodological step forward, but achieving practical quantum advantage will require further advances in state estimation and noise mitigation, with a particular focus on reducing the complexity and computational cost of quantum tomography.

8 Outlook

The remaining limitation identified in this thesis primarily concerns the use of quantum tomography. As applied here, it is not suitable for the advanced state-recycling algorithm, since small reconstruction errors accumulate when the reconstructed state is reused in subsequent steps, which becomes the dominant source of error. The following outlines concrete research directions that could address this issue, together with references to relevant literature that may be adapted for application within this work.

A first approach is to replace full quantum tomography with scalable state representations and measurement compressions. Classical shadow tomography is applicable to a wide range of quantum systems and provides a way to create compact classical sketches that predict large numbers of observables with far fewer measurements than full tomography [95][96][97]. The classical shadow is an efficient method for creating a precise yet minimal classical description of quantum states. Rather than attempting a full reconstruction of the state, which becomes exponentially demanding with system size, this technique focuses on gathering sufficient data to predict a large number of state properties. The creation of a classical shadow is an iterative process composed of a few simple steps. The process begins with an unknown quantum state, denoted as ρ . A randomly selected unitary transformation, U , is applied to the state $\rho \mapsto U\rho U^\dagger$, which is followed by the measurement of each qubit in the computational basis. From the measurement outcome and the known unitary transformation U , a single classical snapshot of the state is computed and stored. This process is repeated N times, resulting in a collection of N independent snapshots, which is referred to as the classical shadow of the original state ρ . The primary advantage of the classical shadow over traditional quantum tomography lies in its efficiency. Conventional quantum state tomography scales exponentially with the number of qubits to fully reconstruct an n -qubit state, whereas the classical shadow approach needs only $\mathcal{O}(\log M)$ measurements to accurately predict M different properties of the state, making it far more practical for large-scale quantum systems [98].

Next, we introduce a method of particular interest, as it is highly efficient especially for low-rank matrices. Since the state obtained from a quantum circuit is pure and therefore of rank one, compressed sensing tomography can be used to reconstruct it from far fewer measurements by exploiting rank sparsity. This approach uses the fact that many practically relevant states, such as the output states of quantum circuits, are pure or nearly pure, meaning that their density matrices have a rank close to one. Instead of determining all degrees of freedom of the matrix, only a random subset of simple observables is measured. From these measurements, a convex optimization problem is solved that minimizes the trace norm, which naturally favors low-rank solutions and thus enables a reliable reconstruction of the desired density matrix from limited data. Since nearly pure states encode their information in far fewer parameters, significantly fewer measurements are required. Theoretically, instead of $\mathcal{O}(d^2)$, only about $\mathcal{O}(rd \log^2 d)$ measurements for a $d \times d$ matrix are sufficient, where r denotes the rank of the matrix. For pure states with $r = 1$, this yields roughly $\mathcal{O}(d \log^2 d)$. In practice, it reduces the measurement effort for large systems from a quadratic to an almost linear scaling, potentially mitigating the drift observed in Figure 6.22 [99].

A particularly powerful approach is neural-network-based quantum state tomography, which enables the reconstruction of complex, high-dimensional quantum states from a manageable number of experimental data. While traditional methods become infeasible due to the sheer number of measurements required

for large quantum systems, this technique uses machine learning to find a more efficient solution. Machine learning techniques can be used to perform tomography of highly entangled states with more than a hundred qubits, to a high degree of accuracy [100]. Instead of measuring an exponentially growing number of observables, this method uses an artificial neural network to learn a compact representation of the quantum state. The network is trained on a limited set of measurement outcomes, or “snapshots”, of the system. This approach is promising since machine learning is particularly powerful at compressing high-dimensional data into low-dimensional representations. The principle has already been successfully applied to other quantum problems like classifying phases of matter and simulating many-body systems [100]. By compressing the state’s information into an efficient model, neural-network tomography drastically reduces the number of measurements needed as well as the overall computational cost. This approach positions research at the forefront of two rapidly developing fields, quantum technologies and artificial intelligence, where advances in efficient state reconstruction can have far-reaching implications for fundamental science and practical applications.

In conclusion, traditional quantum tomography faces major challenges in scaling due to its exponential resource requirements. The methods discussed in this outlook, including classical shadow tomography, compressed sensing, and neural-network-based approaches, provide promising alternatives. Each method relies on a different principle, such as efficiently predicting properties, exploiting the low-rank structure of pure states, or using the data-compression capabilities of machine learning, to greatly reduce the experimental and computational effort. Although these approaches are still active areas of research, they point toward a future in which quantum state tomography can move beyond small-scale systems and become a practical tool for reliably characterizing complex and highly entangled states with unprecedented accuracy. Continued research in this field will be essential for validating and fully utilizing advanced quantum resources.

A Appendix

A.1 Derivation of $S_\alpha^\dagger(\omega) = S_\alpha(-\omega)$

We compute the adjoint of $S_\alpha(\omega)$ from Equation 4.10. Using the standard property $(XY)^\dagger = Y^\dagger X^\dagger$ for two operators X and Y , and the fact that the eigenstates satisfy $|m\rangle^\dagger = \langle m|$, we have:

$$\begin{aligned} S_\alpha^\dagger(\omega) &= \left[\hbar \sum_{mn} \delta(E_n - E_m - \hbar\omega) \langle m| S_\alpha |n\rangle |m\rangle \langle n| \right]^\dagger \\ &= \hbar \sum_{mn} \delta(E_n - E_m - \hbar\omega) \langle n| S_\alpha^\dagger |m\rangle |n\rangle \langle m|. \end{aligned} \quad (\text{A.1})$$

Renaming m as n and n as m , and using $S_\alpha^\dagger = (a^\dagger a)^\dagger = a^\dagger a = S_\alpha$, we arrive at:

$$\begin{aligned} S_\alpha^\dagger(\omega) &= \hbar \sum_{mn} \delta(E_m - E_n - \hbar\omega) \langle m| S_\alpha^\dagger |n\rangle |m\rangle \langle n| \\ &= \hbar \sum_{mn} \delta(E_n - E_m + \hbar\omega) \langle m| S_\alpha |n\rangle |m\rangle \langle n| = S_\alpha(-\omega). \end{aligned} \quad (\text{A.2})$$

In the second last equality, we used the property that $\delta(\alpha x) = \frac{1}{|\alpha|} \delta(x)$, which yields $\delta(-x) = \delta(x)$ for $\alpha = -1$.

A.2 Derivation of $C_\alpha(-\tau) = C_\alpha^*(\tau)$

The Hamiltonian of an open quantum system is given in Equation 2.1 as $H = H_S + H_B + H_{SB}$. For the transformation of an operator B_α from the Schrödinger picture into the interaction picture, we thus obtain:

$$\tilde{B}_\alpha(t) = e^{\frac{i}{\hbar} H_0 t} B_\alpha e^{-\frac{i}{\hbar} H_0 t}, \quad (\text{A.3})$$

where we defined $H_0 \equiv H_S + H_B$. According to Equation 4.18, the bath correlation function is defined as:

$$C_\alpha(-\tau) \equiv \langle \tilde{B}_\alpha(0) \tilde{B}_\alpha(\tau) \rangle = \text{Tr}_B \left[\tilde{B}_\alpha(0) \tilde{B}_\alpha(\tau) \rho_B \right], \quad (\text{A.4})$$

where ρ_B is the density matrix of the bath, and Tr_B denotes the trace over the bath degrees of freedom. Next, we show the Hermiticity of $\tilde{B}_\alpha(t)$. We thus recall from Equation 2.14:

$$H_{SB} = \sum_{m=1}^N \sum_{\xi} \hbar \omega_{\xi m} d_{\xi m} (b_{\xi m}^\dagger + b_{\xi m}) \otimes |m\rangle \langle m|, \quad (\text{A.5})$$

and read off $B_\alpha = \hbar \omega_{\xi m} d_{\xi m} (b_{\xi m}^\dagger + b_{\xi m})$. Since, obviously, $B_\alpha = B_\alpha^\dagger$, we obtain:

$$\tilde{B}_\alpha^\dagger(t) = \left(e^{\frac{i}{\hbar} H_B t} B_\alpha e^{-\frac{i}{\hbar} H_B t} \right)^\dagger = e^{\frac{i}{\hbar} H_B t} B_\alpha^\dagger e^{-\frac{i}{\hbar} H_B t} = \tilde{B}_\alpha(t), \quad (\text{A.6})$$

and $\tilde{B}_\alpha(t)$ is Hermitian. Next, we consider the complex conjugate of $C_\alpha(-\tau)$. Since the trace is essentially a sum of the diagonal elements of a matrix, it holds that:

$$C_\alpha^*(-\tau) = \text{Tr}_B \left[\tilde{B}_\alpha(0) \tilde{B}_\alpha(\tau) \rho_B \right]^* = \text{Tr}_B \left[\left(\tilde{B}_\alpha(0) \tilde{B}_\alpha(\tau) \rho_B \right)^\dagger \right]. \quad (\text{A.7})$$

Taking the Hermitian conjugate of the product in the trace, and using that $\tilde{B}_\alpha(t) = \tilde{B}_\alpha^\dagger(t)$ and $\rho_B = \rho_B^\dagger$, we have:

$$C_\alpha^*(-\tau) = \text{Tr}_B \left[\left(\tilde{B}_\alpha(0) \tilde{B}_\alpha(\tau) \rho_B \right)^\dagger \right] = \text{Tr}_B \left[\rho_B \tilde{B}_\alpha(\tau)^\dagger \tilde{B}_\alpha(0)^\dagger \right] = \text{Tr}_B \left[\rho_B \tilde{B}_\alpha(\tau) \tilde{B}_\alpha(0) \right]. \quad (\text{A.8})$$

Using the cyclic property of the trace, we obtain:

$$C_\alpha^*(-\tau) = \text{Tr}_B \left[\rho_B \tilde{B}_\alpha(\tau) \tilde{B}_\alpha(0) \right] = \text{Tr}_B \left[\tilde{B}_\alpha(\tau) \tilde{B}_\alpha(0) \rho_B \right] = C_\alpha(\tau). \quad (\text{A.9})$$

Applying the complex conjugate yields $C_\alpha(-\tau) = C_\alpha^*(\tau)$.

A.3 Proof that Every Kraus Operator is a Contraction

In this proof, we follow the supplementary information of reference [85]. An operator A is a contraction if

$$\|A\| = \sup \frac{\|Av\|}{\|v\|} \leq 1, \quad (\text{A.10})$$

i.e., if it shrinks or preserves the norm of any vector. To preserve the probability, the Kraus operators must fulfill the property:

$$\sum_k M_k^\dagger M_k = I, \quad (\text{A.11})$$

where I is the identity. Suppose an arbitrary Kraus operator M_i is not a contraction. Then, there exists a vector v_i such that $\frac{\|M_i v_i\|}{\|v_i\|} = \sqrt{\frac{\mathbf{v}_i^\dagger M_i^\dagger M_i \mathbf{v}_i}{\|\mathbf{v}_i\|^2}} > 1$. This implies:

$$\sqrt{\frac{\mathbf{v}_i^\dagger \sum_k M_k^\dagger M_k \mathbf{v}_i}{\|\mathbf{v}_i\|^2}} > \sqrt{\frac{\mathbf{v}_i^\dagger M_i^\dagger M_i \mathbf{v}_i}{\|\mathbf{v}_i\|^2}} > 1, \quad (\text{A.12})$$

since each $M_k^\dagger M_k$ is positive-semidefinite, with $\mathbf{v}_i^\dagger M_k^\dagger M_k \mathbf{v}_i \geq 0$. However, due to $\sum_k M_k^\dagger M_k = I$, it also follows that:

$$\sqrt{\frac{\mathbf{v}_i^\dagger \sum_k M_k^\dagger M_k \mathbf{v}_i}{\|\mathbf{v}_i\|^2}} = \sqrt{\frac{\mathbf{v}_i^\dagger \mathbf{v}_i}{\|\mathbf{v}_i\|^2}} = 1. \quad (\text{A.13})$$

Hence, by contradiction, each M_k is a contraction.

B Appendix

B.1 Spectral Density

The spectral density describes the influence of the bath on the system at a certain frequency ω . It contains all the information on the bath, and hence must consider all bath modes ξ with frequency ω_ξ , which is expressed by the sum in Equation B.1. The bath couples to the system only when the frequency of the bath modes is resonant with the transition frequency between excitons in the system, which invokes a delta function. The strength of the coupling for a bath mode ξ is expressed by g_ξ . We thus introduce a general form of the spectral density [54]:

$$J(\omega) = \sum_{\xi} g_{\xi}^2 \delta(\omega - \omega_{\xi}). \quad (\text{B.1})$$

A large number of resonant bath modes and a strong coupling yield a higher intensity of the spectral distribution. Usually, the spectral density is assumed to be independent of the site, which yields $J(\omega) = J_i(\omega)$. The exact shape of the spectral density strongly depends on the system of interest. In general, finding an appropriate shape is a very delicate endeavor, as is argued by Kell et al. [101]. Since, for the FMO complex, no specific form for the spectral density is known [102], it is most commonly chosen to be of the Drude-Lorentz form [103]:

$$J(\omega) = 2\nu\lambda \frac{\omega}{\nu^2 + \omega^2}, \quad (\text{B.2})$$

which corresponds to an overdamped Brownian oscillator model [86]. Here, ν is the bath relaxation rate representing the width of the spectral density. It depends on the bath correlation time by $\nu = 1/\tau_c$, which is the timescale over which the fluctuations of the bath remain correlated. Therefore, a decreasing ν increases the non-Markovianity. Moreover, ν can be viewed as the characteristic frequency of the bath, which fulfills:

$$\frac{dJ(\omega)}{d\omega}(\omega = \nu) = 0. \quad (\text{B.3})$$

The reorganization energy λ is the energy dissipated by the nuclei when they relax to their new equilibrium position after an excitation to the electronic system. Hence, it is a measure for the coupling strength between the bath and the system. For further details on the reorganization energy, I refer to Section 2.2.2.

The action of the oscillators of the bath on the system can be interpreted as a kind of quantum mechanical friction. This leads to the classical dissipation of energy into the bath, and to a quantum mechanical effect which is called decoherence. The decoherence affects the quantum states of the system and is caused by a change in the relative phase of the states, leading to a loss of superposition. The phase change occurs as the result of a change in the site energy, which is caused by the excitation or deexcitation of the vibrational states of the bath [55].

The Drude-Lorentz spectral density is shown in Figure B.1, where $\lambda = 35\text{cm}^{-1}$ and $\nu = 106.18\text{cm}^{-1}$. It is a very convenient way of describing the influence of the bath on the system, since only the parameters λ and ν must be specified, and no specific definition of the bath degrees of freedom is necessary.

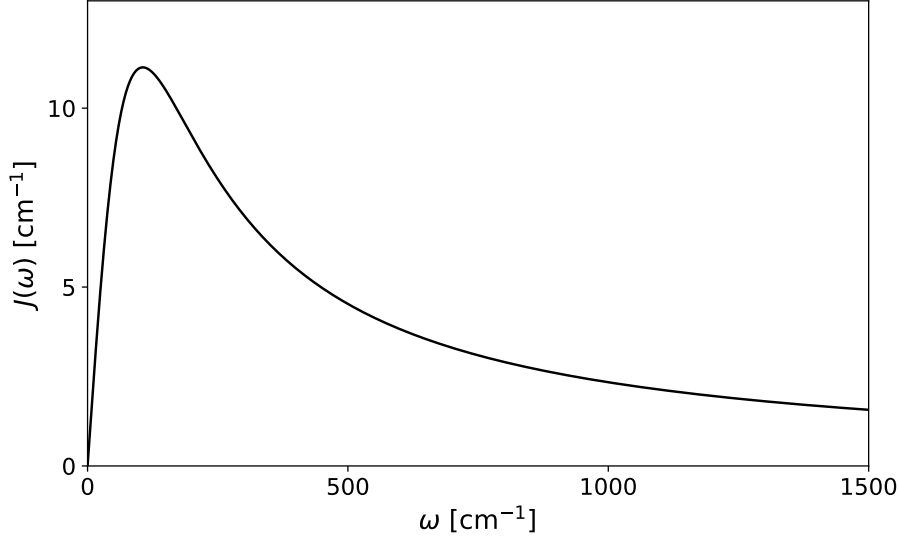


Figure B.1: Drude-Lorentz spectral density $J(\omega)$ as a function of the frequency ω . It describes the action of the bath oscillators on the system and shows high values for a strong coupling between the system and the bath.

B.2 Matsubara Series

The Matsubara series is obtained by approximating the integral of the bath correlation function in Equation 4.81. To this end, we follow reference [55] and start the derivation by rewriting Equation 4.81 in a more convenient form for a contour integration:

$$\begin{aligned}\alpha(t) &= \int_0^\infty J(\omega) \left[\coth\left(\frac{\beta\hbar\omega}{2}\right) \cos(\omega t) - i \sin(\omega t) \right] d\omega \\ &= \int_0^\infty \frac{2\lambda\nu\omega}{\nu^2 + \omega^2} \left[\coth\left(\frac{\beta\hbar\omega}{2}\right) \cos(\omega t) - i \sin(\omega t) \right] d\omega,\end{aligned}\tag{B.4}$$

where we inserted the Drude-Lorentz spectral density from Equation B.2. Expanding the sine and cosine functions in the exponentials, and rearranging terms, we get:

$$\begin{aligned}\alpha(t) &= 2\lambda\nu \int_0^\infty \frac{\omega}{\nu^2 + \omega^2} \left[\coth\left(\frac{\beta\hbar\omega}{2}\right) \cos(\omega t) - i \sin(\omega t) \right] d\omega \\ &= 2\lambda\nu \int_0^\infty \frac{\omega}{\nu^2 + \omega^2} \left[\frac{1}{2} \coth\left(\frac{\beta\hbar\omega}{2}\right) (e^{i\omega t} + e^{-i\omega t}) - \frac{i}{2i} (e^{i\omega t} - e^{-i\omega t}) \right] d\omega \\ &= \lambda\nu \int_0^\infty \frac{\omega}{\nu^2 + \omega^2} \left[\left(\coth\left(\frac{\beta\hbar\omega}{2}\right) - 1 \right) e^{i\omega t} + \left(\coth\left(\frac{\beta\hbar\omega}{2}\right) + 1 \right) e^{-i\omega t} \right] d\omega \\ &= \frac{\lambda\nu}{2} \int_{-\infty}^\infty \frac{\omega}{\nu^2 + \omega^2} \left[\left(\coth\left(\frac{\beta\hbar\omega}{2}\right) - 1 \right) e^{i\omega t} + \left(\coth\left(\frac{\beta\hbar\omega}{2}\right) + 1 \right) e^{-i\omega t} \right] d\omega.\end{aligned}\tag{B.5}$$

In the last equality, we used that $\alpha(t)$ is an even function, which yields $\int_0^\infty d\omega [\dots] = \frac{1}{2} \int_{-\infty}^\infty d\omega [\dots]$. Replacing ω by $-\omega$ in the first summand of the last expression of Equation B.5, we can simplify as follows: $(\coth(-x) - 1)(-\omega)e^{-i\omega t} = (-\coth x - 1)(-\omega)e^{-i\omega t} = \omega(1 + \coth x)e^{-i\omega t}$, where $x = \frac{\beta\hbar\omega}{2}$. We then have:

$$\begin{aligned}
\alpha(t) &= \lambda\nu \int_{-\infty}^{\infty} \frac{\omega}{\nu^2 + \omega^2} \left(1 + \coth\left(\frac{\beta\hbar\omega}{2}\right)\right) e^{-i\omega t} d\omega \\
&= \lambda\nu \int_{-\infty}^{\infty} \left(\frac{\omega e^{-i\omega t}}{\nu^2 + \omega^2} + \frac{\coth\left(\frac{\beta\hbar\omega}{2}\right) \omega e^{-i\omega t}}{\nu^2 + \omega^2} \right) d\omega \\
&= \int_{-\infty}^{\infty} f(\omega) d\omega.
\end{aligned} \tag{B.6}$$

The poles of the function $f(\omega)$ can be directly identified, which allows us to evaluate the integral via contour integration. For the first term, the poles are located at $\omega = \pm i\nu$. For the second term, the poles occur at $\omega = \pm i\nu$, as well as at $\omega = \pm \frac{i2\pi k}{\beta\hbar}$, where $k \in \mathbb{Z}$, since $\sinh\left(\frac{\beta\hbar\omega}{2}\right) = 0$ at these points, causing $\coth\left(\frac{\beta\hbar\omega}{2}\right)$ to diverge. Only $\alpha(t)$ with $t > 0$ yields physical results, and hence we must close the contour with a semiarc in the lower half plane to prevent the integral from diverging. Thus, we choose $\omega = -i\nu$, as well as $\omega = -i2\pi k/\beta\hbar$, and need an additional minus sign before the integral, since we run through the contour clockwise. Using the residue theorem and letting $R \rightarrow \infty$, we must evaluate:

$$\alpha(t) = -2\pi i \left(\text{Res}_{\omega=-i\nu} f(w) + \sum_{k=1}^{\infty} \text{Res}_{\omega=-i2\pi k/\beta\hbar} f(w) \right). \tag{B.7}$$

We start with the evaluation of the first residuum at $\omega = -i\nu$, and use $\coth(-ix) = \frac{\cosh(-ix)}{\sinh(-ix)} = \frac{e^{-ix} + e^{ix}}{e^{-ix} - e^{ix}} = i \frac{\cos(x)}{\sin(x)} = i \cot(x)$ in the last equality:

$$\begin{aligned}
\text{Res}_{\omega=-i\nu} f(w) &= \lim_{w \rightarrow -i\nu} (\omega + i\nu) f(w) = \lambda\nu \lim_{w \rightarrow -i\nu} (\omega + i\nu) \left[\frac{\omega e^{-i\omega t}}{(\omega + i\nu)(\omega - i\nu)} + \frac{\coth(\beta\hbar\omega/2) \omega e^{-i\omega t}}{(\omega + i\nu)(\omega - i\nu)} \right] \\
&= \lambda\nu \left[\frac{-i\nu e^{-\nu t}}{-2i\nu} - \frac{\coth(-i\beta\hbar\nu/2) i\nu e^{-\nu t}}{-2i\nu} \right] = \frac{\lambda\nu}{2} \left[1 + i \cot(\beta\hbar\nu/2) \right] e^{-\nu t}.
\end{aligned} \tag{B.8}$$

For the second residuum at $\omega = -i2\pi k/\beta\hbar$, we define the Matsubara frequencies as $\nu_k = 2\pi k/\beta\hbar$, and obtain:

$$\sum_{k=1}^{\infty} \text{Res}_{\omega=-i2\pi k/\beta\hbar} f(w) = \lambda\nu \sum_{k=1}^{\infty} \lim_{w \rightarrow -i\nu_k} \frac{w}{w^2 + \nu^2} e^{-i\omega t} \frac{2}{\beta\hbar} = \sum_{k=1}^{\infty} \frac{2i\nu\lambda\nu_k}{\hbar\beta(\nu_k^2 - \nu^2)} e^{-\nu_k t}, \tag{B.9}$$

where the residuum of $\coth\left(\frac{\beta\omega}{2}\right)$ is evaluated to $\frac{2}{\beta\hbar}$. The first term of $f(\omega)$ vanishes, since it does not contain the corresponding residue. Inserting the results for the residue in Equation B.7, we find the following series representation for the bath correlation function $\alpha(t)$:

$$\begin{aligned}
\alpha(t) &= -2\pi i \left(\frac{\lambda\nu}{2} \left[1 + i \cot(\beta\hbar\nu/2) \right] e^{-\nu t} + \sum_{k=1}^{\infty} \frac{-2i\nu\lambda\nu_k}{\hbar\beta(\nu_k^2 + \nu^2)} e^{-\nu_k t} \right) \\
&= \pi\lambda\nu \left[\cot(\beta\hbar\nu/2) - i \right] e^{-\nu t} + \sum_{k=1}^{\infty} \frac{4\pi\nu\lambda\nu_k}{\hbar\beta(\nu_k^2 + \nu^2)} e^{-\nu_k t}.
\end{aligned} \tag{B.10}$$

Equation B.10 is called the Matsubara series, and consists of a sum of decaying exponentials. The $\nu_k = 2\pi k/\beta\hbar$ are called the Matsubara frequencies, where the integer k determines the magnitude of the contribution to $\alpha(t)$ for each exponential. If these contributions are negligible for larger k , it is justified to truncate the series after a few exponentials.

B.3 Padé Frequency

In the following, the Padé series will be introduced very briefly. For details, I refer to Reference [55]. The Padé series is, similarly to the Matsubara series, an approximation for the bath correlation function in Equation 4.81. To this end, we follow Reference [55], and rewritten Equation 4.81 as:

$$\alpha(t) = 2\lambda\nu \int_{-\infty}^{\infty} d\omega \frac{\omega e^{-i\omega t}}{\nu^2 + \omega^2} \frac{1}{1 - e^{-\beta\hbar\omega}}, \quad (\text{B.11})$$

where we used the Drude-Lorentz spectral density from Equation B.2. We aim to express the integral in Equation B.11 as a fast converging sum. To proceed, we notice that Equation B.11 contains the term $\frac{1}{1 - e^{-\beta\hbar\omega}}$, which can be rewritten as:

$$\frac{1}{1 - e^{-x}} = \frac{e^{x/2}}{e^{x/2} - e^{-x/2}} = \frac{1}{2} \left(\frac{e^{x/2} + e^{-x/2}}{e^{x/2} - e^{-x/2}} + \frac{e^{x/2} - e^{-x/2}}{e^{x/2} - e^{-x/2}} \right) = \frac{1}{2} + \frac{1}{2} \frac{e^{x/2} + e^{-x/2}}{e^{x/2} - e^{-x/2}} \quad (\text{B.12})$$

$$= \frac{1}{2} + \frac{1}{2} \frac{\cosh(x/2)}{\sinh(x/2)} = \frac{1}{2} + \frac{1}{2} \coth(x/2). \quad (\text{B.13})$$

Using the expansion of the tanh, we get;

$$\tanh(y) = \frac{y}{1 + \frac{y^2}{3 + \frac{y^2}{5 + \ddots}}} \quad (\text{B.14})$$

and thus

$$\coth(y) = \frac{1}{y} + \frac{y}{3 + \frac{y^2}{5 + \frac{y^2}{\ddots}}} \quad (\text{B.15})$$

which yields

$$\frac{1}{1 - e^{-x}} = \frac{1}{2} + \frac{1}{2} \frac{2}{x} + \frac{1}{2} \frac{x/2}{3 + \frac{x^2/4}{5 + \frac{x^2/4}{\ddots}}} = \frac{1}{2} + \frac{1}{x} + x \frac{1/4}{3 + \frac{x^2/4}{5 + \frac{x^2/4}{\ddots}}}. \quad (\text{B.16})$$

So, we end up with the following approximation:

$$\frac{1}{1 - e^{-x}} \approx \frac{1}{2} + \frac{1}{x} + x\mathcal{C}_{2N}(x^2), \quad (\text{B.17})$$

where $\mathcal{C}_{2N}(x^2) = \sum_{j=1}^N \frac{2\eta_j}{x^2 + \xi_j^2}$, and

$$\eta_j = \left(N^2 + \frac{3}{2}N\right) \frac{\prod_{k=1}^{N-1} (\zeta_k^2 - \xi_j^2)}{\prod_{k \neq j}^N (\xi_k^2 - \xi_j^2)}, \quad j = 1, 2, \dots, N. \quad (\text{B.18})$$

Here, we define $\xi_j = \frac{2\pi}{c_j}$, and $\zeta_j = \frac{2\pi}{\tilde{c}_j}$, where c_j are the positive eigenvalues of the matrix Λ , and $\tilde{c}_j$ are the positive eigenvalues of the matrix $\tilde{\Lambda}$. The matrix Λ has dimension $2N \times 2N$, while $\tilde{\Lambda}$ has dimension $(2N - 1) \times (2N - 1)$. Their elements are given by:

$$\Lambda_{mn} = \frac{\delta_{m,n\pm 1}}{\sqrt{(2m+1)(2n+1)}}, \quad \tilde{\Lambda}_{mn} = \frac{\delta_{m,n\pm 1}}{\sqrt{(2m+3)(2n+3)}}. \quad (\text{B.19})$$

We have N parameters ξ_j , and $N - 1$ parameters ζ_j . Inserting Equation B.17 into Equation B.11 gives:

$$\alpha(t) = 2\lambda\nu \int_{-\infty}^{\infty} d\omega \frac{\omega e^{-i\omega t}}{\nu^2 + \omega^2} \left(\frac{1}{2} + \frac{1}{\beta\hbar\omega} + \frac{2\omega}{\beta\hbar} \sum_{j=1}^N \frac{\eta_j}{\nu_j^2 + \omega^2} \right), \quad (\text{B.20})$$

where the $\nu_j = \xi_j/\beta\hbar$ are the Padé frequencies. This integral can be evaluated with the same contour used to derive the Matsubara series:

$$\alpha(t) = \frac{2\lambda\pi}{\beta\hbar} \left(1 - \sum_{j=1}^N \frac{2\eta_j\nu^2}{\omega_j^2 - \nu^2} - \frac{i\nu\hbar}{\beta} \right) e^{-\nu t} + \frac{2\pi\nu}{\beta\hbar} \sum_{j=1}^N \frac{2\eta_j\omega_j\nu}{\omega_j^2 - \nu^2} e^{-\nu_j t}. \quad (\text{B.21})$$

Equation B.21 is called the Padé series, and consists of a sum of decaying exponentials. Similar to the Matsubara series, the Padé series must be truncated at some point. Since it converges faster to the bath correlation function, fewer summands need to be considered.

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